arXiv:submit/8213697 [math.NT] 12 Oct 2026

IMPROVED ZERO-FREE HALF-PLANES FOR THE QUASI-RIEMANN HYPOTHESIS

BAIYING LIU

ABSTRACT. Recently, a uniform $\frac{7}{8}$ zero-free half-plane for finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and its transfer to Dirichlet L-functions was given by OpenAI in [Ope26a]. In its proof, there are several parameters $b = \frac{1}{8}, \ell = \frac{1}{6}$, and κ. In this paper, via varying b, ℓ, and κ, following the same strategy, we slightly improve the bound $\frac{7}{8} = 0.875$ to

$$B_{\mathrm{new}} = \frac{18\xi_{\mathrm{new}} + 7}{18\xi_{\mathrm{new}} + 9} = 0.874957019420098946\ldots,$$

where ξ_{new} is the unique root in $(0.38858, 0.38859)$ of $1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0$. Explicitly, with $\arccos \in [0, \pi]$,

$$\xi_{\mathrm{new}} = \frac{20}{33} + \frac{\sqrt{1391}}{33}\cos\left(\frac{1}{3}\arccos\frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right).$$

The limiting parameters are

$$b_{\mathrm{new}} = \frac{8(3\xi_{\mathrm{new}} - 1)(12\xi_{\mathrm{new}}^2 - 40\xi_{\mathrm{new}} + 15)}{9(2\xi_{\mathrm{new}} + 1)(24\xi_{\mathrm{new}}^2 - 20\xi_{\mathrm{new}} + 5)},$$

$$\ell_{\mathrm{new}} = \frac{5 - 6\xi_{\mathrm{new}}}{9 + 18\xi_{\mathrm{new}}}, \qquad \kappa_{\mathrm{new}} = \frac{18\xi_{\mathrm{new}} + 5}{18\xi_{\mathrm{new}} + 9} = 2B_{\mathrm{new}} - 1.$$

We also prove its optimality within the explicitly specified exponent comparison system. This result has been formalized by Lean [Liu26].

CONTENTS

Introduction 2

Part I. Summary of the 7/8 proof 6
1. The original argument and its two-stage reduction 6
2. Arithmetic conventions 8
3. A criterion for zero-free continuation 11
4. Cubic theta series and the smoothed average 12
5. Zero detection by two character polynomials 19
6. The preliminary 11/12 bound 21
7. The modified average and its direct estimate 22
8. Local cancellation and principal residues 23
9. Moment estimates for the two detecting polynomials 26
10. Counting elements from simultaneous large values 29
11. Contour estimates and the endpoint inequality 31
12. Auxiliary parameter choices and the uniform comparison estimate 33
13. Mellin continuation and proof of the Hecke bound 34
14. Transfer to Dirichlet L-functions 35

Part II. A rational extension of the bound 35
15. The rational zero-free bound and its parameters 35

Date: 10 October 2026.

16. The direct estimate for balanced lengths 37
17. Euler products and extraction of the principal term 38
18. Character counts and nonprincipal exponents at the fixed moment parameter 40
19. Polynomial verification of the rational saving and the remaining norm ranges 43
20. Proof of the rational zero-free bound by the continuation criterion 45

Part III. An algebraic extension of the bound 47
21. The algebraic zero-free bound and its parameters 47
22. The direct estimate with free total length 47
23. Comparison of the character-count estimates 49
24. A polynomial proof of the algebraic endpoint inequality 50
25. An algebraic lower bound for the comparison constraints 51
26. Auxiliary estimates and proof of the algebraic zero-free bound 52

Part IV. A moment-parameter extension and the optimal comparison bound 54
27. The limiting zero-free bound and its parameters 54
28. Extending the unweighted fourth moment below three quarters 54
29. Counting by simultaneous large values with a variable moment parameter 58
30. Balanced averaging lengths and the critical counting class 59
31. A uniform proof of the endpoint inequality 60
32. The one-step zero-free bound via the continuation criterion 63
33. Proof of the limiting zero-free bound by iteration 65
34. Proof of optimality in the full comparison system 67
Appendix A. Theta coefficient conventions 74
Appendix B. Index of cited results 75
References 76

Introduction

In this paper, we study how varying the parameters in the cubic-theta argument of [Ope26a] improves its zero-free half-plane. After recalling the original argument, we derive three successive improvements and determine the optimal boundary within the specified system of estimates.

The quasi-Riemann hypothesis asks whether there is a fixed $B < 1$ such that $\zeta(s) \neq 0$ for $\operatorname{Re} s > B$, with its pole at $s = 1$ allowed. The uniform character version asks for the same B independently of conductor and height. A fixed half-plane supplies a uniform gap between zeros and the line $\operatorname{Re} s = 1$; classical zero-free regions instead narrow towards that line as conductor or height increases. For general analytic background, see [IK04, Chapter 5]; for the Dedekind zeta function of an abelian extension, and hence its Hecke L-function factors, a precise zero-free region allowing an exceptional real zero is [TZ19, Theorem 3.1]. Zero-density estimates bound the number of zeros in a region; for example, see [GM26, Theorem 1.2].

We follow the 30 September 2026 preprint [Ope26a], retaining its numbering in all citations to it. The companion [Ope26b] gives a different 11/12 argument. OpenAI's public `math` repository has a formalization scope note [Ope26c, Scope] that lists the 7/8 statements for zeta, Dirichlet and finite-order Hecke functions, with principal poles excluded; the paper's later applications are outside that scope.

Let $F = \mathbb{Q}(\sqrt{-3})$. Let $\mathcal{Z}_F$ be the set of nontrivial zeros of all primitive finite-order Hecke L-functions over F, of arbitrary conductor; poles are excluded. Put

$$\beta_* := \sup\big(\{1/2\} \cup \{\operatorname{Re}\rho : \rho \in \mathcal{Z}_F\}\big).$$

We study the dependence of the cubic-theta argument on its length parameters and on the moment parameter κ. The stated 7/8 bound of [Ope26a] is the starting point for three deductions from its intermediate estimates. The strongest deduction is the following.

Main theorem. *The family supremum satisfies*

$$\beta_* \le B_{\mathrm{new}} := \frac{18\xi_{\mathrm{new}} + 7}{18\xi_{\mathrm{new}} + 9} = 0.874957019420098946\ldots,$$

where $\xi_{\mathrm{new}} \in (0.38858, 0.38859)$ *is the unique root there of*

$$1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0.$$

Consequently every finite-order Hecke L-function over F *and every Dirichlet L-function, including* ζ*, is zero-free for* $\operatorname{Re} s > B_{\mathrm{new}}$*. Principal poles at* $s = 1$ *are allowed.*

The main theorem is proved in Theorem 27.1. In the original proof of OpenAI, there are several variables b, ℓ, and $3/4 \le \kappa \le 1$. Part II first varies ℓ while retaining both b and $3/4 \le \kappa \le 1$, obtaining

$$B_r = \frac{34999}{40000} = 0.874975.$$

Part III then varies both variables b and ℓ while retaining $3/4 \le \kappa \le 1$, obtaining

$$B_{\mathrm{alg}} = \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\ldots.$$

Part IV extends the unweighted fourth-moment estimate to smaller values of κ, reoptimizes the lengths and iterates the resulting bounds to obtain B_{new}. We prove that B_{new} is optimal in the extended system, including variation of its auxiliary parameters, in Theorem 34.4.

The choices of b *and* ℓ*.* In the original 7/8 argument the lengths are chosen with $b = 1/8$ and $\ell = 1/6$. Here $b = l_y - l_x$ is the difference between the two summation exponents, and ℓ is the total exponent of the selected prime lengths. Let $Z \ge 2$ be a real variable tending to infinity. With balanced averaging lengths,

$$X = Z^{l_x}, \qquad Y = Z^{l_y}, \qquad l_x = \frac{1 - b - \ell}{2}, \qquad l_y = \frac{1 + b - \ell}{2}.$$

The prime-norm parameters P_i satisfy $\prod_i P_i = Z^{\ell}$. For the admissible parameter choices used below and every $\varepsilon > 0$, the direct bound and the exponent of the principal Mellin term are respectively

$$|J_\eta(Z)| \ll_\eta Z^{l_x/2 + b/12 + \varepsilon}, \qquad C_b(s) = s - \frac{4 + b}{6},$$

where J_η is the normalized cubic-theta average constructed in Part I and generalized in Part II. Matching these exponents gives

$$C_b(B) = \frac{l_x}{2} + \frac{b}{12} \quad \Longleftrightarrow \quad B = B(\ell) := \frac{11}{12} - \frac{\ell}{4}. \tag{0.1}$$

Thus increasing ℓ lowers the comparison boundary. The allowable increase is constrained by the nonprincipal terms, whose exponents also depend on b. For the following parameter comparisons, write $E(b, \ell; \delta, x) = E_{B(\ell)}(h)$ for the relative nonprincipal exponent in (18.5), with $h = (1 + b + 3\ell)/2$ and the character-count exponent R_* of (18.2). Here $\delta = 2a - 1$ is the zero-class parameter, and $x = \bar{g}/\delta$ measures the normalized prime-sum size, as

defined in Section 10; the polynomial comparison uses $(\delta, x) \in [0, 5/6] \times [0, 1/2]$. These comparisons use the fixed moment parameter $\kappa = 3/4$. The notation $\partial_\ell E$ denotes partial differentiation of $E(b, \ell; \delta, x)$ with respect to ℓ, holding b, δ, x fixed; the dependence of $B(\ell)$ and $h = (1 + b + 3\ell)/2$ on ℓ is included in this differentiation.

For the first improvement we keep the original length difference and increase the total prime length by $1/10000$:

$$b_{\mathrm{r}} = \frac{1}{8}, \qquad \ell_{\mathrm{r}} = \frac{1}{6} + \frac{1}{10000}, \qquad B_{\mathrm{r}} = B(\ell_{\mathrm{r}}) = \frac{7}{8} - \frac{1}{40000} = \frac{34999}{40000}.$$

The rational choice is deliberately conservative: b_{r} retains the original difference between the summation exponents, while $1/10000$ is a convenient perturbation rather than a distinguished or maximal value. Lemma 19.1 establishes the uniform bounds

$$E(1/8, 1/6; \delta, x) \le -\frac{1}{5000}, \qquad \partial_\ell E(1/8, \ell; \delta, x) \le \frac{45}{32}.$$

Consequently, at $\ell = \ell_{\mathrm{r}}$,

$$E(1/8, \ell_{\mathrm{r}}; \delta, x) \le -\frac{1}{5000} + \frac{45}{32 \cdot 10000} = -\frac{19}{320000} < 0.$$

Thus the improvement retains a positive saving in the error exponent. Part II proves these bounds by an exact calculation in the Bernstein basis polynomials (defined in (19.3), in the proof of Lemma 19.1; see also [Ber12, Far12]) and verifies the remaining contour and length conditions.

For the second improvement we vary both parameters:

$$b_{\mathrm{alg}} = -\frac{4}{29} + \frac{230\sqrt{921}}{26709}, \qquad \ell_{\mathrm{alg}} = \frac{33 + 8\sqrt{921}}{1653},$$
$$B_{\mathrm{alg}} = B(\ell_{\mathrm{alg}}) = \frac{1507 - 2\sqrt{921}}{1653}.$$

The algebraic values arise from the critical error class and a tangency condition. At $x = 1/2$, the count exponent is

$$R_*(\delta, 1/2) = \frac{185 - 288\delta + 96\delta^2}{185 - 138\delta}, \qquad \partial_b E = \frac{R_*}{2} - \frac{1}{3}.$$

Hence a class with $R_* = 2/3$ imposes a constraint that cannot be relaxed by changing b. Its admissible parameter is the smaller root

$$288\delta^2 - 588\delta + 185 = 0, \qquad \delta_{\mathrm{c}} = \frac{49 - \sqrt{921}}{48}.$$

In particular, $\sqrt{921}$ comes from the discriminant $588^2 - 4 \cdot 288 \cdot 185 = 144 \cdot 921$, rather than from an arbitrary choice. At this class, $E = 0$ determines

$$\ell = \frac{5 - 6\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}} = \ell_{\mathrm{alg}},$$

and the stationary condition

$$\partial_\delta E(b, \ell_{\mathrm{alg}}; \delta_{\mathrm{c}}, 1/2) = 0$$

determines $b = b_{\mathrm{alg}}$. These values balance the direct comparison with the largest non-principal exponent. The balancing equations locate the candidate; their solution alone does not establish the estimate for other classes. Part III derives the critical zero class, proves the endpoint inequality for every permitted zero and amplitude class by an exact quadratic identity, and verifies all auxiliary strict auxiliary inequalities. The positive linear combination of inequalities in Proposition 25.1 explains the algebraic obstruction

suggested by these parameter choices. The notation alg distinguishes this bound from the rational one.

Varying the moment parameter. We vary the parameter κ governing the admissible lengths in the mixed fourth-moment estimate for unweighted character polynomials. Part IV extends the unweighted moment from $3/4 \le \kappa \le 1$ to $13/18 \le \kappa \le 1$, retaining the indispensable hypothesis

$$\beta_* \le (1+\kappa)/2 \qquad (\kappa < 1).$$

The bound from Part III therefore permits $\kappa = 2B_{\mathrm{alg}} - 1 < 3/4$. Decreasing κ increases the available selected-prime length in the unweighted moment, improves the nonprincipal count, and permits a further increase of ℓ. The direct boundary $B(\ell)$ would remain B_{alg} if ℓ_{alg} were kept fixed. Thus the improvement uses changed length parameters as well as a changed moment parameter.

For a critical class $\delta = \xi$ and $x = 1/2$, the count equation $R_\kappa = 2/3$ becomes

$$(216\kappa - 18)\xi^2 - (468\kappa - 57)\xi + 150\kappa - 20 = 0.$$

The equations $E = 0$ and $\partial_\delta E = 0$ then give

$$\ell(\xi) = \frac{5 - 6\xi}{9 + 18\xi}, \qquad b(\xi) = \frac{8(3\xi - 1)(12\xi^2 - 40\xi + 15)}{9(2\xi + 1)(24\xi^2 - 20\xi + 5)}, \qquad B(\xi) = \frac{18\xi + 7}{18\xi + 9}.$$

Part IV proves the nonprincipal inequality uniformly in every zero and amplitude class, checks the remaining strict analytic inequalities, and obtains a new bound from each previously established moment hypothesis. Iteration reaches the limiting equality $\kappa_{\mathrm{new}} = 2B_{\mathrm{new}} - 1$; eliminating κ gives the cubic in the main theorem. Only the resulting real upper bounds are passed to their limit, so uniform analytic constants over the iteration are not required.

The limiting root has the explicit form

$$\xi_{\mathrm{new}} = \frac{20}{33} + \frac{\sqrt{1391}}{33} \cos\left(\frac{1}{3}\arccos\frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right), \qquad \arccos \in [0, \pi].$$

Substitution in the balanced length relations gives the exact limiting parameters

$$b_{\mathrm{new}} = \frac{8(3\xi_{\mathrm{new}} - 1)(12\xi_{\mathrm{new}}^2 - 40\xi_{\mathrm{new}} + 15)}{9(2\xi_{\mathrm{new}} + 1)(24\xi_{\mathrm{new}}^2 - 20\xi_{\mathrm{new}} + 5)} = 0.123405685419541002\ldots,$$

$$\ell_{\mathrm{new}} = \frac{5 - 6\xi_{\mathrm{new}}}{9 + 18\xi_{\mathrm{new}}} = 0.166838588986270882\ldots,$$

$$\kappa_{\mathrm{new}} = \frac{18\xi_{\mathrm{new}} + 5}{18\xi_{\mathrm{new}} + 9} = 2B_{\mathrm{new}} - 1 = 0.749914038840197892\ldots.$$

The root formula is proved in Lemma 33.1; the parameter identities follow from (30.2) and (33.3). In particular,

$$B_{\mathrm{alg}} - B_{\mathrm{new}} = 5.037906979388974\ldots \times 10^{-8}.$$

The main structural contribution of Part IV is Theorem 34.4: it determines the exact optimum of the explicitly specified exponent comparison system, allowing free total length, every permitted detector cutoff and contour choice, and the full moment range $13/18 \le \kappa \le 1$. The proof isolates a critical class for which every permitted detector cutoff gives an upper-bound exponent at least $2/3$ in the counting estimate, forcing three affine inequalities for any putative boundary $B < B_{\mathrm{new}}$. An explicit positive weighted combination eliminates the length parameters from these inequalities; the moment compatibility condition $B \le (1+\kappa)/2$ then forces $B \ge B_{\mathrm{new}}$, and the limiting construction shows that this lower bound is sharp. This assertion concerns the supplied uniform upper bounds and does not exclude improvements from stronger analytic estimates. The limiting pair

$(b_{\mathrm{new}}, \ell_{\mathrm{new}})$ consists of cubic irrationals, whereas the rational parameter choice does not attain the limit.

All three improvements retain the exact arithmetic formulas and principal residues of [Ope26a]. Their Euler correction is shown to be holomorphic below 7/8. In Parts II and III, the original conclusion $\beta_* \le 7/8$ permits $\kappa = 3/4$, removing the loss associated with the original varying parameter. In Part IV each new moment hypothesis is supplied by the preceding proved bound. The strict exponent comparisons then give the same Mellin-continuation contradiction as in Part I. The analytic dependencies are indexed in Appendix B.

Here is the structure of the paper. Part I recalls the 7/8 argument and fixes the arithmetic and analytic notation used throughout. Part II establishes the shared changed-parameter estimates and gives the simpler rational extension B_r. Its calculation in the Bernstein basis polynomials is retained as a separate rational example; this bound is superseded by the stronger one in Part III. Part III varies both length parameters, proves the error-exponent inequality at the algebraic parameter choices, and discusses the resulting algebraic lower bound B_{alg}. Part IV proves the smaller-κ moment estimate, derives the changed count and length parameters, and iterates the analytic comparison to B_{new}. All three extensions use Proposition 3.1 and the Dirichlet transfer of Corollary 14.1.

Acknowledgments. OpenAI Codex (GPT 6.1 Sol Medium-Ultra) assisted with drafting, checking references, and the search for improved bounds. Anthropic Claude (Opus 5.5 Medium) assisted with auditing. The zero-free conclusions for B_{r}, B_{alg} and B_{new} have Lean 4 formalizations for the finite-order Hecke family over F and its Dirichlet and zeta consequences. The Lean formalization, exact-arithmetic certificates and verification instructions are available in the accompanying GitHub repository [Liu26].

Part I. Summary of the 7/8 proof

1. The original argument and its two-stage reduction

In this section, we briefly describe the two representations of the cubic-theta average, i.e., the smoothed arithmetic average $I_\eta(X, Y, Z)$ defined in (4.5), formed from the Fourier coefficients of the cubic theta function, and the two-stage reduction leading to the 7/8 bound. The first stage proves 11/12 using the unmodified average; the second introduces selected prime factors and stronger moment estimates to reach 7/8.

The argument compares two descriptions of one cubic-theta average: the divisibility-restricted form of the cubic-theta functional equation (4.2) ([Ope26a, Corollary 14.1]) gives the theta mean-square estimate (7.4). Together with the additive mean-square estimate (7.5), the Cauchy–Schwarz inequality (4.7) gives the direct bound (7.3). Poisson summation (4.9), applied to the modification (7.2), isolates the principal Mellin integral (8.8) involving $1/L_F(s, \eta)$. The simultaneous large values (5.5) forced by a zero, together with the moment bounds (9.1) and (9.4), control the remaining Poisson terms. Part I makes those dependencies, the common-character hypotheses and the numerical exponent comparisons explicit. Its statements record the original argument using the proved results indexed in Appendix B. The accompanying explanations retain the decisive identities, exponent comparisons and contradiction, including calculations used in Parts II–IV. Complete proofs of the imported results are given at the cited locations in [Ope26a]. Numbered statements have this paper's numbering; locators inside citations always refer to the cited work.

Theorem 1.1 (Main statement of [Ope26a, Theorem 1.1]). *Every finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ has no zeros in*

$$\operatorname{Re} s > 7/8. \tag{1.1}$$

The same holds for every Dirichlet L-function, including ζ. Principal poles at $s = 1$ are allowed; the boundary $\operatorname{Re} s = 7/8$ is not included.

The Hecke and Dirichlet conclusions are proved below in Theorem 13.1 and Corollary 14.1. Here $L_F(s, \eta)$ denotes the Hecke L-function over F, and $\zeta(s)$ the Riemann zeta function; η denotes the character fixed in each application.

For the primitive Hecke family define

$$\beta_* = \sup\left(\{1/2\} \cup \left\{\operatorname{Re}\rho : \begin{array}{l} L_F(\rho,\eta) = 0, \quad 1/2 \le \operatorname{Re}\rho \le 1, \\ \eta \text{ primitive and finite-order} \end{array}\right\}\right). \tag{1.2}$$

Poles are excluded. The inserted value $1/2$ gives $\beta_* \ge 1/2$; absolute convergence of the Euler product gives $\beta_* \le 1$. The family in (1.2) comprises *all primitive finite-order Hecke characters over F, of arbitrary conductor*. Indeed, Poisson summation produces denominators $L_F^S(s, \eta\overline{\chi_\bullet(u)})$ in (4.9). The primitive inducing characters of these twists belong to that family, so the same β_* controls their zeros.

The two stages. Stage 1 proves $\beta_* \le 11/12$. Stage 2 assumes, for contradiction, $\beta_* > 7/8$ and sets

$$\Delta = \beta_* - 7/8 \in (0, 1/24], \qquad \kappa = 2\beta_* - 1 = 3/4 + 2\Delta \le 5/6. \tag{1.3}$$

Group the nonprincipal terms by the location of the zeros of their associated twists; these groups are the zero classes of Section 5. In the second stage they satisfy

$$\delta = 2a - 1 \le \kappa \le \alpha := 5/6.$$

This restriction is essential in the original paper's character-count argument: the long-polynomial bound contains $(\alpha - \delta)r$, and its comparison with $r \le t$ uses $\alpha - \delta \ge 0$. Thus the preliminary theorem cannot be omitted from this proof without replacing that estimate.

The parameter choice giving $7/8$. The normalized modified average has a direct bound $Z^{L+\varepsilon}$, while its principal residue has Mellin exponent $C(s)$, where

$$L = \frac{l_x}{2} + \frac{b}{12}, \qquad b = l_y - l_x, \qquad C(s) = s + \frac{l_x}{3} - \frac{5}{6} + \frac{\ell}{6}.$$

Under $l_x + l_y = 1 - \ell$, matching these exponents gives

$$C(B) = L \quad \Longleftrightarrow \quad B = \frac{11}{12} - \frac{\ell}{4}.$$

The mean-square calculation using the functional equation in [Ope26a, Lemma 15.1, proof] contains the algebraic excess

$$\left(\frac{1 + 3\ell' - 2M'}{4}\right)_+ = \left(\frac{d_J + 5\ell - 1}{4}\right)_+, \qquad \ell' = \ell - d_J, \quad M' = 1 - \ell - 2d_J.$$

For $0 \le d_J \le \ell$, this term vanishes for every subset exactly when $6\ell \le 1$. The convenient choice $\ell = 1/6$ therefore gives $B = 7/8$ in the exponent matching above. With the paper's parameter choices (7.1),

$$B = 7/8, \qquad C(s) = s - 11/16, \qquad C(7/8) = L = 3/16.$$

This exponent matching motivates the parameters; proving a strict saving for every nonprincipal contribution requires both large values from the zero detector and the two moment estimates. That saving continues every reciprocal $1/L_F(s,\eta)$ to the left of β_*, producing the contradiction.

Sections 2–4 introduce the characters, continuation criterion and theta average. Sections 5–6 give the detector and the 11/12 stage. Sections 7–12 develop the modified average, moments, counts and exponent savings. Sections 13–14 complete the continuation and Dirichlet transfer.

2. Arithmetic conventions

In this section, we fix the arithmetic conventions used in every subsequent construction, including character presentations, zero extensions and reciprocity phases. The family zero supremum β_* used in the continuation argument is defined in (2.3). These choices determine the local factors of the theta average in Section 4 and must be retained when its parameters vary.

Set

$$\omega = e^{2\pi i/3}, \quad \mathcal{O} = \mathbb{Z}[\omega], \quad \lambda = \sqrt{-3}, \quad q_a = |a|^2, \quad q_{\mathfrak{a}} = N\mathfrak{a}, \quad \varpi(a) = a/|a|.$$

$\mathcal{O}$ is norm-Euclidean. Ideals coprime to 3 have unique primary generators $a \equiv 1 \pmod 3$. Ideal sums use these generators; sums over elements may also include units. Unless stated otherwise, ideal sums run over nonzero integral ideals. A sum over $x \bmod c$ runs over residue classes in $\mathcal{O}/(c)$. For an integral ideal $\mathfrak{a}$, $N\mathfrak{a} = |\mathcal{O}/\mathfrak{a}|$. Write $v_p(a)$ for its prime valuation, $\mathrm{rad}(a) = \prod_{p|a} p$, and μ for the ideal Möbius function. For a nonzero integral ideal,

$$\mu(\mathfrak{a}) = \begin{cases} 1, & \mathfrak{a} = \mathcal{O}, \\ (-1)^k, & \mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \text{ with distinct prime ideals } \mathfrak{p}_j, \\ 0, & \mathfrak{p}^2 \mid \mathfrak{a} \text{ for some prime ideal } \mathfrak{p}. \end{cases}$$

When an ideal is represented by a primary generator n, we write $\mu(n) = \mu((n))$. Thus Möbius coefficients restrict a sum to squarefree ideals and assign the sign $(-1)^k$ according to the number of their prime factors. The notation sf in a sum means squarefree, and 1_E is the indicator of a condition E. Thus $(a,b) = 1$ means that the corresponding ideals are coprime. A prime outside S is called good.

Unless specified otherwise, $Z \geq 2$ is a real variable tending to infinity; $\varepsilon, \varepsilon_1 > 0$ are arbitrarily small losses. For $a \in \mathbb{C}$ and $b \geq 0$, the notation $a \ll b$, read as "a is bounded by a constant times b", means $|a| \leq Cb$ for some $C > 0$. The absolute value is unnecessary when a is already nonnegative. A subscript specifies allowed dependence of the constant: for example,

$$a \ll_\varepsilon b \quad \Longleftrightarrow \quad |a| \leq C_\varepsilon b,$$

where $C_\varepsilon > 0$ may depend on ε but is independent of the other varying quantities; the notation $\ll_S$ has the analogous meaning for the fixed prime set S. For nonnegative a, b, the notation $a \gg b$, read as "a is bounded below by a positive constant times b", means $b \ll a$. The notation $a \asymp b$, read as "a is comparable to b", means

$$cb \leq a \leq Cb \qquad \text{for some } c, C > 0.$$

These constants are independent of the varying quantities, subject to any indicated dependence; comparability does not assert that a/b is constant. We also write $x_+ = \max(x, 0)$.

Throughout the paper, we often express an actual summation length $N > 0$ as a power of the common real variable Z, and use its *length exponent* r:

$$N = Z^r, \qquad r = \frac{\log N}{\log Z}. \tag{2.1}$$

For instance, a sum supported on norms $q_n \asymp N$ is then written as supported on $q_n \asymp Z^r$. Products of lengths correspond to sums of their exponents:

$$Z^{r_1} Z^{r_2} = Z^{r_1+r_2}.$$

In estimates stated in this notation, "length r" means the length exponent; the actual norm parameter is Z^r.

Auxiliary letters are used independently in separately stated lemmas: in particular, K in (4.4) is a norm bound, whereas K in Section 7 counts prime intervals. For $c \in \mathbb{R}$, the notation $\int_{(c)} F(s)\,ds$ denotes integration upwards along the vertical line $\operatorname{Re} s = c$, interpreted as the limit of the integrals from $c-iT$ to $c+iT$ as $T \to \infty$. The test functions used to restrict norms are smooth functions supported on fixed compact subintervals of $(0,\infty)$; their logarithmic derivatives are bounded. Constants A, J in height estimates denote finite nonnegative exponents or derivative orders fixed before the height cutoff; their values may differ from one estimate to another.

Fix a finite set S containing the prime ideals above 6, all prime ideals dividing the defining moduli of the fixed characters, and all prime ideals of norm at most P_0, where $P_0 \in \mathbb{R}_{>0}$ is the fixed prime-norm cutoff. Unless stated otherwise, ideal sums are restricted to ideals coprime to every prime ideal in S. These arithmetic data are fixed independently of Z and the summation variables; they may depend on an already fixed character η. Denote these fixed moduli, characters, ray group, exclusions and normalizations collectively by $\mathcal{A}$.

Symbol	Meaning and scope
β_*, Δ, κ	Family zero supremum; $\Delta = \beta_* - 7/8$; $\kappa = 2\beta_* - 1$.
$\alpha = 5/6,\ \varpi(a)$	Fixed amplification slope; unit phase $a/\lvert a\rvert$ (written $\alpha(a)$ in [Ope26a]).
$\epsilon_{\mathrm{b}},\ \varepsilon$	Fixed zero-class width (written e in [Ope26a]); arbitrarily small exponent loss.
$\mathfrak{a}(r)$	Amplified moment exponent $\max\{1, (1+5r)/6\}$ (written $e(r)$ in [Ope26a]).
$Z, U = Z^d, T_1$	Real variable tending to infinity; dyadic norm parameter; zero-detection height.
l_x, l_y, b, h, ℓ	Averaging exponents; $b = l_y - l_x$; combined frequency exponent; total auxiliary prime length.
z_{cap}, z_0	Upper bound for total prime length in (9.2); real contour value $z_0 = 17/50$ in Section 11.
$\mathfrak{r}$	Coprimality modulus in (9.2) (written R in [Ope26a]); distinct from a counting exponent R.
$\mathcal{R}, \mathcal{M}$	Reciprocity phase; Mellin transform of the Fourier-transformed weight.
H_η, J_η, f_η	Nonvanishing correction; normalized average; principal-residue Mellin integral.
$\Gamma_1(3), \Gamma_2;\ \Gamma(s)$	Matrix groups; Euler gamma function.

Other indices and length parameters are local to the statement in which they are introduced. A *zero class* is a set of elements satisfying the same pair of zero-location inequalities (5.1).

Hecke characters, moduli and conductors. For a nonzero integral ideal $\mathfrak{f}$, called the modulus, the ray class group is $I(\mathfrak{f})/P_1(\mathfrak{f})$: $I(\mathfrak{f})$ consists of fractional ideals coprime to $\mathfrak{f}$, and $P_1(\mathfrak{f})$ of principal ideals generated by elements congruent to 1 modulo $\mathfrak{f}$. A finite-order Hecke character is a homomorphism from this finite group to roots of unity. Its conductor $\mathfrak{f}_*$ is the smallest modulus through which it factors. A *character presentation* specifies a defining modulus $\mathfrak{f}$ divisible by $\mathfrak{f}_*$, together with extension by zero on integral ideals not coprime to $\mathfrak{f}$. The presentation is primitive when $\mathfrak{f} = \mathfrak{f}_*$ and imprimitive otherwise. More precisely, a character ψ with defining modulus $\mathfrak{f}$ is induced by a unique primitive character ψ^* of conductor $\mathfrak{f}_* \mid \mathfrak{f}$. If $\mathrm{Cl}_{\mathfrak{f}}(F)$ denotes the ray class group above, the natural surjection gives

$$\pi : \mathrm{Cl}_{\mathfrak{f}}(F) \longrightarrow \mathrm{Cl}_{\mathfrak{f}_*}(F), \qquad \psi = \psi^* \circ \pi. \tag{2.2}$$

Thus $\psi(\mathfrak{a}) = \psi^*(\mathfrak{a})$ whenever $(\mathfrak{a}, \mathfrak{f}) = 1$. Primitivity means that ψ^* does not factor through the ray class group of any proper divisor of $\mathfrak{f}_*$. The zero extensions can nevertheless differ: $\psi(\mathfrak{a}) = 0$ when $(\mathfrak{a}, \mathfrak{f}) \neq 1$, including when $(\mathfrak{a}, \mathfrak{f}_*) = 1$. Thus the defining modulus records the coprimality restriction attached to this presentation, including primes absent from the conductor; see [Ope26a, Section 4]. Initially,

$$L_F(s, \eta) = \sum_{\mathfrak{a} \neq 0} \eta(\mathfrak{a})(N\mathfrak{a})^{-s} = \prod_{p} (1 - \eta(p) q_p^{-s})^{-1}, \qquad \operatorname{Re} s > 1;$$

outside this region the notation refers to its meromorphic continuation. A norm twist multiplies a character by $\mathfrak{a} \mapsto (N\mathfrak{a})^{it}$, $t \in \mathbb{R}$.

For a prime $p \notin S$, the sextic residue character is

$$\chi_p(a) = (a/p)_6, \qquad \chi_p(a) \equiv a^{(q_p - 1)/6} \pmod{p} \quad (p \nmid a), \qquad \chi_p(a) = 0 \quad (p \mid a).$$

For a primary element c, meaning $c \equiv 1 \pmod 3$, write $c = \prod_p p^{v_p(c)}$ with primary prime generators p, and put $\chi_c(a) = \prod_{p \mid c} \chi_p(a)^{v_p(c)}$. Throughout, character powers are first taken on invertible residue classes and then extended by zero on the remaining classes. This convention applies to every integer exponent, including zero, negative exponents and multiples of six:

$$\chi_c(a)^6 = 1_{(a,c)=1}. \tag{2.3}$$

Passing to a primitive inducing character therefore requires retaining the coprimality indicator for the original defining modulus. Coefficients and such conditions are independent of the variable being averaged, except for the explicitly displayed residue symbols.

The additive normalization is

$$e(z) = \exp\bigl(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda)\bigr), \qquad d\mu(z) = \frac{2}{\sqrt{3}}\, d\xi_1\, d\xi_2.$$

Here $z = \xi_1 + i\xi_2$ with $\xi_1, \xi_2 \in \mathbb{R}$; for $z \in \mathbb{C}$ the trace in $e(z)$ means $z/\lambda + \overline{z/\lambda}$. The pairing $(z_1, z_2) \mapsto e(z_1 z_2)$ makes $\mathcal{O}$ self-dual. For squarefree c define

$$\gamma_j(c) = q_c^{-1/2} \sum_{v \bmod c} \chi_c(v)^j e(v/c).$$

Fixed-numerator symbols $n \mapsto \chi_n(u)$ are finite-order ray characters [Ope26a, Lemma 4.1]. Prime Gauss identities and sextic reciprocity provide root-of-unity-valued functions $\mathcal{R}, G$ that depend only on fixed ray classes. Precisely, $\mathcal{R}$ is the bicharacter on residue square classes modulo 4 defined in [Ope26a, Lemma 4.4]; on coprime primary pairs it satisfies

$$\chi_b(a) = \mathcal{R}(a, b)\chi_a(b).$$

The same lemma defines G on all primary indices, with $|G| = 1$. Its value depends only on the ray class of its argument in a fixed finite ray quotient; on squarefree c, $G(c) =$

$\overline{\chi_c(4)}\gamma_3(c)$. Its extension to nonsquarefree indices is that finite-ray function, rather than a Gauss sum, and satisfies

$$G(AB) = G(A)G(B)\mathcal{R}(A,B)$$

on the admissible primary elements [Ope26a, Lemmas 4.2–4.4]. These functions cancel the reciprocity factors linking distinct primes, so that the coefficients can be expressed as an Euler product in Section 4.

3. A criterion for zero-free continuation

In this section, we isolate the analytic implication that converts the two estimates for an average into a zero-free half-plane. Proposition 3.1 compares the direct bound (3.3) with the approximation by the principal Mellin integral in (3.4). All four parts use this criterion; their remaining task is to construct the average and verify its hypotheses with a saving independent of the target character.

For a finite excluded set,

$$L_F^S(s,\eta) = L_F(s,\eta)\prod_{p\in S}(1-\eta(p)q_p^{-s}).$$

The deleted factors are nonzero for $\operatorname{Re} s > 0$.

The formulation below is taken from [Ope26a, Proposition 2.1]. Its proof uses the classical principle that a growth bound for an inverse Mellin transform yields holomorphic continuation of a reciprocal L-function, and hence zero-freeness; compare Riesz's criterion [Rie16, Sections 3–4]. Here the positive saving must be uniform over the target characters, whereas the implied constants may depend on the character.

Proposition 3.1 (Continuation criterion, [Ope26a, Proposition 2.1])**.** *Fix $1/2 < \sigma_0 < 1$ and $C(s) = s + c$, $c \in \mathbb{R}$. Suppose that, whenever $\beta_* > \sigma_0$, the following construction is available. For every primitive η, suppose H_η is holomorphic on $\operatorname{Re} s > \sigma_0$ and*

$$\sup_{\operatorname{Re} s>\sigma_0} |H_\eta(s) - 1| \le 1/2. \tag{3.1}$$

Define

$$f_\eta(Z) = \frac{1}{2\pi i}\int_{(2)} Z^{C(s)} e^{(s-5/6)^2}\frac{H_\eta(s)}{L_F^S(s,\eta)}\,ds. \tag{3.2}$$

Here (v) denotes a vertical line of real part v. Assume there is a normalized average J_η satisfying

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega_0}, \qquad 0 < \omega_0 < \beta_* - \sigma_0, \tag{3.3}$$

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma}, \qquad \sigma > 0, \tag{3.4}$$

The positive numbers ω_0, σ are independent of η; constants, finite excluded sets and lower thresholds may depend on η. Then $\beta_* \le \sigma_0$.

Idea of the continuation argument. Assume $\beta_* > \sigma_0$ and put

$$\epsilon_c = \min\{\beta_* - \sigma_0 - \omega_0,\ \sigma,\ (\beta_* - \sigma_0)/2\} > 0.$$

The estimates (3.3) and (3.4) give $f_\eta(Z) \ll_\eta Z^{C(\beta_*)-\epsilon_c}$ for $Z \ge 1$. Moving (3.2) to any fixed line $\operatorname{Re} s = b_c > 2$ gives $f_\eta(Z) \ll_{\eta,b_c} Z^{b_c+c}$ for $0 < Z \le 1$. Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s-c}\,\frac{dZ}{Z}$$

is holomorphic on $\operatorname{Re} s > \beta_* - \epsilon_c > \sigma_0$. Mellin inversion and analytic continuation identify

$$F_\eta(s) = e^{(s-5/6)^2} H_\eta(s)/L_F^S(s,\eta) \qquad (\operatorname{Re} s > 1).$$

Since $|H_\eta| \geq 1/2$, the function

$$e^{-(s-5/6)^2} F_\eta(s)/H_\eta(s)$$

is a holomorphic continuation of $1/L_F^S(s,\eta)$ to $\operatorname{Re} s > \beta_* - \epsilon_c$, for every primitive η. The supremum definition supplies some η_0 and a zero ρ there; attainment of β_* is unnecessary. Locally,

$$L_F^S(s,\eta_0) = (s-\rho)^{k_\rho} g(s), \qquad k_\rho \geq 1, \quad g(\rho) \neq 0.$$

Its reciprocal has a pole, contradicting this holomorphic continuation. Neither the Gaussian nor H_{η_0} can cancel it. Principal poles, in contrast, give zeros of the reciprocal and cause no obstruction.

The two applications in Part I use

σ_0	$C(s)$	$C(\sigma_0)$
11/12	$s - 2/3$	1/4
7/8	$s - 11/16$	3/16

and different normalized averages.

4. Cubic theta series and the smoothed average

In this section, we construct the cubic-theta average whose two representations enter Proposition 3.1. The cubic-theta functional equation (4.2) supplies the mean-square estimate (4.3) for the direct bound, while Poisson summation (4.9) exposes the reciprocal L-function and the nonprincipal terms to be controlled in the following sections. These representations first give the 11/12 bound and then, after the modification in Section 7, the 7/8 bound; Parts II–IV retain this construction with changed parameters.

4.1. **The cubic theta function and its Fourier coefficients.** In this subsection, we introduce the cubic theta function and the Fourier coefficients used to construct the smoothed average. The transformation law leads to the functional equation (4.2) for the theta sums, while the explicit coefficients connect those sums to the cubic Gauss sums in (4.1).

The function used here is *Kubota's cubic metaplectic theta function* θ_{cub} on $\mathbb{H}^3 = \mathbb{C} \times \mathbb{R}_{>0}$. Kubota developed the underlying metaplectic construction [Kub69]; Patterson determined the explicit cubic theta coefficients [Pat77, Theorem 8.1]. We use the cubic metaplectic theta function in the normalization of Dunn–Radziwiłł [DR24, Section 5.1 and Appendix A], recording below the transformation law and Fourier coefficients needed in our argument, in the form stated in [Ope26a, Proposition 5.1]. Here *cubic* refers to the cubic residue symbol appearing in the transformation law of θ_{cub} displayed below. Set

$$\Gamma_1(3) = \{g \in \mathrm{SL}_2(\mathcal{O}) : g \equiv I \pmod 3\}, \qquad \Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma_1(3) \rangle.$$

For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{C})$, its action on $w = (z,v) \in \mathbb{H}^3$ is

$$g(z,v) = \left(\frac{(az+b)\overline{(cz+d)} + a\bar{c}\, v^2}{|cz+d|^2 + |c|^2 v^2}, \frac{v}{|cz+d|^2 + |c|^2 v^2} \right).$$

The transformation law is

$$\theta_{\mathrm{cub}}(gw) = \kappa_{\mathrm{cub}}(g)\theta_{\mathrm{cub}}(w), \qquad \kappa_{\mathrm{cub}}(g) = \begin{cases} (c/a)_3, & g \in \Gamma_1(3),\ c \neq 0, \\ 1, & g \in \Gamma_1(3),\ c = 0, \text{ or } g \in \mathrm{SL}_2(\mathbb{Z}). \end{cases}$$

The factor $\kappa_{\mathrm{cub}}(g)$ extends to Γ_2 as a homomorphism [Pat77, Section 2]; the action and transformation law above are [DR24, equations (5.1)–(5.4)]. The symbol $(c/a)_3$ is the cubic residue symbol, multiplicative in its primary denominator; at good primes $(x/p)_3 = \chi_p(x)^2$. The notation κ_{cub} distinguishes this transformation factor from the real zero-bound parameter κ.

The Fourier expansions needed in the proof are those at the three cusps represented by

$$H_j = \begin{pmatrix} 1 & 0 \\ t_j & 1 \end{pmatrix}, \qquad (t_0, t_1, t_2) = (0, \omega^2, \omega), \qquad j = 0, 1, 2.$$

Each H_j sends ∞ to a boundary cusp in $F \cup \{\infty\}$; expanding $\theta_{\mathrm{cub}} \circ H_j$ gives its Fourier expansion there. We take the complex conjugate so that the Fourier coefficients involve the cubic Gauss sums $\widetilde{g}_3(c)$ used below, rather than their complex conjugates; Appendix A gives the precise coefficient conversion. With $\check{e}(x) = \exp(2\pi i(x + \bar{x}))$, these expansions are

$$\overline{\theta_{\mathrm{cub}}(H_j(z, v))} = C_j(v) + \sum_{\substack{\mu \in \lambda^{-4}\mathcal{O} \\ \mu \neq 0}} d_j(\mu)\, v K_{1/3}(4\pi|\mu|v)\check{e}(\mu z).$$

Here

$$C_0(v) = \frac{3^{5/2}}{2} v^{2/3}, \qquad C_1(v) = C_2(v) = 0.$$

The symbol d_j denotes the fixed arithmetic Fourier-coefficient function, and $K_{1/3}$ the modified Bessel function of the second kind. The precise correspondence with the notation of [DR24] is given in Appendix A. In particular, for primary squarefree c and primary n,

$$d_0(\lambda^{-3} c n^3) = 3^{5/2}|n|\, \widetilde{g}_3(c), \qquad \widetilde{g}_3(c) = q_c^{-1/2} \sum_{x \bmod c} (x/c)_3 \check{e}(x/c).$$

When $(c, S) = 1$ this becomes

$$\widetilde{g}_3(c) = \chi_c(\lambda)^{-2} \gamma_2(c).$$

Thus the squarefree Gauss coefficients and the cube multiples in the following Dirichlet series come directly from these Fourier coefficients. The automorphy law supplies the functional equation; its explicit coefficients supply the mean-square estimates. These are the theta identities and estimates used below; Appendix A checks the constants directly from [DR24, equations (5.6), (5.7), (5.15), (5.16)].

4.2. **Cubic theta sums, their functional equation and mean-square estimates.** In this subsection, we define the theta sums including all cube multiples and record their functional equation and mean-square estimates for the direct bound. The sum (4.1) supplies the theta factor in (4.6); its mean-square estimate is combined with the additive estimate by (4.7). The estimate (4.3) is used for the preliminary 11/12 bound, while the divisibility-restricted estimates described below are used for the modified average and the subsequent bounds.

For a ray character Ψ with the conventions of Section 2, let $V \in C^\infty(0, \infty)$ satisfy, for every $C > 0$ and $j \geq 0$,

$$(x\partial_x)^j V(x) = O_{C,j}(x^C) \quad (x \to 0), \qquad (x\partial_x)^j V(x) = O_{C,j}(x^{-C}) \quad (x \to \infty).$$

The indices c, n below represent nonzero integral ideals coprime to S, using their primary generators; c is squarefree and n is unrestricted. Put

$$D(s,\Psi) = \sum_{c \text{ sf}} \gamma_2(c)\overline{\varpi(c)}\Psi(c)q_c^{-s},$$
$$L_0(s,\Psi) = \sum_{n} \overline{\varpi(n)}^3\Psi(n)^3 q_n^{-3s+1/2},$$
$$T(s,\Psi) = D(s,\Psi)L_0(s,\Psi).$$

The series $T(s,\Psi)$ is the character-twisted Dirichlet series built from the theta coefficients after the fixed rescaling and phase conversion in [Ope26a, Proposition 5.1]; it converges absolutely for $\operatorname{Re} s > 1$. Its smoothed expression is

$$\mathcal{C}_V(X;\Psi) = \sum_{c \text{ sf},n} \frac{\gamma_2(c)\overline{\varpi(cn^3)}\Psi(cn^3)}{\sqrt{q_c}\, q_n} V(q_c q_n^3/X). \tag{4.1}$$

No condition $(c,n) = 1$ is imposed. The cube factor is included because the natural theta support consists of the indices cn^3; thus the sum includes all cube multiples of each squarefree index c. For a primary A,

$$A = cn^3 \quad \Longleftrightarrow \quad v_p(A) \equiv 0 \text{ or } 1 \pmod 3 \quad \text{for every } p,$$

with c squarefree and c, n uniquely determined primary generators. The gamma factors enter separately through the functional equation.

Proposition 4.1 (Cubic theta functional equation, [Ope26a, Proposition 5.1])**.** *Fix a finite family of ray characters with conductors supported on S, and a common periodicity modulus L_{per} supported on S, divisible by 18 and their defining moduli, as specified in* [Ope26a, Proposition 5.1, hypotheses]. *Write $\Psi = \Psi_0 \prod_{p\in\mathcal{P}} \chi_p^{j_p}$ on primary elements, with $\mathcal{P}$ a finite set of distinct primes outside S and $j_p \in \{0,\dots,5\}$; zero extensions are retained. The product $T(s,\Psi)$ continues to an entire function with polynomial growth on each fixed vertical strip, with constants depending on the fixed arithmetic data and the strip. For $X > 0$ and the test V specified above, the precise functional-equation identity is*

$$\mathcal{C}_V(X;\Psi) = \sum_{\mathfrak{t}} \zeta_{\mathfrak{t}} \sum_{0\neq\mu\in\lambda^{-4}\mathcal{O}} \frac{d_{\mathfrak{t}}(\mu)\varpi(\mu)\vartheta_{\mathfrak{t}}(\lambda^4\mu)}{\sqrt{q_\mu}} \prod_{p \text{ active}} B_p(\lambda^4\mu) V^\sharp(q_\mu X/q_{c_{\mathfrak{t}}}^2). \tag{4.2}$$

Here $\mathfrak{t}$ runs over the terms indexed by $h_0 \bmod L_{\mathrm{per}}$ and an active subset of $\mathcal{P}$ [Ope26a, Proposition 5.1, item (1)]. *Every prime with $j_p \neq 0$ is active. The denominator is $c_{\mathfrak{t}} = c_F \prod_{p \text{ active}} p$, where c_F is the reduced denominator of $\lambda^2 h_0/L_{\mathrm{per}}$, with the normalizing unit fixed as in that item. The function $d_{\mathfrak{t}}$ is one of d_0, d_1, d_2; $\vartheta_{\mathfrak{t}}$ is an additive character modulo a fixed modulus depending on L_{per}; the scalar $\zeta_{\mathfrak{t}}$ is independent of μ and has $|\zeta_{\mathfrak{t}}| \le 1/81$. Their exact phases are specified in* [Ope26a, Proposition 5.1, items (2)–(3)]. *In the following local factors, every character value is extended by zero on noninvertible residue classes:*

$$B_p(x) = \begin{cases} \chi_p(x)^{-j_p-2}, & j_p \neq 0, 4, \\ q_p^{-1/2}(-1 + q_p 1_{p\mid x}), & j_p = 4, \\ q_p^{-1/2}\chi_p(x)^{-2}, & j_p = 0 \text{ and } p \text{ active}. \end{cases}$$

The function $V^\sharp$ appearing in the functional equation is obtained from V by the integral transform

$$V^\sharp(x)=\frac{1}{2\pi i}\int_{(v)}\widehat{V}(-t)\prod_{\pm}\frac{\Gamma(1+t\pm 1/6)}{\Gamma(1-t\pm 1/6)}\left(\frac{(2\pi)^4x}{27}\right)^{-t}dt,\qquad v\ge 0,$$

where Γ is the Euler gamma function and $\widehat{V}(t)=\int_0^\infty V(x)x^t\,dx/x$. The dual coefficients have support $u\lambda^k cn^3$, where $u\in\mathcal{O}^\times$ is a unit, $k\in\mathbb{Z}$, $k\ge -4$, c is squarefree, and c,n are primary. At such an index each of the three fixed cusp coefficient functions d_j of Section 4.1 has absolute value $\le 27\,3^{k/6}|n|$. The formula retains the local character zeros, including the coprimality conditions of imprimitive presentations.

The quadratic large sieve over number fields is [GL13, Definition 1 and Theorem 1.1].

Lemma 4.2 (Cubic theta mean square, [Ope26a, Lemma 5.8])**.** *Fix the arithmetic data, bounded ranges for M,N and dyadic support comparisons. For every $\varepsilon>0$, there exist finite seminorm orders A,J such that, for all sufficiently large Z and every nonempty powerful-part dyadic range O, one has*

$$\sum_{\substack{0<q_m\ll Z^M\\ q_{m_{\mathrm{pow}}}\asymp Z^O}}|\mathcal{C}_V(Z^N;\nu\chi_\bullet(m))|^2\ll Z^{O/2+\max(M-O,2M-O-N)+\varepsilon}\mathfrak{M}(V)^2. \tag{4.3}$$

Here m ranges over nonzero elements of $\mathcal{O}$, including all unit associates. The parameters $M,O\ge 0$ and $N\in\mathbb{R}$ are norm-range exponents, ν is a fixed finite ray character, and $\chi_\bullet(m)$ means $n\mapsto\chi_n(m)$. The ideal $m_{\mathrm{pow}}=\prod_{v_p(m)\ge 2}p^{v_p(m)}$ is the powerful part. The seminorm is $\mathfrak{M}=\mathfrak{M}_{A,J}$ for sufficiently large fixed A,J, where [Ope26a, Proposition 5.1] *defines*

$$\mathfrak{M}_{A,J}(V)=\sup_{-A\le\sigma\le 1/4}\int_{\mathbb{R}}(1+|t|)^J|\widehat{V}(\sigma+it)|\,dt.$$

Decomposition into the fixed ray classes and decay of the integral transform defining $V^\sharp$ justify the specialization to the Gaussian test [Ope26a, Lemmas 5.2–5.7].

For the later divisibility restrictions, the corresponding mean-square estimate is [Ope26a, Lemma 14.3]. Its arithmetic ingredient [Ope26a, Lemma 14.2] combines the quadratic sieve [GL13, Theorem 1.1] and the cubic sieve [HB00, Theorem 2].

Lemma 4.3 (Quadratic–cubic sieve, [Ope26a, Lemma 14.2])**.** *Let K,N,B,L be independent positive real norm parameters:*

$$K,N,B,L\ge 1,\qquad q_k\ll K,\quad q_n\asymp N,\quad q_b\asymp B,\quad q_P\asymp L.$$

Here k,n,P are squarefree primary ideals and b is any primary ideal. Thus $KNBL$ is the product of the four norm parameters, not a further variable. The indices k,P avoid S; n,b may meet S except at λ, as specified in the cited lemma. For every $\varepsilon>0$,

$$\sum_{k\ \mathrm{sf},\ q_k\ll K}\left|\sum_{P\ \mathrm{sf},\ q_P\asymp L}\frac{a(P)}{\sqrt{q_P}}1_{(P,k)=1}\sum_{\substack{n\ \mathrm{sf},\ q_n\asymp N\\ q_b\asymp B}}\beta(n,b)\chi_k(nb)^3\chi_P(n)^{-2}1_{(P,b)=1}\right|^2 \tag{4.4}$$
$$\ll(KNBL)^\varepsilon(K+NB)B\big[N+L+(NL)^{2/3}\big].$$

The coefficient functions satisfy $|a(P)|,|\beta(n,b)|\le 1$; $a(P)$ is independent of k,n,b, and $\beta(n,b)$ of k,P.

The ideal P need not be prime. Inclusion-exclusion removes the moving coprimality condition before applying the sieves. The full estimate obtained from the functional equation follows by dyadic decomposition, the local theta coefficients, and decay of $V^\sharp$.

4.3. **The smoothed average and its additive and theta factors.** In this subsection, we define the smoothed average $I_\eta(X,Y,Z)$ in (4.5) and decompose it into the two factors in (4.6). The Cauchy–Schwarz inequality (4.7) then reduces its direct bound to their mean-square estimates; the next subsection gives its Poisson representation, which contains the reciprocal L-function needed in Proposition 3.1.

Choose a fixed finite ray class group $\mathcal{T}$ through which the functions $\mathcal{R}$, G, and the fixed root-of-unity factors arising from the primary-generator convention and sextic reciprocity factor [Ope26a, Lemma 4.4]. Thus these factors depend only on the corresponding ray classes; the fixed congruence conditions involved are supported at primes above 2 and 3. Here $\widehat{\mathcal{T}}$ is its character group, and $1_{\mathcal{T}}$ its identity ray class. Membership $p \in 1_{\mathcal{T}}$ means that the primary generator of p has trivial image in $\mathcal{T}$. Write $\Theta = \langle \eta, \widehat{\mathcal{T}} \rangle$ for the finite group of Hecke characters generated by these characters; all its conductor primes are in S [Ope26a, Section 8]. Let b_* generate the squarefree product of primes in S and choose a primitive character ξ mod b_* of order dividing six, nonprincipal at each local prime. For a residue character ψ modulo a primary modulus c and $h \in \mathcal{O}$, set

$$g_\psi(c,h) = \sum_{d \bmod c} \psi(d) e(hd/c), \quad \tau_* = q_{b_*}^{-1/2} g_\xi(b_*,1), \quad |\tau_*| = 1, \quad \Xi(A) = \xi(A)\chi_A(b_*).$$

Here $A \in \mathcal{O}$ ranges over primary elements, so $A \equiv 1 \pmod{3\mathcal{O}}$. For the primary averaging variable v let

$$\Psi_{m,v,\eta}(A) = \eta(A)\Xi(A)^{-1}\mathcal{R}(A,v)\chi_A(m)$$

on primary elements coprime to S, extended by zero elsewhere, and define

$$T_{m,v,\eta}(t) = \sum_{c \text{ sf},n} \gamma_2(c)\overline{\varpi(cn^3)}\Psi_{m,v,\eta}(cn^3)\overline{G(cn^3)}q_c^{-1/2-t}q_n^{-1-3t}.$$

For fixed nonnegative nonzero test functions W_0, W_1 and $\Phi(t) = e^{t^2}$, define the *smoothed average* $I_\eta(X,Y,Z)$ used in the original paper by

$$\begin{aligned} I_\eta(X,Y,Z) = \frac{1}{Y} \sum_{\substack{v \text{ primary} \\ (v,S)=1}} \frac{W_1(q_v/Y)\chi_v(b_*)}{\tau_*\xi(v)\sqrt{q_{b_*}q_vX}} \sum_{m \in \mathcal{O}\setminus\{0\}} \xi(m) q_v^{-1/2} g_{\chi_v}(v,-m) \\ \times W_0(q_m/(q_{b_*}q_vX)) \frac{1}{2\pi i} \int_{(4)} Z^t \Phi(t) T_{m,v,\eta}(t)\,dt. \end{aligned} \tag{4.5}$$

In (4.5), v runs over the primary generators of nonzero integral ideals coprime to S, with no squarefree restriction, and m runs over nonzero elements of $\mathcal{O}$, including all unit associates. The factor $\xi(m)$ vanishes when $(m,b_*) \neq 1$; the compact supports of W_0, W_1 make both sums finite and exclude $m = 0$. These same domains are used in the factors A_{m,σ,t_0} and $B_{m,\sigma}$ below; $v \in \sigma$ additionally restricts v to the indicated ray class. Here *smoothed* means that the summation ranges are restricted by smooth functions rather than by characteristic functions of intervals: $W_1(q_v/Y)$ restricts the v-sum and $W_0(q_m/(q_{b_*}q_vX))$ restricts the m-sum. The function V_Φ defined below similarly smooths the theta sum. The term *average* refers to the weighted summation over m, v, with the normalization $1/Y$ and the other factors in (4.5). The smooth functions permit Mellin inversion and Poisson summation with decay estimates for their transforms. The m, v sums are finite, and the theta series including all cube multiples converges absolutely on the initial line.

Expand $\overline{G(A)} = \sum_{\theta\in\widehat{\mathcal{T}}} a_\theta\theta(A)$, where $\sum|a_\theta| \le |\mathcal{T}|^{1/2}$. To relate (4.5) directly to the theta sum (4.1), define

$$V_\Phi(x) = \frac{1}{2\pi i}\int_{(4)} \Phi(t)x^{-t}\,dt = \frac{1}{2\sqrt{\pi}}\exp\left(-\frac{(\log x)^2}{4}\right), \qquad x>0.$$

The coefficient definitions and Mellin inversion give

$$T_{m,v,\eta}(t) = \sum_{\theta\in\widehat{\mathcal{T}}} a_\theta T(t+1/2, \Psi_{m,v,\eta}\theta),$$

$$\frac{1}{2\pi i}\int_{(4)} Z^t\Phi(t)T_{m,v,\eta}(t)\,dt = \sum_{\theta\in\widehat{\mathcal{T}}} a_\theta \mathcal{C}_{V_\Phi}(Z;\Psi_{m,v,\eta}\theta).$$

Thus (4.5) is a weighted average over m and v of finite linear combinations of the cubic theta sums (4.1), with the character depending on m, v, η. For a class $\sigma\in\mathcal{T}$ and representative v_σ, put $\nu_\sigma(A) = \eta(A)\Xi(A)^{-1}\mathcal{R}(A, v_\sigma)$ and

$$B_{m,\sigma}(Z) = \sum_\theta a_\theta \frac{1}{2\pi i}\int_{(4)} Z^t\Phi(t)T(t+1/2, \nu_\sigma\theta\chi_\bullet(m))dt,$$

$$A_{m,\sigma,t_0}(Y) = \frac{1}{Y}\sum_{v\in\sigma}\frac{W_1(q_v/Y)\chi_v(b_*)}{\tau_*\xi(v)}(q_v/Y)^{-1/2+it_0}q_v^{-1/2}g_{\chi_v}(v,-m).$$

Writing $Q = q_{b_*}XY$, Mellin inversion separates the average:

$$I_\eta = \frac{Q^{-1/2}}{2\pi}\sum_{\sigma\in\mathcal{T}}\int_{\mathbb{R}}\widehat{W}_0(it_0)\sum_{m\ne0}\Omega(q_m/Q)\xi(m)(q_m/Q)^{-it_0}A_{m,\sigma,t_0}(Y)B_{m,\sigma}(Z)\,dt_0. \tag{4.6}$$

Here $t_0\in\mathbb{R}$ is the Fourier integration variable, $\widehat{W}_0(it_0) = \int_0^\infty W_0(r)r^{it_0}\,dr/r$, and Ω is a fixed smooth cutoff equal to one on the support of W_0 [Ope26a, Section 6]. Applying the Cauchy–Schwarz inequality to (4.6) gives

$$\begin{aligned}|I_\eta| \ll Q^{-1/2}\sum_{\sigma\in\mathcal{T}}\int_{\mathbb{R}}|\widehat{W}_0(it_0)|&\left(\sum_{0<q_m\ll Q}|A_{m,\sigma,t_0}(Y)|^2\right)^{1/2}\\ &\times\left(\sum_{0<q_m\ll Q}|B_{m,\sigma}(Z)|^2\right)^{1/2}dt_0.\end{aligned} \tag{4.7}$$

The fixed bound for Ω is absorbed into the implied constant. Thus the direct estimate reduces to the two displayed mean squares.

4.4. **Poisson summation and the scalar Euler factors.** In this subsection, we apply Poisson summation to the smoothed average and express the resulting terms through their scalar Euler factors. The term $u=1$ yields the principal Mellin integral involving $H_\eta(s)/L_F^S(s,\eta)$ in (3.2); bounding the remaining terms gives the comparison estimate (3.4). Together with the direct mean-square bound from Section 4, this comparison allows Proposition 3.1 to be applied at 11/12 in Section 6. The 7/8 argument uses the same Poisson representation after inserting the selected prime factors of Section 7, with the refined counts of Section 10 controlling the remaining terms.

Let $\widetilde{W}_0(|y|^2)$ be the Fourier transform on $\mathbb{C}\cong\mathbb{R}^2$ of the function $x\mapsto W_0(|x|^2)$, with the pairing and self-dual measure of Section 2:

$$\widetilde{W}_0(|y|^2) = \int_{\mathbb{C}} W_0(|x|^2)e(-xy)\,d\mu(x).$$

For a fixed theta index $A = cn^3$ and primary v coprime to S, the arithmetic coefficient of the m-sum in (4.5) is periodic modulo vb_*A, even when $(v, A) \neq 1$. Define its finite Fourier coefficient by

$$\mathcal{F}_{v,A}(H) = \sum_{\substack{d \bmod v \\ H \equiv b_* A d \pmod v}} \chi_v(d)\, g_{\xi\chi_A}\left(b_*A, \frac{H - b_*Ad}{v}\right), \qquad H \in \mathcal{O}.$$

Expanding $g_{\chi_v}(v, -m)$ gives

$$\sum_{r \bmod vb_*A} \xi(r)\chi_A(r) q_v^{-1/2} g_{\chi_v}(v, -r) e\left(\frac{rH}{vb_*A}\right) = q_v^{1/2}\mathcal{F}_{v,A}(H).$$

Poisson summation on $\mathcal{O}$ therefore gives the explicit identity

$$\begin{aligned} &\sum_{m \in \mathcal{O}\setminus\{0\}} \xi(m)\chi_A(m) q_v^{-1/2} g_{\chi_v}(v, -m) W_0\left(\frac{q_m}{q_{b_*} q_v X}\right) \\ &\qquad = \frac{X q_v^{1/2}}{q_A} \sum_{H \in \mathcal{O}} \mathcal{F}_{v,A}(H) \widetilde{W}_0\left(\frac{X q_H}{q_A}\right). \end{aligned} \tag{4.8}$$

Here W_0 is extended by zero at 0. The normalization follows from $q_{vb_*A} = q_v q_{b_*} q_A$ and the dilation factor $q_{b_*} q_v X$; see [Ope26a, Section 7, Poisson summation]. Primitivity of ξ at the primes in S gives $\mathcal{F}_{v,A}(H) = 0$ unless $(H, S) = 1$, and in particular removes $H = 0$. Every remaining frequency is uniquely $H = ua^6$, where $u \in \mathcal{O}\setminus\{0\}$ includes its unit factor and has prime valuations at most five, while a is a primary generator. Mellin inversion of the two test functions, followed by the local calculation below, converts (4.8) into (4.9). Set

$$\mathcal{M}(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr, \quad \widehat{W}_1(w) = \int_0^\infty W_1(r) r^{w-1} dr,$$
$$\mathcal{W} = X^{1/2-z} Z^{s+z-1} Y^{w-1} \Phi(s + z - 1)\mathcal{M}(z)\widehat{W}_1(w).$$

The complete local calculation [Ope26a, Lemma 7.1] gives

$$I_\eta = \frac{1}{(2\pi i)^3} \int_{(3)} \int_{(3)} \int_{(2)} \mathcal{W} \sum_u^{(6)} q_u^{-z} \overline{\xi(u)} \frac{\zeta_F^S(6z) L_F^S(w, \chi_\bullet(u))}{L_F^S(s, \eta\overline{\chi_\bullet(u)})} \mathcal{H}_{\eta,u}(s, w, z)\, dz\, dw\, ds. \tag{4.9}$$

Here $\chi_\bullet(u) : n \mapsto \chi_n(u)$ and $\sum^{(6)}$ means sixth-power-free nonzero elements coprime to S.

Let $\zeta_F^S(s) = L_F^S(s, 1)$ be the Dedekind zeta function with the primes of S deleted. The local coefficient $P_p = P_p(s, w, z; \eta, u)$ is the complete valuation series computed in [Ope26a, Lemma 7.1], immediately before and in its local identity: $v_p(c) \in \{0, 1\}$ and $v_p(n), v_p(v), v_p(a) \geq 0$ are summed, with $v_p(u) \in \{0, \dots, 5\}$ fixed. This summary's s is the variable called x in that lemma. For its scalar Euler factors write

$$V_p = q_p^{-6z}, \quad W_p = \chi_p(u) q_p^{-w}, \quad D_p = \eta(p)\overline{\chi_p(u)} q_p^{-s}, \qquad H_p = P_p \frac{(1 - V_p)(1 - W_p)}{1 - D_p}. \tag{4.10}$$

Then $\mathcal{H}_{\eta,u} = \prod_{p \notin S} H_p$. Gauss identities, reciprocity and the inserted phases factor the coefficient into these local terms. The six cases $v_p(u) = 0, \dots, 5$ are summed before extracting the scalar quotient. The correction is holomorphic in the two regions

$$\mathcal{D}_1 : \quad s_r \geq 51/100, \quad z_r \geq 17/50, \quad w_r \geq -1/100, \quad s_r + w_r \geq 1 + \varepsilon_0, \tag{4.11}$$

$$\mathcal{D}_2 : \quad s_r \geq 7/8, \quad z_r \geq 33/200, \quad w_r \geq 19/20, \tag{4.12}$$

where $s_r = \operatorname{Re} s$, $w_r = \operatorname{Re} w$, $z_r = \operatorname{Re} z$, and $\varepsilon_0 > 0$ is the fixed gap above 1 in the condition $s_r + w_r \geq 1 + \varepsilon_0$, with $\mathcal{H}_{\eta,u} \ll_\varepsilon q_u^\varepsilon$ uniformly in heights.

The numerator character $\chi_\bullet(u)$ in (4.9) is principal only when $u = 1$: a good prime dividing u forces nontrivial ramification, and a nontrivial unit gives a nontrivial Kummer character. At $u = 1$ the poles are $w = 1, z = 1/6$. Define

$$H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6), \qquad c_S = \widehat{W}_1(1)\mathcal{M}(1/6)\frac{(\operatorname{Res}_{v=1}\zeta_F^S(v))^2}{6} > 0. \tag{4.13}$$

The factor $1/6$ comes from taking the residue of $\zeta_F^S(6z)$ at $z = 1/6$:

$$\operatorname{Res}_{z=1/6}\zeta_F^S(6z) = \frac{1}{6}\operatorname{Res}_{s=1}\zeta_F^S(s).$$

A positive Gaussian-integral representation proves $\mathcal{M}(1/6) > 0$. Since $(s, 1, 1/6) \in \mathcal{D}_2$ for $\operatorname{Re} s > 7/8$, (4.12) and (4.13) give holomorphy of H_η there. The uniform local estimate $H_\eta(s) = 1 + O(P_0^{-4/5})$ [Ope26a, Lemma 7.1] allows P_0 to be chosen sufficiently large that

$$\sup_{\operatorname{Re} s > 7/8} |H_\eta(s) - 1| \le 1/2. \tag{4.14}$$

The choice of P_0 is uniform in the fixed target characters and root-of-unity factors; see also [Ope26a, Section 11, principal correction].

5. Zero detection by two character polynomials

In this section, we associate the nonprincipal terms in the Poisson representation (4.9) of Section 4.4 with zero classes and detecting character polynomials. The zero classes are defined in (5.1). A selected zero forces the two detecting sums M_r, S_m, with the same character presentation and norm-twist height, to satisfy the simultaneous large-value bounds (5.5). The zero classes and the large value of M_r are first used with the sextic sieve in Section 6 to obtain the preliminary $11/12$ bound. For the sharper $7/8$ bound, Sections 9–10 combine both large values with the two moment estimates to obtain a stronger count of elements in each class.

For the zero-detection method, see [MP24, Appendix C, Zero detecting setup]. The simultaneous Hecke-character version used here is [Ope26a, Proposition 8.3].

For nonzero sixth-power-free elements $u \in \mathcal{O}$ coprime to S, including all unit associates, with $q_u \asymp U = Z^d$, consider

$$\mathcal{X}_u = \{\nu(n)\chi_n(u)^{\pm 1} : \nu \in \Theta\}.$$

For a presentation ψ, let ψ^* denote its primitive inducing character as in (2.2): it has the minimal defining modulus and agrees with ψ on ideals coprime to the defining modulus of ψ. Let $L(s, \psi)$ be the Hecke series of ψ with the original zero extension; thus $L(s, \psi^*) = L_F(s, \psi^*)$. Primitive conductors have norm $\ll_{\mathcal{A}} U$; any Euler factors omitted by the original presentations occur only at prime ideals in S or dividing (u). Choose $0 < \epsilon_{\mathrm{b}} < 10^{-3}$ and $T_1 = Z^\tau \le U^{1/100}$. Set $I_{\epsilon_{\mathrm{b}}} = \lceil 1/\epsilon_{\mathrm{b}} \rceil + 2$. With $a_0 = 51/100$ and $j \in \{1, \dots, I_{\epsilon_{\mathrm{b}}}\}$, let

$$M_j(u) = \max\big(\{a_0\} \cup \{\operatorname{Re}\rho : \psi \in \mathcal{X}_u,\ L(\rho, \psi^*) = 0,\ \operatorname{Re}\rho \ge a_0,\ |\operatorname{Im}\rho| \le 3jT_1\}\big).$$

A pigeonhole argument gives $1 \le i < I_{\epsilon_{\mathrm{b}}}$ and $a \in (a_0 + \epsilon_{\mathrm{b}}\mathbb{Z}_{\ge 0}) \cap [a_0, 1]$ such that

$$a \le M_i(u) < a + \epsilon_{\mathrm{b}}, \qquad M_{i+1}(u) < a + 2\epsilon_{\mathrm{b}} \qquad \text{[Ope26a, Lemma 8.1]}. \tag{5.1}$$

Hence the associated functions have a zero-free rectangle to the right of $a + 2\epsilon_{\mathrm{b}}$ at the needed height. Concretely,

$$\mathcal{U}_{a,i} = \{s \in \mathbb{C} : a + 2\epsilon_{\mathrm{b}} \le \operatorname{Re} s \le 2,\ |\operatorname{Im} s| \le 3(i+1)T_1\}$$

contains no zero of $L_F(s,\psi^*)$ for $\psi \in \mathcal{X}_u$, by (5.1) and the definition of M_{i+1}. If $a > a_0$, a selected zero $\rho = \sigma + i\gamma$ satisfies $a \le \sigma < a + \epsilon_{\mathrm{b}}$, $|\gamma| \le 3iT_1$. Since these are actual family zeros, $a \le \beta_*$.

Let $\mathcal{B}$ denote a subset of sixth-power-free elements $q_u \asymp U$ in one fixed (i,a) class; later it may also fix prime-sum size ranges. All class counts are finite for fixed parameters [Ope26a, Lemma 8.1]. For one presentation $\psi \in \mathcal{X}_u$, real length exponents $r, m \ge 0$, and test functions W_M, W_S, define

$$M_r = U^{-r/2} \sum_n \mu(n)\psi(n) W_M(q_n/U^r), \qquad S_m = U^{-m/2} \sum_n \psi(n) W_S(q_n/U^m).$$

Here n runs over nonzero integral ideals, represented by their primary generators where a residue symbol is used; the character values retain their stated zero extensions. The test functions may include factors q_n^{it} with $t \in \mathbb{R}$; these are the norm twists specified in Section 2. Polynomial bounds on fixed vertical strips, logarithmic bounds in zero-free disks and deleted Euler estimates [Ope26a, Lemmas 4.8–4.10] yield

$$|M_r|^2 \ll U^{\delta r+\varepsilon}, \qquad |S_m|^2 \ll U^{\delta \min(m,1-m)+\varepsilon}, \qquad \delta = 2a-1, \tag{5.2}$$

when $(1+T_1)^A \le U^{\varepsilon/10}$ [Ope26a, Lemma 8.2]. Here T_1 is the height cutoff used in (5.1), and $A \ge 0$ is a fixed finite exponent for its polynomial dependence, as in Section 2. The second estimate takes the better of the direct bound and the bound obtained by applying the Hecke functional equation. For the classical functional equation in this setting, see [GZ20, equation (1.1)]; the zero-class bounds with deleted Euler factors are those of [Ope26a, Lemma 8.2].

Proposition 5.1 (Simultaneous large values, [Ope26a, Proposition 8.3])**.** *Fix $\varepsilon > 0$, a class (i,a) from (5.1) with $a > a_0$, and a sixth-power-free u of norm $\asymp U$ in that class. Assume $T_1 \le U^{1/100}$ and the height bound $(1+T_1)^A \le U^{\varepsilon/10}$ from [Ope26a, Lemma 8.2], after choosing its preliminary losses sufficiently small. For every $t \in [1,3/2]$ and all $U \ge U_0(\mathcal{A},\varepsilon)$, there exist a character presentation $\psi \in \mathcal{X}_u$, a zero ρ of $L_F(s,\psi^*)$, where ψ^* is the primitive character inducing ψ, and two sums M_r, S_m of the form defined above, both using this same presentation (including its defining modulus and zero extension) and the same norm-twist height, such that*

$$r \le t + O(1/\log U), \qquad r + m \ge t - O(1/\log U), \tag{5.3}$$

$$t - \tfrac{1}{2} - O(\varepsilon) \le r, \qquad 0 \le m \le \tfrac{1}{2} + O(\varepsilon), \tag{5.4}$$

$$|M_r|^2 \gg U^{\delta r-\varepsilon}, \qquad |S_m|^2 \gg U^{\delta m-\varepsilon}. \tag{5.5}$$

Idea of zero detection. To obtain them, put

$$C_\psi(s) = \sum_l \mu(l)\psi(l) V_{\le}(q_l/U^t) q_l^{-s}, \qquad V_{\le} = 1 \text{ on } [0,1], \quad V_{\le} = 0 \text{ on } [2,\infty),$$

where $V_{\le}$ is a fixed smooth cutoff. Here l ranges over nonzero integral ideals and t is the real truncation exponent. The polynomial C_ψ is a truncated version of the Möbius series for $1/L(s,\psi)$ in $\operatorname{Re} s > 1$. Insert it into

$$\frac{1}{2\pi i} \int_{(2)} U^{20z} \Gamma(z) L(\rho+z,\psi) C_\psi(\rho+z)\, dz.$$

Moving the contour to $\operatorname{Re} z = -1/4$ gives a small integral; the residue at $z=0$ vanishes because $L(\rho,\psi) = 0$. Expansion of LC_ψ cancels the ideal coefficients up to length U^t except the term of the unit ideal. Splitting the remaining ideal norms into dyadic intervals therefore gives a product of detecting polynomials with $|M_r S_m|^2 \gg U^{\delta(r+m)-\varepsilon}$. Comparison with (5.2) forces $m \le 1/2 + O(\varepsilon)$ and then both individual lower bounds. At

$a = a_0$, the class definition (5.1) need not provide a selected zero, so this lowest class is bounded by counting all elements in its norm range.

6. The preliminary 11/12 bound

In this section, we first apply Proposition 3.1 at 11/12, using the unmodified average (4.5) and the preliminary character count (6.2). This bound restricts the possible zero classes and supplies the starting point for the refined 7/8 argument in the remaining sections of Part I.

Take $X = Y = Z^{1/2}$ and $Q = q_{b_*}Z$. For the number-field additive large sieve, see [Hux68]; its exact planar form and application here are [Ope26a, Lemmas 6.1–6.2]. It gives

$$\sum_{q_m \ll Q} |A_{m,\sigma,t_0}(Y)|^2 \ll (Q + Y^2)/Y \ll Q/Y.$$

Equation (4.3), at $M = N = 1$, gives $\sum |B_{m,\sigma}(Z)|^2 \ll Z^{1+\varepsilon}$. Thus (4.6) implies

(6.1)
$$|I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll Q^{-1/2}(Q/Y)^{1/2}Z^{1/2+\varepsilon} = Z^{1/4+\varepsilon} \qquad \text{[Ope26a, Proposition 6.3].}$$

The underlying estimates are the cubic sieve [HB00, Theorem 2], the quadratic sieve [GL13, Theorem 1.1], and the higher-order norm recursion [BGL14, Theorem 1.3 and Section 3]. An Eisenstein-integer sextic formulation is also recorded in [GZ24, Lemma 2.9]; [Ope26a, Lemma 9.1] gives the primary-generator convention used here.

Lemma 6.1 (Sextic sieve, [Ope26a, Lemma 9.1]). *Let $K, N_{\mathrm{s}} \geq 1$ bound the norms of squarefree primary elements k, n coprime to S, and let c_n be finitely supported and independent of k. For every $\varepsilon > 0$,*

$$\sum_{k \text{ sf},\ q_k \leq K} \left| \sum_{n \text{ sf},\ q_n \leq N_{\mathrm{s}}} c_n \chi_n(k)^{\pm 1} \right|^2 \ll (KN_{\mathrm{s}})^{\varepsilon} \left[K + N_{\mathrm{s}} + (KN_{\mathrm{s}})^{2/3}\right] \sum_n |c_n|^2.$$

Proposition 6.2 (Preliminary character count, [Ope26a, Proposition 9.2]). *Let $\mathcal{B}$ be a zero class of sixth-power-free elements $q_u \asymp U$, defined by (5.1) in Section 5, with the detector height assumptions and $\delta = 2a - 1$. For every $\varepsilon > 0$ and sufficiently large U,*

$$\#\mathcal{B} \ll U^{R_{\mathrm{I}}(\delta)+\varepsilon}(1 + T_1)^A, \quad R_{\mathrm{I}}(\delta) = \min\{1, \max(1 - \delta/2, 4/3 - \delta)\}. \tag{6.2}$$

Here T_1 is the zero-detection height cutoff of Section 5; $A \geq 0$ is a fixed finite exponent controlling the height dependence, as in Section 2.

Proposition 6.3 (Preliminary zero-free bound, [Ope26a, Theorem 3.1]). *The family supremum in (1.2) satisfies*

$$\beta_* \leq 11/12. \tag{6.3}$$

The preliminary exponent comparison. Assume $\beta_* > 11/12$ and put $\Delta_1 = \beta_* - 11/12 > 0$. The principal residue in (4.9) has exponent $X^{1/3}Z^{s-5/6} = Z^{s-2/3}$. Hence

$$C_{\mathrm{I}}(s) = s - 2/3, \qquad J_{\mathrm{I},\eta}(Z) = I_\eta(Z^{1/2}, Z^{1/2}, Z)/c_S,$$

with $f_{\mathrm{I},\eta}$ defined by (3.2). The preliminary character count (6.2), the contour estimates [Ope26a, Lemmas 10.2–10.6], and the height choice in [Ope26a, Proposition 11.2] give

$$|J_{\mathrm{I},\eta} - f_{\mathrm{I},\eta}| \ll_\eta Z^{C_{\mathrm{I}}(\beta_*)-1/4800}.$$

Apply Proposition 3.1 with

$$\sigma_0 = 11/12, \qquad c = -2/3, \qquad \omega_0 = \Delta_1/2, \qquad \sigma = 1/4800.$$

The correction hypotheses follow from (4.12), (4.13) and (4.14). The displayed comparison bound and (6.1), with its loss chosen below $\Delta_1/2$, verify the two estimates, since $C_{\mathrm{I}}(11/12) = 1/4$. Thus the preliminary argument reaches the contradiction through the same continuation criterion used for the 7/8 bound.

7. The modified average and its direct estimate

In this section, we modify the average used for the preliminary 11/12 bound in Section 6. There the lengths are $X = Y = Z^{1/2}$ and the direct estimate is $Z^{1/4+\varepsilon}$, as in (6.1). Here we use unequal lengths and apply the selected-prime subtractions in (7.2) to obtain the direct estimate $Z^{3/16+\varepsilon}$ in (7.3). The corresponding principal exponent changes from $C_{\mathrm{I}}(s) = s - 2/3$ to $C(s) = s - 11/16$, so that $C_{\mathrm{I}}(11/12) = 1/4$ and $C(7/8) = 3/16$. The next section computes this principal term and the effect of the subtractions on the nonprincipal terms.

Parts II–IV retain this modified average and vary its parameters. Theorem 34.4 proves that further variation within the explicitly specified system of estimates cannot give a bound below B_{new}. A different modification of the average or a stronger estimate for its terms lies outside that optimality statement.

Assume $\beta_* > 7/8$ and use (1.3). Choose the averaging lengths and their exponents as follows:

$$\begin{gathered} b = 1/8, \quad h = 13/16, \quad \ell = 1/6, \quad l_x = 17/48, \quad l_y = 23/48, \\ X = Z^{l_x}, \quad Y = Z^{l_y}, \quad l_x + l_y = 5/6, \quad b = l_y - l_x = 1/8, \quad h = 1 - l_x + \ell. \end{gathered} \tag{7.1}$$

Choose a positive integer K and K disjoint prime intervals with smooth nonnegative weights W_i, norms $q_p \asymp P_i = Z^{\ell_i}$, and $\sum_i \ell_i = \ell$. The allowed set $\mathcal{P}_i$ also imposes $p \notin S$ and $p \in 1_{\mathcal{T}}$. The underlying intervals are disjoint before these restrictions.

For a squarefree ideal D outside S, let $I_{\eta;D}$ denote (4.5) with $1_{D|cn^3}$ inserted. Write $J^c = \{1, \dots, K\} \setminus J$ and $p_J = \prod_{i\in J} p_i$ (empty product 1). All tuple sums have $p_i \in \mathcal{P}_i$. Define

$$\begin{aligned} I_{\eta,\mathrm{mod}}(Z) = \sum_{(p_i)} \prod_i W_i(q_{p_i}/P_i) \sum_{J\subseteq\{1,\dots,K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \\ \times I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \end{aligned} \tag{7.2}$$

The one-prime operation is $\overline{\eta(p)} I_{\eta;p}(X, Y, Zq_p) - q_p^{-3/2} I_\eta(X/q_p, Y/q_p, Z)$. Its effect on the local Euler factors is computed in (8.1): the subtraction cancels an unwanted scalar Euler contribution and produces sextic-character prime sums, together with the local errors bounded in (8.3). These prime sums enter the estimates for the nonprincipal terms, whereas their principal contribution is nonzero by (8.4) and gives the normalization (8.5). These substitutions change the averaging lengths X, Y, while each selected prime p_i remains in its original set $\mathcal{P}_i$, with norm $q_{p_i} \asymp P_i = Z^{\ell_i}$ and test function $W_i(q_{p_i}/P_i)$, using the original value of Z.

Proposition 7.1 (Modified direct bound, [Ope26a, Proposition 15.3])**.** *For the fixed arithmetic data, prime intervals and smooth weights specified above, with the parameter choices* (7.1)*, one has*

$$|I_{\eta,\mathrm{mod}}(Z)| \ll_{\eta,\varepsilon} Z^{3/16+\varepsilon}. \tag{7.3}$$

For a fixed J put

$$d_J = \sum_{i\in J} \ell_i, \quad X' = X/q_{p_J}, \quad Y' = Y/q_{p_J}, \quad M' = 5/6 - 2d_J, \quad Q = q_{b_*} X'Y'.$$

Let $B^J_{m,\sigma}$ be the theta factor including all cube multiples in the separated expression, with the primes $i \notin J$ already summed and their divisibility restrictions imposed. The theta functional equation with the divisibility restrictions [Ope26a, Corollary 14.1] and the resulting mean-square bound give the following estimate.

Lemma 7.2 (Modified theta mean square, [Ope26a, Lemma 15.1]). *For every $\varepsilon > 0$, fixed subset J and ray class σ, uniformly in the fixed tuple $(p_i)_{i\in J}$, with J and the prime sets as in (7.2),*

$$\sum_{q_m \ll Q} |B^J_{m,\sigma}|^2 \ll Z^{M'+((d_J-1/6)/4)_+ +\varepsilon} = Z^{M'+\varepsilon}. \tag{7.4}$$

The extra term vanishes because $0 \le d_J \le \ell = 1/6$. The remaining prime sums are included in $B^J_{m,\sigma}$.

Proposition 7.3 (Additive mean square, [Ope26a, Proposition 15.2]). *Fix $S, \mathcal{T}, b_*, \xi, W_1$ and a ray class σ in the finite ray group $\mathcal{T}$ of Section 4.3. Let $Q, Y' \ge 1$ range over fixed polynomial ranges in Z, and suppose $P_a = Y'^2/Q \ge 1$. For every $\varepsilon > 0$ and real t_0, with the coefficient A defined in Section 4,*

$$\sum_{q_m \ll Q} |A_{m,\sigma,t_0}(Y')|^2 \ll (1+|t_0|)^{J_{\mathrm{Gr}}} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'}\right) Z^{\varepsilon}. \tag{7.5}$$

Here J_{Gr} is a fixed seminorm order, independent of the subset J, varying norm parameters Q, Y' and t_0.

Idea of the additive estimate. Expansion of the square and Poisson summation reduce (7.5) to complete finite correlations. The common-factor identities and local prime-power formulas are [Ope26a, Lemmas 13.2–13.4]; the final mean-square estimate is [Ope26a, Proposition 15.2]. Both error terms $P_a^{1/6}$ and P_a^2/Y' in (7.5) are retained. Their comparison for the chosen parameter values is the calculation needed below.

The direct exponent calculation. With the fixed parameter choices (7.1),

$$P_a = q_{b_*}^{-1} Z^{1/8}, \qquad l_y - d_J - 11b/6 \ge 1/12, \qquad P_a^2/Y' \ll P_a^{1/6}.$$

Thus $P_a \ge 1$ once $Z^{1/8} \ge q_{b_*}$, which holds for all sufficiently large Z with the arithmetic data fixed. Equations (7.4)–(7.5) and the Cauchy–Schwarz inequality (4.7), applied after fixing J and $(p_i)_{i\in J}$ and summing the complementary prime variables in (7.2), therefore bound each separated term by $(X')^{1/2} P_a^{1/12} Z^{\varepsilon}$. The tuples $(p_i)_{i\in J}$ have total weight

$$Z^{d_J+\varepsilon} O(Z^{-3d_J/2}), \qquad (X')^{1/2} \asymp X^{1/2} Z^{-d_J/2}.$$

Consequently the exponent in (7.3) is $l_x/2 + b/12 = 3/16$, since $d_J - 3d_J/2 - d_J/2 = -d_J \le 0$. The asymmetry in (7.1) was chosen to obtain this exponent while keeping the additive error under control.

8. Local cancellation and principal residues

In this section, we compute the effect of the modification in Section 7 on the local Euler factors (8.1) and the principal coefficient obtained from (8.4). The normalized average $J_\eta(Z)$ is defined in (8.5). Its principal contribution is the integral $f_\eta(Z)$ in (8.8), whose numerator contains the correction $H_\eta(s)$ defined in (4.13); the estimate (4.14) gives $|H_\eta(s)| \ge 1/2$ for $\operatorname{Re} s > 7/8$. We also prove the joint local error estimate (8.3), used in Section 11. The same local cancellation and normalization are retained in Parts II–IV, with the extended domain proved in Section 17.

8.1. **Selected-prime Euler factors and joint local errors.** In this subsection, we compute the local cancellation caused by the selected prime factors: (8.1) gives the local factor and (8.2) defines the resulting holomorphic correction. The joint local error is bounded in (8.3). These bounds ensure that the correction remains holomorphic on the contours used in Section 11 and that the modification does not introduce an error as large as the principal term.

Let P_p be the local valuation series introduced in Section 4.4 immediately before (4.10), and let P_p^* consist of its terms satisfying $v_p(c) + 3v_p(n) > 0$, equivalently $p \mid cn^3$. The selected-prime operation replaces P_p by

$$q_p^{z-1}\big[\overline{\eta(p)}q_p^s P_p^* - q_p^{-w}P_p\big].$$

After extracting the same scalar Euler factors as in (4.9), define

$$G_p = \big[\overline{\eta(p)}q_p^s P_p^* - q_p^{-w}P_p\big]\frac{(1-V_p)(1-W_p)}{1-D_p}. \tag{8.1}$$

The local identity of [Ope26a, Lemma 7.1] shows that G_p continues holomorphically in $\mathcal{D}_1, \mathcal{D}_2$. It does not require division by P_p. The correction is

$$\mathfrak{H}_{\eta,u,Z} = \sum_{(p_i)}\prod_i \big[W_i(q_{p_i}/P_i)q_{p_i}^{z-1}G_{p_i}\big] \prod_{\substack{p\notin S\\ p\notin\{p_i\}}} H_p. \tag{8.2}$$

Replacing $\mathcal{H}$ by $\mathfrak{H}$ in (4.9) gives the exact modified average.

On the contours used for the nonprincipal contributions, $s_r = a + 16\epsilon_{\mathrm{b}}, w_r = 1 - a - 6\epsilon_{\mathrm{b}}, z_r = 17/50$, [Ope26a, Section 16] decomposes

$$\mathfrak{H} = \mathcal{H}\prod_i(\mathcal{Q}_i + \mathcal{E}_i), \qquad \mathcal{Q}_i = -\sum_{p\in\mathcal{P}_i} W_i(q_p/P_i)q_p^{z-1}\overline{\chi_p(u)}.$$

For each selected prime set $\mathcal{P}_i$, define its exact sum $\mathcal{B}_i$ and error sum $\mathcal{E}_i$ on these contours by

$$\mathcal{B}_i = \sum_{p\in\mathcal{P}_i} W_i(q_p/P_i)q_p^{z-1}\frac{G_p}{H_p}, \qquad \mathcal{E}_i = \mathcal{B}_i - \mathcal{Q}_i.$$

Proposition 8.1 (Joint local error, [Ope26a, Proposition 16.1])**.** *Let $a_0 \le a \le 1$, $0 < \epsilon_{\mathrm{b}} \le 10^{-3}$, $q_u \asymp U$, with u sixth-power-free and $(u, S) = 1$. Put $\delta = 2a - 1$. Suppose the primitive character ψ_u^* inducing $\chi_\bullet(u)$ is nonprincipal. On the displayed contours restrict to $|\Im w| \le T_1$, and assume $|L_F(1-w, \overline{\psi_u^*})| \ll U^{\varepsilon}$ uniformly on the points under consideration. For sufficiently large prime-norm cutoff P_0, as defined in Section 2, the local H_p are nonzero on these contours. Moreover, for every subset $I \subseteq \{1, \dots, K\}$, one has*

$$\Big|L_F^S(w, \chi_\bullet(u))\prod_{i\in I}\mathcal{E}_i\Big| \ll U^{\delta/2+O(\epsilon_{\mathrm{b}})+\varepsilon}(3+T_1)^A\prod_{i\in I}P_i^{17/50-1/2+O(\epsilon_{\mathrm{b}})+\varepsilon}. \tag{8.3}$$

Here $A \ge 0$ is a fixed finite exponent controlling the dependence on the height cutoff T_1, as in Section 2. Recall that $P_i = Z^{\ell_i}$ is the norm parameter of the selected prime set $\mathcal{P}_i$ in Section 7, and ϵ_{b} is the zero-class width in (5.1).

Let $\mathfrak{f}_u$ be the conductor ideal of ψ_u^*, and I_0 any set of distinct selected primes dividing u. With $j_p = v_p(u) \in \{1, \dots, 5\}$,

$$q_{\mathfrak{f}_u} \ll_S q_{\mathrm{rad}(u)} \le q_u\prod_{p\in I_0} q_p^{-(j_p-1)}.$$

The displayed conductor bound saves the product $\prod_{p\in I_0} q_p^{j_p-1}$ relative to q_u. Applying this saving once in the functional-equation estimate for $L_F(w,\psi_u^*)$ controls the local errors at all selected ramified primes simultaneously, as required in (8.3). Absolute bounds on the other contours use [Ope26a, Lemma 16.2].

8.2. **The principal coefficient and the normalized average.** In this subsection, we evaluate the principal contribution using the prime-sum asymptotic (8.4) and use (8.5) to normalize the modified average for Proposition 3.1. The normalized average is $J_\eta(Z)$, and its principal integral is $f_\eta(Z)$ in (8.8), with exponent $C(s)$ in (8.7).

At $u=1$ in $\mathcal{D}_2$, $G_p/H_p=-1+O(q_p^{-7/8})$ uniformly in heights.

Lemma 8.2 (Prime asymptotic in a fixed ray class, [Ope26a, Lemma 13.1])**.** *Fix the ray quotient $\mathcal{T}$ of Section 4.3, the excluded set S of Section 2, and a nonnegative nonzero test function W_i. As $P_i\to\infty$, with $\mathcal{P}_i$ restricted to the identity class of $\mathcal{T}$ and primes outside S, one has*

$$S_i(Z):=\sum_{p\in\mathcal{P}_i} W_i(q_p/P_i)q_p^{-5/6}\sim \frac{P_i^{1/6}}{|\mathcal{T}|\log P_i}\int_0^\infty W_i(y)y^{-5/6}dy>0. \tag{8.4}$$

Fixed-ray prime counting follows by applying Chebotarev to the fixed ray-class field; see [TZ19, Theorem 1.1]. Smooth partial summation gives (8.4).

For large Z set

$$A_{\mathcal{T}}(Z)=(-1)^K Z^{-\ell/6}\prod_i S_i(Z),\qquad J_\eta(Z)=\frac{I_{\eta,\mathrm{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}. \tag{8.5}$$

Then $|A_{\mathcal{T}}|\asymp(\log Z)^{-K}$ and $|A_{\mathcal{T}}|^{-1}\ll_\varepsilon Z^\varepsilon$, so

$$|J_\eta(Z)|\ll_{\eta,\varepsilon} Z^{3/16+\varepsilon}. \tag{8.6}$$

At $w=1, z=1/6$ the correction equals

$$H_\eta(s)Z^{\ell/6}A_{\mathcal{T}}(Z)\big[1+O(Z^{-\kappa_P})\big],\qquad 0<\kappa_P<(7/8)\min_i \ell_i.$$

The residue exponent is

$$C(s)=s+l_x/3-5/6+\ell/6=s-11/16,\qquad C(7/8)=3/16, \tag{8.7}$$

and the resulting integral is

$$f_\eta(Z)=\frac{1}{2\pi i}\int_{(2)} Z^{s-11/16}e^{(s-5/6)^2}\frac{H_\eta(s)}{L_F^S(s,\eta)}ds. \tag{8.8}$$

Crossing $w=1$ on the way to $19/20$, then $z=1/6$ on the way to $33/200$, gives principal remainder exponents

$$C(\beta_*)+(1+h)\epsilon_{\mathrm{b}}-m_w+\varepsilon,\quad C(\beta_*)+\epsilon_{\mathrm{b}}-m_z+\varepsilon,\quad C(\beta_*)+\epsilon_{\mathrm{b}}-\kappa_P+\varepsilon,$$

where

$$m_w=23/960,\qquad m_z=13/9600\qquad [\text{Ope26a, Lemma 10.5}].$$

Throughout these contour shifts, the denominator $L_F^S(s,\eta)$ is evaluated on $\operatorname{Re} s=\beta_*+\epsilon_{\mathrm{b}}$, to the right of all family zeros.

9. Moment estimates for the two detecting polynomials

In this section, we record the two moment estimates for the detecting polynomials of Section 5, one with Möbius coefficients and one without. The length restrictions in the hypotheses of the moment estimates (9.1) and (9.4) are combined in Section 10 to bound each zero and prime-sum size class for the 7/8 argument. Parts II and III reuse these moments at $\kappa = 3/4$; Part IV extends the unweighted fourth moment to smaller κ while retaining the Möbius-weighted estimate.

The lengths in this section are expressed by their exponents in base Z, as in (2.1): length r means a norm parameter Z^r. Products of auxiliary prime polynomials use disjoint prime intervals and one common character presentation. A mean square is a sum $\sum_u |F_u|^2$; the fourth moment arises from $\sum_u |S_u S_u|^2 = \sum_u |S_u|^4$.

9.1. Möbius-weighted second moment.

In this subsection, we state the second-moment estimate for the Möbius-weighted detecting polynomial in (9.1), including its admissible length ranges. Its version with auxiliary prime factors is (9.3). Together with the large-value bound for M_r in (5.5), this supplies one of the two counting estimates in (10.4); the unweighted moment supplies the other.

Fix a ray character ν and a sign $\varepsilon_\chi \in \{-1, 1\}$, and put $\psi_u(n) = \nu(n)\chi_n(u)^{\varepsilon_\chi}$. Here $\chi_n(u)$ is the sextic residue character defined in Section 2, with the primary generator n as modulus and u as argument. The estimates below hold for either choice, with the same sign retained for every n, u and prime factor in the polynomials. Character powers retain the zero-extension convention of (2.3), including for the negative sign. Here n runs over nonzero integral ideals, represented by their primary generators, with the fixed coprimality restrictions and zero extensions retained. The factor $\mu(n)$ makes nonsquarefree ideals contribute zero, and $W(q_n/Z^r)$ restricts the norm to $q_n \asymp Z^r$. Let I be a fixed finite index set. For each $i \in I$, let $\mathcal{P}_i$ be a set of prime ideals outside S, represented by primary generators, with norms $q_p \asymp Z^{z_i}$. The sets $\mathcal{P}_i$ are pairwise disjoint and independent of u; a product over the empty index set equals 1. Define

$$M_u(Z^r) = Z^{-r/2} \sum_n \mu(n)\psi_u(n) W(q_n/Z^r), \qquad Q_u = \prod_{i\in I} Z^{-z_i/2} \sum_{p\in\mathcal{P}_i} a_i(p)\psi_u(p) W_i(q_p/Z^{z_i}).$$

Lemma 9.1 (Möbius-weighted second moment, [Ope26a, Lemma 17.1]). *Let $r, m_0, z_i \geq 0$ be norm exponents: the ideal variable n has norm $\asymp Z^r$, $u \in \mathcal{O} \setminus \{0\}$ ranges over $0 < q_u \ll Z^{m_0}$, including all unit associates, and the ith prime set has norms $\asymp Z^{z_i}$. The coefficients $a_i(p)$ are bounded and independent of u. Fix bounded ranges for these lengths and the number of prime factors. Writing $z_{\mathrm{p}} = \sum_{i\in I} z_i$, assume for a fixed $c_m > 0$ that*

$$r + 2z_{\mathrm{p}} \leq m_0 - c_m, \qquad 2r + 8z_{\mathrm{p}} \leq 3m_0 - c_m.$$

Then for every $\varepsilon > 0$ and all sufficiently large Z,

$$\sum_{0<q_u\ll Z^{m_0}} |M_u(Z^r)Q_u|^2 \ll Z^{m_0+\varepsilon}. \tag{9.1}$$

The implied constant may depend on c_m, the bounded ranges and finitely many smooth seminorms, but is uniform in prime supports and bounded coefficients. There is no restriction on individual prime lengths in this lemma.

Structure of the inverse-moment induction. For the recursive Poisson large-sieve method, compare [HB95], [HB00, Section 4], and [GL13, Sections 4–7], as acknowledged in [Ope26a, Section 17, canonical estimate]. The precise moment estimates with the displayed auxiliary restrictions are those of [Ope26a, Sections 17–18]. One initial Poisson

transform reduces the assertion to [Ope26a, Lemma 17.2]:

(9.2)

$$Z^{-V}\sum_{f\ \mathrm{sf},\ q_f\asymp Z^V} w(f)\sum_{0<q_k\ll Z^M}\left|Z^{-N/2}\sum_{n\ \mathrm{sf}}\overline{\varpi(n)}\gamma_2(n)\nu(n)1_{(n,\mathfrak{r})=1}\chi_n(kf^4)c_{\mathrm{sel}}(n)W(q_n/Z^N)\right|^2$$
$$\ll Z^{F_0+\varepsilon},\qquad F_0=N+V,$$

under

$$q_{\mathfrak{r}}\le Z^{F_0-M-z_{\mathrm{cap}}-c_*},\qquad 3M+6z_{\mathrm{cap}}+c_*\le 4F_0,\qquad c_*>0.$$

The parameters $M,N,V,z_{\mathrm{cap}}\ge 0$ in this lemma are local norm exponents: f ranges over squarefree integral ideals coprime to S with norm $\asymp Z^V$, k over nonzero elements of $\mathcal{O}$ with norm $\ll Z^M$, and n over squarefree integral ideals with norm $\asymp Z^N$, using primary generators and the stated zero extensions. The squarefree ideal $\mathfrak{r}$ is the coprimality modulus, independent of k,f; $c_*>0$ is a fixed positive gap in the two displayed length inequalities, and z_{cap} an upper bound for total prime length. Here $w(f)$ depends on f alone and is nonnegative and bounded by a fixed power of the divisor function. Also,

$$c_{\mathrm{sel}}(n)=\sum_{(p_i)}\prod_i a_i(p_i)1_{p_i|n}$$

has total prime length at most z_{cap}: its tuples run over disjoint fixed prime sets of norms $\asymp Z^{z_i}$, with $\sum_i z_i\le z_{\mathrm{cap}}$. The coefficients a_i are bounded, with data independent of k,f. No additional arbitrary coefficient depending on k is admitted.

The strict length inequalities in Lemma 9.1 give the fixed positive gaps required for the induction on the outer range exponent M in (9.2). Two finite Poisson transforms then decrease M while preserving the common character and coprimality condition, and keeping $4F_0-3M-6z_{\mathrm{cap}}$ bounded below by a fixed positive constant [Ope26a, Lemma 17.3]. The mean-square bound obtained from the functional equation [Ope26a, Lemma 14.3] handles the final range of the induction, and [Ope26a, Corollary 17.4] bounds the seminorm and height orders. These strict length inequalities are needed for the later counts.

Lemma 9.2 (Sixth-power amplification, [Ope26a, Lemma 17.6])**.** *Fix a ray character ν, a common sign $\varepsilon_\chi\in\{-1,1\}$, and a test function W, with the zero extensions of Lemma 9.1. Put $\psi_u(n)=\nu(n)\chi_n(u)^{\varepsilon_\chi}$ and use M_u as defined above, without auxiliary prime factors. Let $r\ge 0$ range over a fixed bounded interval and $q_u\asymp U$, with u sixth-power-free. For every $\varepsilon>0$ and sufficiently large U,*

$$\sum_{q_u\asymp U}^{(6)}|M_u(U^r)|^2\ll U^{\mathfrak{a}(r)+\varepsilon}(1+T_1)^A,\qquad \mathfrak{a}(r)=\max\{1,(1+5r)/6\}. \tag{9.3}$$

The estimate permits a smooth weight depending on the element u $W(y)y^{-\sigma_u+it_u}$, with σ_u in a fixed compact interval and $|t_u|\le T_1$. The factor $(1+T_1)^A$ accounts for its smooth seminorms; A depends only on the fixed ranges and arithmetic data.

Idea of sixth-power amplification. Here "amplification" refers to the additional averaging over a after replacing u by ua^6. Since the sextic residue character satisfies $\chi_n(ua^6)=\chi_n(u)$ when $(n,a)=1$, this replacement preserves the character values on those arguments. Inclusion–exclusion accounts for the remaining coprimality restrictions, and averaging over a allows a mean-square estimate over a larger norm range to bound the original sum over u. For a polynomial length $D_{\mathrm{amp}}\ge 1$ and primary a, inclusion-exclusion gives

$$M_u(D_{\mathrm{amp}})=\sum_{d|\mathrm{rad}\,a}\mu(d)\psi_u(d)q_d^{-1/2}M_{ua^6}(D_{\mathrm{amp}}/q_d).$$

Average over $q_a \le P_{\mathrm{amp}} = (H_{\mathrm{amp}}/U)^{1/6}$ with $H_{\mathrm{amp}} = \max(2U, D_{\mathrm{amp}}^{1+c})$. The map $(u,a) \mapsto ua^6$ is injective on sixth-power-free u, and the enlarged mean-square bound contributes the factor $H_{\mathrm{amp}}/P_{\mathrm{amp}} = U^{1/6}H_{\mathrm{amp}}^{5/6}$. Here $c > 0$ is a small fixed enlargement parameter. Choosing c small proves (9.3), including $r = 1$ without letting the positive gap in the length restriction tend to zero.

9.2. **Fourth moment of unweighted character polynomials.** In this subsection, we state the fourth-moment estimate for products of unweighted character polynomials in (9.4) and explain the cancellation in its proof. Applied to the unweighted detecting polynomial of Section 5, this supplies the second bound in (10.4). Its dependence on κ is the estimate extended in Section 28 to obtain the final improvement.

Here "unweighted" means that the character sums contain no Möbius coefficients $\mu(\mathfrak{a})$, in contrast to the sum M_u in Section 9. The smooth test functions W_j are still present.

Lemma 9.3 (Unweighted product moment, [Ope26a, Lemma 18.1])**.** *Let ν_{fix} be a finite-order Hecke character presentation with the fixed ray and residue-symbol factors allowed in* [Ope26a, Lemma 18.1]*, independent of the summation variable k. Put $\psi_k(n) = \nu_{\mathrm{fix}}(n)\chi_n(k)$. Let $m_0 \ge 0$ specify the range $0 < q_k \ll Z^{m_0}$. Let $q_0 \ge 0$ be an exponent such that the product of the distinct prime ideals supporting the displayed variable residue-symbol factors has norm at most Z^{q_0}. These supporting primes are counted before any cancellation of factors or reduction of exponents modulo six. Put $M = m_0 + q_0$. Allow a common coprimality condition with modulus of norm $O(Z^B)$ for a fixed B, independent of k and common to every factor. Prime coefficients are fixed finite combinations of characters in Θ. For $\psi = \psi_k$, define*

$$S_\psi(n_j) = Z^{-n_j/2}\sum_a \psi(a)W_j(q_a/Z^{n_j}), \qquad Q_k = \prod_i Z^{-z_i/2}\sum_{p\in\mathcal{P}_i} a_i(p)\psi_k(p)W_i(q_p/Z^{z_i}),$$

where a runs over nonzero integral ideals, using primary generators and the stated zero extensions, and $n_1, n_2 \ge 0$ are length exponents, not ideals, and $z_{\mathrm{p}} = \sum_i z_i$. Every factor includes the common coprimality indicator just specified. All real length parameters range over prescribed bounded sets. Let $3/4 \le \kappa \le 1$; if $z_{\mathrm{p}} > 0$ and $\kappa < 1$, assume $\beta_* \le (1+\kappa)/2$. For every $\varepsilon > 0$ there exists $\ell_{\mathrm{mesh}} > 0$, depending only on these ranges and ε, such that $z_i \le \ell_{\mathrm{mesh}}$ for every i implies

$$\sum_{k\in\mathcal{R}_{z_{\mathrm{p}}}} |S_{\psi_k}(n_1)S_{\psi_k}(n_2)Q_k|^2 \ll Z^{M+\varepsilon}. \tag{9.4}$$

Here $\mathcal{R}_{z_{\mathrm{p}}}$ consists of the elements $0 < q_k \ll Z^{m_0}$ including all unit associates, whose primitive inducing characters satisfy the exclusions below:

	length condition	*excluded inducing characters*
$z_{\mathrm{p}} = 0$	n_1, n_2 *arbitrary bounded*	*principal*
$z_{\mathrm{p}} > 0$	$n_1 + n_2 + 6\kappa z_{\mathrm{p}} \le M$	Θ

The permitted ℓ_{mesh} is independent of the number of prime factors and uniform for $\kappa \in [3/4, 1]$; constants and finite seminorm orders may depend on the fixed number of factors.

In this application, the choice $\kappa = 2\beta_* - 1$ in (1.3) gives $\beta_* = (1+\kappa)/2$, which satisfies the lemma's hypothesis $\beta_* \le (1+\kappa)/2$.

Cancellation in the unweighted-moment argument. Applying the ordinary Hecke functional equation first reduces the length ranges. Two finite transforms then reduce the combined range exponent M. For transformed characters in Θ, the sums over the two ideal variables may have main terms proportional to the product of their summation lengths. For larger values of $n_1 + n_2 + z_{\mathrm{p}}$ the original paper subtracts two sums whose

length products agree, using fixed ideals $\mathbf{b} = (b_1, b_2)$, primary ideal variables l_1, l_2, and positive real length parameters X_i, Y_i:

$$D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^{2} V_i(q_{b_i} q_{l_i}/X_i) - \prod_{i=1}^{2} V_i(q_{b_i} q_{l_i}/Y_i), \qquad X_1 X_2 = Y_1 Y_2. \tag{9.5}$$

The two transforms preserve, on each side of the squared norm, one character, one coprimality condition and one norm twist on the full product of prime variables and unrestricted ideal variables [Ope26a, Lemma 18.2]. For an inducing character in Θ, fixing the prime variables leaves $\vartheta(l_1)\vartheta(l_2)1_{(l_1 l_2, \mathfrak{r}_*)=1} q_{l_1 l_2}^{i\tau_{\mathrm{tw}}} D_{\mathbf{b}}(l_1, l_2)$.

Here $\vartheta \in \Theta$, $\mathfrak{r}_*$ is a squarefree coprimality modulus with $q_{\mathfrak{r}_*} \le Z^{B_{\mathrm{cop}}}$ for fixed B_{cop}, and $\tau_{\mathrm{tw}} \in \mathbb{R}$ is the common norm-twist height. The V_i are the two test functions of ideal norms; the prime-interval weights remain W_i. By [Ope26a, Lemma 18.3], the two product main terms have a common coefficient, independent of X_i, Y_i, multiplying respectively $(X_1 X_2)^{1+i\tau_{\mathrm{tw}}}$ and $(Y_1 Y_2)^{1+i\tau_{\mathrm{tw}}}$. Hence

$$(X_1 X_2)^{1+i\tau_{\mathrm{tw}}} - (Y_1 Y_2)^{1+i\tau_{\mathrm{tw}}} = 0,$$

which is the decisive cancellation in (9.5). The cited lemma supplies the uniform Poisson remainder. Direct volume bounds suffice first in the uncentered range $n_1 + n_2 + z_{\mathrm{p}} \le 5M/6$; the cancellation handles the remaining ranges. Induction on the combined range exponent proves all cases with $z_{\mathrm{p}} = 0$ before those with $z_{\mathrm{p}} > 0$. Bounding the combined support errors uniformly by $O_K(1/\log Z)$ keeps the maximum-length restriction independent of K. This cancellation is what completes the inductive estimate without discarding those transformed characters.

10. Counting elements from simultaneous large values

In this section, we combine the simultaneous large values of Section 5 with the two moments (9.1) and (9.4). The two counting estimates are displayed in (10.4); optimizing the detecting lengths gives the count (10.7), with exponent $\max(R_{\mathrm{short}}(t), R_{\mathrm{long}}(t))$ for each zero and prime-sum size class. This exponent enters the contour estimate in Section 11. The fixed-parameter count in Part II and the variable-parameter count in Part IV repeat this combination with their respective moment hypotheses.

10.1. **Prime-sum size classes and admissible detecting lengths.** In this subsection, we partition the elements according to the sizes of the auxiliary prime sums, using the size parameter in (10.1), and determine the lengths permitted by (10.3). This partition records both the available savings from large prime sums and their contribution to the nonprincipal integrals, so that the counting bounds below and the contour exponent (11.1) use the same size classes.

For a fixed Mellin parameter z on those contours, put

$$Q_i(u; z) = P_i^{-1/2} \sum_{p \in \mathcal{P}_i} \overline{\chi_p(u)} W_i(q_p/P_i)(q_p/P_i)^{z-1}, \qquad \mathcal{Q}_i = -P_i^{z-1/2} Q_i.$$

The logarithmic-derivative estimate in the zero-free rectangle gives $|Q_i| \ll U^{\varepsilon_1} P_i^{\delta/2 + O(\epsilon_{\mathrm{b}})}$ [Ope26a, Lemma 19.1]. Choose a fixed small width for the exponent intervals $\theta_{\mathrm{bin}} > 0$. Partition by real exponents g_i so

$$|Q_i| \le P_i^{g_i + \theta_{\mathrm{bin}}}, \qquad g_i > 0 \Longrightarrow |Q_i| \ge P_i^{g_i}, \qquad 0 \le g_i \le \delta/2.$$

For an error factor set $g_i = 0$. Write

$$\bar{g} = \frac{\sum_i \ell_i g_i}{\ell}, \qquad 0 \le \bar{g} \le \delta/2, \qquad x = \bar{g}/\delta \in [0, 1/2]. \tag{10.1}$$

Ordering positive exponents decreasingly and selecting an initial set of prime sums of total length at most z_{p} gives

$$\left|\prod_{i \text{ selected}} Q_i\right|^2 \geq U^{2\bar{g}z_{\mathrm{p}}-\delta\max_i(\ell_i/d)}, \tag{10.2}$$

up to a fixed small length decrement. Allowing a fraction of the final prime interval would give gain at least $\bar{g}z_{\mathrm{p}}$; retaining only whole intervals loses at most the contribution of one interval, giving the error in (10.2).

After selecting a presentation, take $\psi(n) = \nu(n)\overline{\chi_n(u)}$. The prime coefficient is $\overline{\nu(p)}1_{p\in 1_{\mathcal{T}}}$, a fixed finite combination of Θ-characters. Conjugate the whole unweighted product when using (9.4). If u has a prime divisor outside S, the primitive character induced by ψ is ramified there and hence cannot belong to Θ, whose conductors are supported on S. The required moment hypotheses therefore hold; smooth parameter estimates handle the allowed element-dependent smooth weights with a fixed polynomial dependence on the height.

Measure lengths in base U: the variable u has norm $\asymp U$, so its range exponent is one. The largest total prime lengths allowed by the two moment inequalities are

$$z_M(r) = \frac{1-r}{2}, \qquad z_S(m) = \frac{1-2m}{6\kappa} = \frac{1-2m}{9/2+12\Delta}. \tag{10.3}$$

Using (5.5), (10.2) and the moments,

$$\#\mathcal{B} \ll \min\left\{U^{1-\delta r-2\bar{g}z_M+\varepsilon}, U^{1-2\delta m-2\bar{g}z_S+\varepsilon}\right\}(1+T_1)^A \tag{10.4}$$

where the respective lengths are admissible. The second bound uses two copies of S_m, hence $|S_m|^4$.

10.2. **Optimization of the two counting estimates.** In this subsection, we optimize the two bounds in (10.4) to obtain a uniform count for each zero and prime-sum size class. The resulting exponent (10.7) is substituted into the nonprincipal contour estimate in Section 11.

Proposition 10.1 (Refined character count, [Ope26a, Proposition 19.2])**.** *Assume the Stage-2 setting (1.3), so $\Delta = \beta_* - 7/8 \in (0, 1/24]$ and $\kappa = 3/4 + 2\Delta$. Let $\mathcal{B}$ consist of elements in one fixed zero class of Section 5, defined by (5.1), and one fixed prime-sum size class of Section 10.1, specified by the exponents g_i, with $q_u \asymp U = Z^d$, $a > a_0$, $0 < \delta \leq \alpha$, $0 \leq \bar{g} \leq \delta/2$ and $x = \bar{g}/\delta$. Assume the detector height hypotheses and that the available prime length ℓ/d exceeds $7/37$ by a fixed positive amount. Put*

$$D_x = 3 - 17x/9, \qquad P_x = (2 - 8x/9)(1-x).$$

For $t \in [1, 3/2]$ define

$$R_{\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t). \tag{10.5}$$

$$R_{\mathrm{long}}(t) = 1 - \delta + (\alpha - \delta)(t-1), \qquad \alpha = 5/6. \tag{10.6}$$

Then, for every $\varepsilon > 0$, the length decrements and maximum individual prime length can be chosen so that

$$\#\mathcal{B} \ll U^{\max(R_{\mathrm{short}}(t), R_{\mathrm{long}}(t))+\Delta/4+\varepsilon}(1+T_1)^A. \tag{10.7}$$

Here T_1 is the zero-detection height cutoff, and $A \geq 0$ is a fixed finite exponent measuring polynomial growth in that height. The constants and exponent A are uniform in u; their arithmetic dependence is on the fixed data. With no selected prime sums, $t = 1$ instead gives $\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon}(1+T_1)^A$ without any hypothesis on the available prime length.

The character-count exponent calculation. At $\delta = \alpha$, choose the detector parameter $t = 3/2$, so $r \ge 1 - O(\varepsilon)$. Sixth-power amplification then gives $\#\mathcal{B} \ll U^{1-\delta+\varepsilon}(1+T_1)^A$, which is at least as strong as (10.7); hence assume $\delta < \alpha$. For short inverse length $r < 1$ and $m < 1/2$, compare the unweighted estimate first with its $\Delta = 0$ prime length. Using $m \ge t - r - O(\varepsilon)$ gives

$$A_I(r) = 1 - \delta\big[x + (1-x)r\big], \qquad A_S(r) = 1 - \delta\big[4x/9 + (2 - 8x/9)(t-r)\big].$$

Let

$$r_*(t) = \frac{(2 - 8x/9)t - 5x/9}{D_x}.$$

Then $r_*(t) \ge 23/37$ and $1/3 \le t - r_*(t) \le 1/2$. The optimal common upper exponent is (10.5).

On the inverse side a fixed decrement $\nu_0 > 0$ ensures

$$1 - r - 2z_{\mathrm{p}} \ge 2\nu_0, \qquad 3 - 2r - 8z_{\mathrm{p}} = 4(1 - r - 2z_{\mathrm{p}}) + (2r - 1), \quad 2r - 1 \ge 9/37.$$

Its required prime length is at most $7/37$, whereas $\ell/h = 8/39 > 7/37$. On the unweighted side, restoring the actual length in (10.3) increases the counting exponent by at most

$$\begin{aligned} 2\bar{g}(1-2m)\left(\frac{2}{9} - \frac{1}{9/2 + 12\Delta}\right) &\le \frac{5}{6}\left(\frac{1}{3} + O(\varepsilon)\right)\frac{16}{27}\Delta \\ &\le \frac{40}{243}\Delta + O(\varepsilon) < \Delta/4 + O(\varepsilon). \end{aligned}$$

Here $\bar{g} \le \delta/2 \le 5/12$, $m \ge t - r_*(t) - O(\varepsilon) \ge 1/3 - O(\varepsilon)$, and $\frac{2}{9} - \frac{1}{9/2+12\Delta} \le \frac{16}{27}\Delta$. For $r \ge 1$, (9.3) instead gives (10.6).

If an admissible auxiliary prime length is zero or very small, use the moment estimates without auxiliary prime-polynomial factors. These comparisons give the count exponent in (10.7). With no selected prime sums and $t = 1$, balance $1 - \delta r$ and $1 - 2\delta(1-r)$ at $r = 2/3$ to obtain the stated unweighted count.

11. Contour estimates and the endpoint inequality

In this section, we insert the count (10.7) into the nonprincipal terms $u \ne 1$ of the Poisson integral (4.9), with the modified correction (8.2) and normalization (8.5). We compare their powers of Z with the principal Mellin integral $f_\eta(Z)$ in (8.8), whose exponent $C(s)$ is given in (8.7). The endpoint inequality (11.3) and the bounds for the remaining frequency ranges provide the positive exponent savings assembled in Section 12.

11.1. The exponent of a nonprincipal contribution.

In this subsection, we insert the character count into the contour estimates and compute the exponent of each nonprincipal contribution. Subtracting the principal-term exponent gives $E(d)$ in (11.1); a negative value supplies the saving required by (3.4).

Recall that $q_u \asymp U = Z^d$, $\delta = 2a - 1$ is the zero-class parameter of (5.1), and $\bar{g}$ is the weighted mean of the prime-sum size exponents in (10.1). The length parameters l_y, ℓ, h are those of (7.1). Move the original zero-class integral with the holomorphic correction (8.2) to $s_r = a + 16\epsilon_{\mathrm{b}}, w_r = 1 - a - 6\epsilon_{\mathrm{b}}, z_r = z_0 := 17/50$. Only afterward subdivide pointwise by prime-sum sizes. For a subset with cardinality $\ll U^{R+\varepsilon}(1+T_1)^A$, the factors, before small losses, are

$$\underbrace{X^{1/2-z_0}Z^{a+z_0-1}Y^{-a}}_{\text{Mellin powers}} \quad \underbrace{U^{R+\delta/2-z_0}}_{\text{count and numerator}} \quad \underbrace{Z^{\ell(z_0-1/2)+\bar{g}\ell}}_{\text{prime sums}}.$$

The numerator and all error factors are estimated jointly by (8.3). Subtracting $C(7/8) = 3/16$ gives

(11.1)
$$\begin{aligned} E(d) &= a - \frac{7}{8} + h(z_0 - 1/6) - al_y - \ell/2 + \bar{g}\ell + d(R + \delta/2 - z_0) \\ &= -\frac{1}{48} + \frac{2}{3}\delta + \frac{\bar{g}}{6} - h(1-R) + (d-h)(R + \delta/2 - z_0) \qquad \text{[Ope26a, Lemma 20.1].} \end{aligned}$$

The integral is bounded by the supremum of this exponent over subsets defined by the sizes of the prime sums at each fixed Mellin parameter. The contour shift is made before this subdivision, so no holomorphic continuation of a subset is required.

11.2. **Choice of the detecting cutoff and proof of the endpoint saving.** In this subsection, we choose the zero-detector truncation exponent as in (11.2) and prove the endpoint inequality (11.3). The explicit lower bound (11.5) supplies a positive saving. The endpoint $d = h$ controls the central frequency ranges by the monotonicity of (11.1); the next subsection handles the smaller and larger ranges needed for the full comparison estimate.

Lemma 11.1 (Endpoint saving, [Ope26a, Lemma 20.2])**.** *Let $1/50 < \delta < \alpha = 5/6$ and $0 \le x \le 1/2$, with D_x, P_x as above and the fixed parameter choices (7.1). Choose*

$$\mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \qquad t = 1 + \frac{\delta P_x}{2\mathcal{J}}, \qquad R_* = R_{\text{short}}(t) = R_{\text{long}}(t). \tag{11.2}$$

Let $E_* = E(h)$ in (11.1) with $\bar{g} = x\delta$ and $R = R_*$. Then $-E_* \ge 49/440640$. If the count loss $\Delta/4$ is included and all remaining exponent losses have sum $\varepsilon_{\text{total}}$, then

$$E_{\text{actual}}(h) - \Delta \le -\frac{49}{440640} - \frac{51}{64}\Delta + \varepsilon_{\text{total}}. \tag{11.3}$$

Here E_{actual} denotes the resulting exponent relative to $C(7/8)$.

The endpoint exponent calculation. Here $1 < t < 3/2$ and $35/54 \le \mathcal{J} \le 5/2$. Put

$$y = 1/2 - x \in [0, 1/2], \quad v_y = 51 + 41y, \quad p_y = 7 + 18y + 8y^2,$$
$$j_y = 185 + 170y + (-138 + 12y + 96y^2)\delta.$$

Then $P_x = p_y/9$ and $\mathcal{J} = j_y/108$. The original paper's polynomial identity is

$$\begin{aligned} 10368 v_y \mathcal{J}(-E_*) &= (3+5y)\big[(4v_y\delta - 79)^2 + 49\big] \\ &\quad + 4y\Big[4v_y\delta((1+3y)(15+32y)\delta + 9 - 13y) + 265 + 3485y\Big]. \end{aligned} \tag{11.4}$$

Since $9 - 13y \ge 5/2$ and $v_y \le 17(3+5y)$,

$$-E_* \ge \frac{49}{176256\mathcal{J}} \ge \frac{49}{440640} > \frac{1}{10000}. \tag{11.5}$$

Restoring the $\Delta/4$ in (10.7), the coefficient of R in (11.1) is h. Since $C(\beta_*) - C(7/8) = \Delta$, this gives (11.3). Here E_{actual} includes the $\Delta/4$ count loss, and $\varepsilon_{\text{total}}$ denotes the sum of the small contour, length, size-partition and smoothing losses. The subtraction of Δ is essential.

At $\delta = \alpha$, use $t = 3/2$ and unweighted amplification: $R = 1 - \delta$, whence $E(h) \le -1/48 - \delta/16$. At $a = a_0$ use the direct count $R = 1$, giving $E(h) \le -7/1200$. No selected zero or moment estimate with a length restriction approaching equality is needed in these respective boundary cases.

11.3. **Estimates over all nonprincipal norm ranges.** In this subsection, we combine the endpoint saving with the bounds for small and large norms to control all nonprincipal frequency ranges. This completes the estimates that Section 12 combines into a single saving uniform over all classes.

Let $d_{\min} = 1/100$ and $0 < \zeta < 1/48$. The estimates covering $U = Z^d$ are:

(1) $d < d_{\min}$: absolute tuple bounds on $(s_r, w_r, z_r) = (\beta_* + \epsilon_{\mathrm{b}}, 1/2, 17/50)$ give saving $79/800 - 2d_{\min} = 63/800$ relative to $C(\beta_*)$. Here $2d_{\min}$ bounds the contribution $63d/50$ to the exponent for $d < d_{\min}$.

(2) $d_{\min} \le d \le 1/2$: using $R = 76/75 - 2\delta/3$ (also valid at $a = a_0$),

$$E(d) \le E(1/2) \le -529/2400 + (25/96)\delta \le -49/14400.$$

The slope in d is $R + \delta/2 - z_0 = 101/150 - \delta/6 > 0$, since $\delta \le 5/6$; this justifies $E(d) \le E(1/2)$.

(3) $1/2 \le d \le h$: the total available prime length is $\ell/d \ge 8/39 > 7/37$, and $R + \delta/2 - z_0 \ge 33/50 - \delta/2 \ge 4/25 > 0$. Thus (11.3) controls these ranges.

(4) $h < d \le h + \zeta$: the total available prime length remains $> 1/5 > 7/37$. The derivative of the exponent with respect to d is below two, so its increase is at most 2ζ.

(5) $d > h + \zeta$: the contours $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (2, 2, z_{\mathrm{new}})$, where the series converge absolutely, give

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{53/32 + (h+\zeta)(1+\varepsilon) - \zeta z_{\mathrm{new}} + \varepsilon}.$$

Here U runs over the dyadic parameters indexing the ranges $q_u \asymp U$, and $\mathcal{R}_\eta(U; Z)$ denotes the modified nonprincipal contribution from $q_u \asymp U$ in (4.9), before normalization, on these contours of absolute convergence [Ope26a, Lemma 16.2]. Choose the fixed $z_{\mathrm{new}} > 2$ sufficiently large.

The estimates for the small- and large-norm ranges are [Ope26a, Lemmas 10.6, 16.2]. Only the ranges $d \ge 1/2$ use selected prime sums in the character-count moments.

12. Auxiliary parameter choices and the uniform comparison estimate

In this section, we choose the auxiliary parameters in an order that preserves all strict inequalities and exponent savings obtained in Sections 8–11. The resulting direct and comparison estimates are displayed together in (12.3) and have precisely the uniformity required by Proposition 3.1, whose application completes the Hecke argument in the next section.

Recall that $\Delta = \beta_* - 7/8 > 0$ is the assumed gap in the contradiction argument, and $C(s) = s - 11/16$ is the principal exponent in (8.7). The normalized average J_η and principal integral f_η are given in (8.5) and (8.8).

The lowest zero class $a = a_0 = 51/100$ of (5.1) and the intermediate norm range $d_{\min} \le d \le 1/2$ considered in Section 11 satisfy $E \le -7/1200$ and $E \le -49/14400$, respectively, relative to $C(7/8)$. Relative to $C(\beta_*) = C(7/8) + \Delta$, they therefore save at least Δ. Since $51\Delta/64 < \Delta$, both are already covered by m_{high} below. The common exponent savings are

$$m_{\mathrm{high}} = \min\{51\Delta/64, 63/800\}, \quad m_w = 23/960, \quad m_z = 13/9600.$$

Choose small moment errors and strict length decrements. Recall that ℓ_{mesh} bounds each individual prime-length exponent in the moment estimate (9.4). Obtain the permissible value $\ell_{\mathrm{mesh}} > 0$ from [Ope26a, Lemma 18.1] before choosing K. Then choose a fixed even K with equal $\ell_i = \ell/K$, where $2\ell/K$ is smaller than ℓ_{mesh} and than the allowed exponent

loss from discarding a partial prime interval. Smooth weights in disjoint subintervals of $(1,2)$ give disjoint prime intervals with the common norm parameter $Z^{\ell/K}$. Thus $\ell_i/d \le 2\ell/K$ throughout the selected ranges. Choose $0 < \kappa_P < (7/8)\ell/K$, then the width of the prime-sum size intervals, ϵ_{b}, ζ, z_{new} and P_0 so that the resulting losses are smaller than the remaining positive exponent savings. All these parameters are chosen before fixing η.

One may retain

$$\mu_0 = \frac{1}{4}\min\{m_{\mathrm{high}}, m_w, m_z, \kappa_P\} > 0, \qquad \omega_0 = \Delta/2. \tag{12.1}$$

After fixing η and its arithmetic data, choose all internal seminorm orders. [Ope26a, Lemmas 10.2–10.4] and the preceding estimates give

$$|J_\eta - f_\eta| \ll_{\eta,N} Z^{C(\beta_*)-\mu_0}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}. \tag{12.2}$$

A_η, B_η are finite nonnegative exponents fixed before the external decay order $N \in \mathbb{Z}_{\ge 0}$. The later N_η is a sufficiently large choice of this order. There is a fixed $\tau_{0,\eta} > 0$ ensuring all zero-detection height conditions for $T_1 = Z^\tau$, $0 < \tau \le \min(d_{\min}/100, \tau_{0,\eta})$.

Choose

$$0 < \tau_\eta \le \min\left\{ d_{\min}/100, \tau_{0,\eta}, \frac{\mu_0}{4(A_\eta+1)} \right\}, \qquad B_\eta - N_\eta\tau_\eta < C(\beta_*) - \mu_0/2.$$

The first term in (12.2) saves at least $3\mu_0/4$, and the second saves $\mu_0/2$. Increasing the derivative orders used to obtain decay in the truncated contour tails changes only the seminorm constants and the lower threshold for Z. This is the height choice of [Ope26a, Lemma 11.1]; the full order is [Ope26a, Proposition 20.3]. Together with (8.6),

$$|J_\eta(Z)| \ll_\eta Z^{3/16+\Delta/2}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\mu_0/2}. \tag{12.3}$$

The exponents are independent of η. Neither J_η nor f_η contains T_1.

13. Mellin continuation and proof of the Hecke bound

In this section, we apply Proposition 3.1 to the estimates assembled in (12.3) of Section 12 and conclude the 7/8 bound for the Hecke family. We also extend the conclusion to characters presented modulo a larger modulus than their conductor, retaining their prescribed zero extensions; the Dirichlet consequence follows in Section 14.

Theorem 13.1 (Hecke conclusion, [Ope26a, Theorem 1.1])**.** *The family supremum in (1.2) satisfies $\beta_* \le 7/8$. Consequently every finite-order Hecke L-function over F has no zeros in $\operatorname{Re} s > 7/8$, with principal poles allowed.*

The final continuation contradiction. Assume $\beta_* > 7/8$. Apply Proposition 3.1 with

$$\sigma_0 = 7/8, \qquad c = -11/16, \qquad \omega_0 = \Delta/2, \qquad \sigma = \mu_0/2.$$

Holomorphy of H_η follows from (4.12) and (4.13); the chosen cutoff gives (4.14). Together with (12.3), these verify the criterion's hypotheses; $C(7/8) = 3/16$ and $0 < \omega_0 < \beta_* - 7/8$. Its contradiction therefore gives

$$\boxed{\beta_* \le 7/8}. \tag{13.1}$$

For an imprimitive character η induced by η^*,

$$L_F(s,\eta) = L_F(s,\eta^*) \prod_{\substack{\mathfrak{p}\mid \mathfrak{f}_\eta \\ \mathfrak{p}\nmid \mathfrak{f}_{\eta^*}}} (1 - \eta^*(\mathfrak{p})q_{\mathfrak{p}}^{-s}).$$

Here $\mathfrak{f}_\eta$ is the defining modulus and $\mathfrak{f}_{\eta^*}$ the primitive conductor. Each finite factor is nonzero for $\operatorname{Re} s > 0$, giving the same conclusion for every presentation.

14. Transfer to Dirichlet L-functions

In this section, we transfer the Hecke bound to Dirichlet L-functions by pulling a Dirichlet character back along the ideal norm and factoring its L-function over F. The factorization (14.1) gives the transfer stated in Corollary 14.1. The same argument will transfer each improved Hecke bound in Parts II–IV, so it suffices to give the factorization and pole check once.

Corollary 14.1 (Dirichlet conclusion, [Ope26a, Proposition 11.3])**.** *Every Dirichlet L-function, including ζ, has no zeros in $\operatorname{Re} s > 7/8$, with principal poles at $s = 1$ allowed.*

The Dirichlet factorization. The identity below is the quadratic case of Artin formalism (compatibility with induction and direct sums); see [Mil20, Chapter VIII, Section 10]. For a primitive Dirichlet character χ of conductor q, define $\eta(\mathfrak{a}) = \chi(N\mathfrak{a})$ on ideals coprime to $3q$, and let χ_{-3} be the primitive quadratic Dirichlet character associated with $F/\mathbb{Q}$, given by the Kronecker symbol $\chi_{-3}(n) = (-3/n)$. Deleting primes dividing $3q$ and their prime ideals,

$$L_F^{S_F}(s,\eta) = L^{S_{\mathbb{Q}}}(s,\chi)L^{S_{\mathbb{Q}}}(s,\chi\chi_{-3}). \tag{14.1}$$

The split and inert Euler factors give (14.1) in absolute convergence; meromorphic continuation extends the identity [Ope26a, Proposition 11.3]. Both Dirichlet factors are holomorphic on $7/8 < \operatorname{Re} s \le 1$, $s \ne 1$. A zero of either is therefore a zero of their product, contradicting the Hecke conclusion. Euler convergence handles $\operatorname{Re} s > 1$. At $s = 1$ a possible principal pole pairs only with χ_{-3}, whose value is

$$L(1,\chi_{-3}) = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

Finally, finite Euler deletions transfer the result to all imprimitive Dirichlet presentations. The character modulo one gives ζ.

Part II. A rational extension of the bound

15. The rational zero-free bound and its parameters

In this section, we state the rational improvement and choose its length parameters, as stated in Theorem 15.1 and (15.1). These choices will be verified in Sections 16–20. The estimates established in this part also supply the common analytic construction for Parts III and IV. We vary the lengths in the modified average of Section 7; all arithmetic conventions, zero extensions, smooth tests and character families remain those of Part I. Recall that $F = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathcal{O}_F$, and $q_{\mathfrak{a}} = N\mathfrak{a}$. The finite-order target character is η, and β_* is the supremum of real parts of family zeros defined in (2.3). Throughout Parts II–IV we use the proved intermediate results of [Ope26a]. In particular, we apply Theorem 13.1 before making the improved comparison.

Theorem 15.1 (Rational extension)**.** *Every finite-order Hecke L-function over F is zero-free in*

$$\operatorname{Re} s > B_{\mathrm{r}}, \qquad B_{\mathrm{r}} = 34999/40000.$$

The same assertion holds for every Dirichlet L-function. Principal poles are allowed, and the boundary line is excluded.

For clarity, the parameters that change are displayed together:

$$\begin{gathered}\vartheta_{\mathrm{r}}=\frac{1}{10000},\quad b_{\mathrm{r}}=\frac{1}{8},\quad \ell_{\mathrm{r}}=\frac{1}{6}+\vartheta_{\mathrm{r}}=\frac{5003}{30000},\\ b=b_{\mathrm{r}},\qquad \ell=\ell_{\mathrm{r}},\\ l_x=\frac{1-b-\ell}{2}=\frac{21247}{60000},\quad l_y=\frac{1+b-\ell}{2}=\frac{28747}{60000},\quad h=1-l_x+\ell=\frac{16253}{20000},\\ X=Z^{l_x},\quad Y=Z^{l_y},\quad P_i=Z^{\ell_i},\quad \sum_{i=1}^{K}\ell_i=\ell.\end{gathered}\tag{15.1}$$

The pair $(b_{\mathrm{r}},\ell_{\mathrm{r}})$ specifies the rational extension. We use $b=b_{\mathrm{r}}$ and $\ell=\ell_{\mathrm{r}}$ for these parameter values, and retain b,ℓ for the general parameter formulas shared with Part III. Here $b=l_y-l_x$, while $\ell=\sum_i\ell_i$ is the total selected prime-length exponent, and $h=1-l_x+\ell$ is the endpoint of the central norm ranges. The integer K is fixed and even. Its disjoint prime sets $\mathcal{P}_i$ lie in the class $1_{\mathcal{T}}$ and outside S, as in Section 7; $1_{\mathcal{T}}$ is the identity class of the fixed ray group and S is the excluded prime set of Section 2. The zero-class width ϵ_{b} of (5.1) is independent of ϑ_{r}. We retain

$$a_0=51/100,\quad \delta=2a-1,\quad x=\bar{g}/\delta\in[0,1/2],\quad z_0=17/50,\quad \alpha=5/6.$$

Here a is the real-part parameter of the zero class in (5.1), and $\bar{g}$ is the weighted mean defined in (10.1). Thus x denotes an amplitude ratio, whereas l_x denotes a length exponent. In all applications $\delta\geq 1/50$, so the ratio is defined. Its extension to $\delta=0$ is used only in polynomial inequalities.

The balanced condition $l_x+l_y+\ell=1$ gives

$$C_b(s)=s-\frac{4+b}{6},\quad B(\ell)=\frac{11}{12}-\frac{\ell}{4},\quad C_b(B(\ell))=\frac{l_x}{2}+\frac{b}{12}.\tag{15.2}$$

The shared balanced hypotheses for the extensions are

$$\begin{gathered}l_x+l_y+\ell=1,\qquad b=l_y-l_x>0,\qquad h=1-l_x+\ell,\\ l_x>\ell>0,\qquad l_y>\ell,\qquad l_y-\ell-11b/6>0,\qquad 1-3\ell>0,\qquad \ell<1/5,\\ \ell/h\geq 7/37+\rho_{\sup},\qquad \rho_{\sup}>0,\qquad 5\ell>h,\\ B=B(\ell)=11/12-\ell/4\in[437/500,7/8].\end{gathered}\tag{15.3}$$

The positive gap $\rho_{\sup}$ above $7/37$ in the prime-length ratio ℓ/h is fixed independently of the target. The conditions on rescaled lengths and the additive estimate are separated from the numerical choice of the length parameters. For (15.1), $C_b(s)=s-11/16$ and $B(\ell)=B_{\mathrm{r}}$. The hypotheses are verified by

$$\begin{gathered}l_x-\ell=\frac{3}{16}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\quad l_y-\ell=\frac{5}{16}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\\ l_y-\ell-\frac{11b}{6}=\frac{1}{12}-\frac{3}{2}\vartheta_{\mathrm{r}}>0,\quad 1-3\ell>0,\quad \ell<\frac{1}{5},\\ \frac{\ell}{h}>\frac{8}{39}>\frac{7}{37},\qquad 5\ell-h=\frac{1}{48}+\frac{7}{2}\vartheta_{\mathrm{r}}>0.\end{gathered}\tag{15.4}$$

They ensure that the same finite modification defines an average with the new length parameters, that all rescaled lengths remain positive, and that there are enough prime factors for the moments with selected prime factors.

16. The direct estimate for balanced lengths

In this section, we derive the direct estimate at variable balanced lengths from the general estimate obtained from the functional equation [Ope26a, Lemma 14.3] and the additive estimate (7.5). The resulting bound is (16.2) in Proposition 16.2; it supplies the direct hypothesis of Proposition 20.1; the specialized bound (7.4) alone does not justify changing ℓ.

If $\mathcal{I} \subset \{1, \dots, K\}$ is the subset of rescaled factors, put

$$d_{\mathcal{I}} = \sum_{i \in \mathcal{I}} \ell_i, \quad \ell' = \ell - d_{\mathcal{I}}, \quad M' = 1 - \ell - 2d_{\mathcal{I}}, \quad X' = Z^{l_x - d_{\mathcal{I}}}, \quad Y' = Z^{l_y - d_{\mathcal{I}}}.$$

The symbol $d_{\mathcal{I}}$ is unrelated to the norm exponent d used later. For a fixed ray class σ and rescaled prime tuple, let $B^{\mathcal{I}}_{a,\sigma}$ denote the theta sum $B^{J}_{m,\sigma}$ of (7.4), with $J = \mathcal{I}$, the outer summation variable renamed a, and the summation lengths replaced by X', Y' above. The primes in the sets $\mathcal{P}_i$ with $i \notin \mathcal{I}$ are already summed, with the divisibility restrictions of Section 7 imposed. Put $Q = q_{b_*} X'Y'$.

Lemma 16.1 (Functional-equation estimate for balanced lengths)**.** *Under the coefficient and smooth-weight hypotheses of* [Ope26a, Lemma 14.3]*, the theta sum including all cube multiples, with remaining prime-length exponent ℓ' and outer element-range exponent M' has squared norm*

$$\sum_{q_a \ll Q} |B^{\mathcal{I}}_{a,\sigma}|^2 \ll Z^{M' + (d_{\mathcal{I}} + 5\ell - 1)_+/4 + \varepsilon}. \tag{16.1}$$

Here $(t)_+ = \max(t, 0)$ for a real argument t; the loss $\varepsilon > 0$ is arbitrary.

Proof. We record the exponent calculation, retaining the normalization of (7.4). Write O and H for the exponents for which the norms of the powerful part and remaining squarefree part of the outer element are respectively $\asymp Z^O$ and $\asymp Z^H$, and put $\Delta_H = M' - O - H$; the powerful part is defined after (4.3). The local auxiliary length exponents in [Ope26a, Lemma 14.3] satisfy, up to losses that can be made arbitrarily small,

$$O, \Delta_H, S_0, B_0 \geq 0, \quad 2A_0 \leq O, \quad 0 \leq N_0 \leq A_0, \quad z_a \leq \ell',$$
$$T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - N_0 - 3B_0, \qquad v + 3\ell_b + e_\lambda \leq T_d.$$

Here A_0, N_0, B_0, S_0 describe the local divisor decompositions, z_a is their retained prime-length exponent before applying the functional equation, v, ℓ_b, e_λ are the retained dual length exponents, and T_d is the available dual length exponent. These variables occur only in this functional-equation calculation. Since $M' + \ell' - 1 = -3d_{\mathcal{I}}$,

$$T_d \leq H - 3d_{\mathcal{I}} \leq H.$$

The exponent in the squared-norm estimate supplied by the cited lemma is

$$E_{\mathrm{ref}} = O/2 + H - S_0 - B_0 + z_a - u_{\mathrm{hyb}} - \ell_b - 2e_\lambda/3 \\ - (T_d - v - 3\ell_b - e_\lambda)_+/2, \qquad u_{\mathrm{hyb}} = \min\{v, z_a, (v + z_a)/3\}.$$

If $u_{\mathrm{hyb}} = z_a$, then $E_{\mathrm{ref}} \leq M'$. Otherwise $u_{\mathrm{hyb}} \geq v/2$, and

$$u_{\mathrm{hyb}} + \ell_b + 2e_\lambda/3 + (T_d - v - 3\ell_b - e_\lambda)_+/2 \geq T_d/4.$$

Substituting T_d, using $N_0 \leq A_0$ and $z_a \leq \ell'$, gives

$$E_{\mathrm{ref}} \leq M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H}{4} \leq M' + \frac{d_{\mathcal{I}} + 5\ell - 1}{4}.$$

The maximum of the two bounds proves (16.1). Fixed offsets arising from the support intervals are $O(1/\log Z)$; the Gaussian partition, the losses from making the prime-length

restrictions strict and from logarithmically many dyadic intervals are absorbed in ε. Separation is performed on the joint weight before positivity or squaring, so the separated coefficients remain independent of the outer element. □

Proposition 16.2 (Balanced direct bound)**.** *Assume* $l_x + l_y + \ell = 1$, $b = l_y - l_x > 0$, *and*

$$l_x > \ell > 0, \quad l_y > \ell, \quad l_y - \ell - 11b/6 > 0, \quad 1 - 3\ell > 0, \quad \ell < 1/5.$$

Then

$$|I_{\eta,\mathrm{mod}}(Z)| \ll_{\eta,\varepsilon} Z^{l_x/2+b/12+\varepsilon}. \tag{16.2}$$

Proof. Apply the separation of (4.6) to each term of (7.2). In (7.5) the parameter in the additive estimate is $P_a = q_{b_*}^{-1} Z^b$, and the inequality $l_y - d_{\mathcal{I}} - 11b/6 > 0$ implies $P_a^2/Y' \ll P_a^{1/6}$. The Cauchy–Schwarz inequality (4.7), (7.5) and Lemma 16.1 bound the contribution of each fixed tuple, before multiplication by its coefficient in (7.2), by

$$(X')^{1/2} P_a^{1/12} Z^{(d_{\mathcal{I}}+5\ell-1)_+/8+\varepsilon}.$$

The number of rescaled tuples is $\ll Z^{d_{\mathcal{I}}+\varepsilon}$. Their coefficient is $O(Z^{-3d_{\mathcal{I}}/2})$, and replacing X' by X contributes $Z^{-d_{\mathcal{I}}/2}$. Thus their net additional exponent is

$$-d_{\mathcal{I}} + (d_{\mathcal{I}} + 5\ell - 1)_+ /8 \le -7d_{\mathcal{I}}/8 \le 0.$$

Sum over the finitely many subsets. This proves the claim without invoking the specialized direct bound with altered length parameters. □

17. Euler products and extraction of the principal term

In this section, we extend the holomorphic Euler correction (8.2) and extract the principal Mellin term (17.6) at the changed lengths of Section 16. The domain is specified in (17.1), and (17.2) ensures that the principal correction does not vanish. Proposition 17.2 bounds the difference between the principal contribution and its Mellin integral in (17.7). These estimates are used in the continuation arguments of Parts II–IV.

Let

$$\frac{437}{500} \le B \le \frac{7}{8}, \qquad \mathcal{D}_2(B) = \{\mathrm{Re}\, s \ge B,\ \mathrm{Re}\, w \ge 19/20,\ \mathrm{Re}\, z \ge 33/200\}. \tag{17.1}$$

The first domain (4.11) remains available, with $\mathrm{Re}(s+w)$ bounded above 1 by a fixed positive amount. All Euler products use the same set S as the modified average. A fixed prime cutoff is chosen first; the target's conductor primes may then be added to S.

Lemma 17.1 (Extended principal domain)**.** *The unselected and selected local expressions of Part I are holomorphic on* $\mathcal{D}_2(B)$. *Their full correction* (8.2) *is defined there by the same local holomorphic expressions. For sufficiently large fixed cutoff* P_0, *the principal correction satisfies*

$$H_\eta(s) = 1 + O(P_0^{-4/5}), \qquad \sup_{\mathrm{Re}\, s \ge B} |H_\eta(s) - 1| \le \tfrac{1}{2}. \tag{17.2}$$

At the principal residue,

$$G_p(s, 1, 1/6)/H_p(s, 1, 1/6) = -1 + O(q_p^{-B}). \tag{17.3}$$

The constants are uniform in the imaginary parts of s, w, z *and in the arithmetic factors of absolute value one appearing below.*

Proof. To check the changed boundary, we give the exact local formulas from [Ope26a, Lemma 7.1]. At a good prime put $j = v_p(u) \in \{0, \dots, 5\}$, $\rho_p = \chi_p(u/p^j)$, and

$$a_p = \overline{\varpi(p)}^3 \eta(p)^3, \quad b_p = \overline{\varpi(p)}^2 \eta(p)^2 \overline{\chi_p(4)}, \quad b_p^3 = a_p^2.$$

Here $\varpi(p) = p/|p|$ is defined in Section 2. The quantities a_p, b_p have absolute value one and depend on p; they are unrelated to the zero-class parameter a and the length difference b. Retain V_p, W_p, D_p from (4.10) and put

$$R = a_p^2 q_p^{4-6s-6z}, \qquad W_{\mathrm{loc}} = \rho_p q_p^{-w}.$$

At $p \mid u$, $W_p = D_p = 0$ by zero extension. With $\mathbf{1}$ denoting an indicator, the complete local formulas are

$$P_p^* = \frac{1}{1-R}\left[\frac{R(1-q_p^{-1}) - \eta(p)(q_p-1)q_p^{-s-w}V_p^{\mathbf{1}_{j\le 1}}}{1-V_p} + J_j\right],$$

$$P_p = \frac{1}{1-V_p} + \mathbf{1}_{j=0}\frac{W_{\mathrm{loc}}}{1-W_{\mathrm{loc}}} + P_p^*, \qquad H_p = P_p\frac{(1-V_p)(1-W_p)}{1-D_p},$$

where

j	J_j
0	$-\eta(p)\rho_p^{-1}q_p^{-s} + W_{\mathrm{loc}}R$
1	$\eta(p)q_p^{-s-w}$
2	$a_p\rho_p^{-3}q_p^{3/2-3s}$
3	$-\eta(p)b_p\rho_p^{-2}q_p^{2-3s-w} + b_p^2\rho_p^{-4}q_p^{2-4s}$
4	$-\eta(p)a_p\rho_p^{-3}q_p^{5/2-4s-w}$
5	$-a_p^2q_p^{3-6s}$.

Writing $\mathcal{F}_p = P_p^* + D_p$, cancellation gives

$$H_p - 1 = \frac{D_p(V_p + W_p - V_pW_p) - V_pW_p + (1-V_p)(1-W_p)\mathcal{F}_p}{1-D_p}. \tag{17.4}$$

For $p \nmid u$, the possible exponents are

$$-s_r - w_r, \quad -s_r - 6z_r, \quad -w_r - 6z_r, \quad 4 - 6s_r - 6z_r, \quad 1 - s_r - w_r - 6z_r,$$

where $s_r = \operatorname{Re} s$, and similarly for w_r, z_r. Their maximum on (17.1) is $-B - 47/50 < -9/5$. For $p \mid u$, the table instead gives $H_p - 1 = O(q_p^{1/20-B})$. The denominators $1-R, 1-V_p, 1-D_p$ are uniformly separated from zero; (17.4) has no denominator $1-W_p$. The Euler product over $p \nmid u$ converges locally uniformly, while the finite product over $p \mid u$ is $O_\varepsilon(q_u^\varepsilon)$. For $u = 1$, the tail saves $B - 3/50 > 4/5$, proving (17.2).

The selected factor is defined without dividing by H_p:

$$G_p = \frac{(\overline{\eta(p)}q_p^s - q_p^{-w})(1-V_p)(1-W_p)P_p^* - q_p^{-w}(1-V_pW_p)}{1-D_p}. \tag{17.5}$$

It agrees with (8.1) on contours where the series converge absolutely, and hence continues the finite modification by (8.2). For $u = 1$, the unselected factors H_p are uniformly close to one on (17.1), so G_p/H_p is well defined on the contours used to extract the principal residue. The four error exponents are

$$-s_r, \quad -6z_r, \quad 4 - 5s_r - 6z_r, \quad 1 - w_r - 6z_r,$$

all at most $-B$ for the range (17.1).

This proves (17.3). No nonvanishing of a nonprincipal local correction is needed. □

Recall that c_S is the positive residue constant in (4.13), and define $A_{\mathcal{T}}(Z)$ by (8.5), with the present ℓ_i and the same Mellin transform $\mathcal{M}$ of $\widetilde{W_0}$ defined in Section 4.4. The prime asymptotic (8.4) gives, for even K,

$$A_{\mathcal{T}}(Z) > 0 \quad (Z \text{ sufficiently large}), \qquad |A_{\mathcal{T}}(Z)| \asymp (\log Z)^{-K}.$$

The poles remain $w = 1$ and $z = 1/6$; the latter is independent of ℓ. For arbitrary positive l_x, l_y, ℓ, put

$$h = 1 - l_x + \ell, \qquad C(s) = s + l_x/2 - 1 + h/6,$$

$$J_\eta(Z) = \frac{I_{\eta,\mathrm{mod}}(Z)}{c_S A_{\mathcal{T}}(Z)}, \qquad f_\eta(Z) = \frac{1}{2\pi i}\int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds. \tag{17.6}$$

These are the same constructions as (8.8), with the changed length parameters. Neither function depends on a later height truncation.

Proposition 17.2 (Principal extraction)**.** *Assume (15.3) and $\beta_* \ge B$. The individual prime-length exponents satisfy $\ell_i > 0$ and $\sum_{i=1}^K \ell_i = \ell$, as in (15.1). For every $0 < \kappa_P < B \min_i \ell_i$, the principal term $\mathcal{P}_\eta$ satisfies*

$$\left| \frac{\mathcal{P}_\eta(Z)}{c_S A_{\mathcal{T}}(Z)} - f_\eta(Z) \right| \ll Z^{C(\beta_*)+(1+h)\epsilon_{\mathrm{b}} - l_y/20+\varepsilon} + Z^{C(\beta_*)+\epsilon_{\mathrm{b}}-h/600+\varepsilon} + Z^{C(\beta_*)+\epsilon_{\mathrm{b}}-\kappa_P+\varepsilon}. \tag{17.7}$$

Proof. The identification of the principal frequency in Section 4.4 and [Ope26a, Lemmas 10.2–10.5] is arithmetic and unchanged: the only principal numerator occurs at $u = 1$. For this frequency the scalar quotient is $\zeta_F^S(6z)\zeta_F^S(w)/L_F^S(s,\eta)$. Start with $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$, $\operatorname{Re} w = 1 + \epsilon_{\mathrm{b}}$, $\operatorname{Re} z = 1/6 + \epsilon_{\mathrm{b}}$. Move w to $19/20$, then in its residue move z to $33/200$. Lemma 17.1 justifies both moves. The raw exponent is

$$l_x/2 + \operatorname{Re} s - 1 + h \operatorname{Re} z + l_y(\operatorname{Re} w - 1).$$

The two unresidued integrals therefore save $l_y/20$ and $h/600$. At the double residue, (17.3) and the nonnegative prime tests give

$$\mathfrak{H}_{\eta,1,Z}(s,1,1/6) = H_\eta(s) Z^{\ell/6} A_{\mathcal{T}}(Z)(1 + O(Z^{-\kappa_P})).$$

The scalar residue is exactly c_S, including the factor $1/6$ from $\zeta_F^S(6z)$. Bound the error on the original contour $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$; move only the main Gaussian integral right to $\operatorname{Re} s = 2$. The reciprocal has no poles to the right of β_*, and principal L-poles become reciprocal zeros. Gaussian decay removes horizontal sides. This proves (17.7) without moving an approximation error through possible target zeros. □

18. Character counts and nonprincipal exponents at the fixed moment parameter

In this section, we use the already proved $7/8$ bound to keep $\kappa = 3/4$ fixed while varying the averaging lengths. Proposition 18.1 gives the count (18.3), and Proposition 18.2 gives the nonprincipal exponent (18.4). These are used by the exact rational inequalities in Section 19 and the algebraic optimization in Part III.

Recall that $\delta = 2a - 1$, $x = \bar{g}/\delta$, and $q_u \asymp U = Z^d$ specify the zero, prime-sum size and norm classes; see (5.1) and (10.1). Theorem 13.1 permits the fixed choice

$$\kappa = 3/4, \qquad \beta_* \le (1+\kappa)/2. \tag{18.1}$$

It is important to fix this value rather than substitute $\kappa = 2\beta_* - 1$ below the allowed moment range. All moments below have exactly the common-character, coefficient and smooth-weight hypotheses of Section 9.

Proposition 18.1 (Fixed-parameter count)**.** *Let $U = Z^d$, and let $\mathcal{B}$ be a set of the sixth-power-free summation indices u in (4.9), with $q_u \asymp U$, in one fixed zero class of (5.1) and one fixed prime-sum size class of Section 10.1. The latter is also called an amplitude class below. Suppose $0 < \delta \le \alpha$, $0 \le \bar{g} \le \delta/2$, and $\ell/d \ge 7/37 + \rho_{\sup}$ for a fixed $\rho_{\sup} > 0$. Write $x = \bar{g}/\delta$, and retain $D_x = 3 - 17x/9$ and $P_x = (2 - 8x/9)(1 - x)$ from the refined character count in Section 10. The detector parameter t_* balances the bounds for short and long detecting polynomials; R_* is the resulting counting exponent. Set*

$$\mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \quad t_* = 1 + \frac{\delta P_x}{2\mathcal{J}}, \quad R_* = 1 - \delta + \frac{(\alpha - \delta)\delta P_x}{2\mathcal{J}}. \tag{18.2}$$

For every $\varepsilon > 0$, subdividing the total prime-length exponent ℓ into sufficiently small positive exponents ℓ_i, fixed independently of the target character, gives

$$\#\mathcal{B} \ll U^{R_* + \varepsilon}(1 + T_1)^A. \tag{18.3}$$

Here T_1 is the height cutoff of Section 5, and $A \ge 0$ is a fixed finite height exponent under the conventions of Section 2. Without an assumption on the available prime length, the count without selected prime factors is $\#\mathcal{B} \ll U^{1 - 2\delta/3 + \varepsilon}(1 + T_1)^A$.

Proof. The simultaneously large detecting polynomials of (5.5) have lengths U^r, U^m, $r + m \ge t - O(\varepsilon)$, and satisfy

$$|M_r|^2 \gg U^{\delta r - O(\varepsilon)}, \qquad |S_m|^4 \gg U^{2\delta m - O(\varepsilon)}.$$

Here M_r, S_m are the Möbius-weighted and unweighted detecting polynomials defined in Section 5, with the same character $\psi \in \mathcal{X}_u$, including its chosen defining modulus and zero extension. For a selected product of prime sums whose norm lengths have product U^z, the inverse moment with selected prime factors (9.1) and the unweighted moment (9.4) give the same set two count bounds:

$$U^{1 - \delta r - 2\bar{g}z + \varepsilon}(1 + T_1)^A, \qquad U^{1 - 2\delta m - 2\bar{g}z + \varepsilon}(1 + T_1)^A.$$

For elements with $q_u \asymp U$, the admissible selected-prime exponents satisfy $z \le z_M$ or $z \le z_S$, respectively, where

$$z_M = (1 - r)/2, \qquad z_S = 2(1 - 2m)/9.$$

Ordering the prime sums by decreasing size exponent and taking the first intervals, as in (10.2), achieves gain at least $\bar{g}z$ before discarding a partial interval; discarding the one partially selected prime-polynomial factor loses at most $\delta \max_i(\ell_i/d)$ in the squared lower bound. Decrease the selected prime-length bound by a fixed $\nu > 0$ before applying a moment; when the remaining prime-length bound is nonpositive, use the version without selected prime factors. These errors fit the requested ε.

For $r < 1, m < 1/2$, the affine count bounds without an additional prime-length restriction are

$$A_I(r) = 1 - \delta\big[x + (1 - x)r\big], \qquad S_t(r) = 1 - \delta\big[4x/9 + (2 - 8x/9)(t - r)\big].$$

They cross at

$$r_*(t) = \frac{(2 - 8x/9)t - 5x/9}{D_x}, \quad r_*(1) \ge 23/37, \quad r_*(3/2) = 1.$$

Use the inverse estimate on $r \ge r_*$ and the unweighted estimate on $r \le r_*$. The prime-length bound for the inverse moment is at most $7/37$, the bound for the unweighted moment is at most $2/27 + O(\varepsilon)$; the hypothesis on the available prime length suffices. On the inverse side the second length inequality has the positive gap

$$3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) \ge 9/37,$$

while the first length inequality has the positive gap $1-r-2z \geq 2\nu$.

For short detecting polynomials, the count exponent is consequently

$$R_{\mathrm{short}}(t) = 1-\delta+\frac{\delta P_x}{D_x}(3/2-t).$$

For $r \geq 1$, use the amplification estimate without selected prime factors (9.3):

$$\mathfrak{a}(r)-\delta r \leq R_{\mathrm{long}}(t) = 1-\delta+(\alpha-\delta)(t-1).$$

For $m \geq 1/2$ the unweighted bound without selected prime factors is at most $1-\delta$. Balancing the short and long bounds gives (18.2). Since $D_x - P_x = 1+x-8x^2/9 > 0$,

$$\mathcal{J} \geq 35/54, \qquad 1 \leq t_* \leq 3/2.$$

At $\delta = \alpha$, the balanced values are $t_* = 3/2$ and $R_* = 1-\alpha = 1/6$. At $r = 1$ the inverse prime-length bound is $z_M = 0$; use the estimate without selected prime factors (9.3). For $r \geq 1$, $\mathfrak{a}(r) - \alpha r = 1/6$. Thus the endpoint uses a moment without selected prime factors, rather than a limit of moment estimates with selected prime factors whose strict length gaps tend to zero.

For each character presentation chosen by the zero detector, both moment estimates are applied to the same pair of detecting polynomials. The dyadic choices of their lengths contribute only logarithmic factors. Common smooth-weight separation puts height dependence into $(1+T_1)^A$. There is no $\Delta/4$ term from (10.7): that term compares a varying κ with $3/4$, whereas (18.1) is already exact. The common character can be written as $\psi(n) = \nu(n)\overline{\chi_n(u)}$, and each selected prime coefficient is $\overline{\nu(p)}1_{p\in 1_{\mathcal{T}}}$; it is a fixed finite combination of Θ-characters. Conjugate the whole unweighted product to match (9.4). Elements divisible by a good prime divisor induce outside Θ, as in Section 10; unit elements are covered by the small-norm estimate below.

With no selected factors, $t = 1$ and the crossing $r = 2/3$ give the last assertion. □

Proposition 18.2 (Exponent for arbitrary length parameters)**.** *Let $B \leq \beta_*$, and assume (17.1), positive lengths and the unchanged joint-error and moment hypotheses of Part I. Here $C(s) = s - 5/6 + l_x/3 + \ell/6$ is the principal exponent in (17.6), and $h = 1 - l_x + \ell$. For a class with element-norm exponent d, count exponent R, zero-class parameter a of (5.1) and prime-sum size mean $\bar{g}$ of (10.1), the normalized nonprincipal contribution is, for each prescribed loss $\varepsilon > 0$ and sufficiently small fixed contour and zero-class widths, bounded by*

$$Z^{C(B)+E_B(d)+\varepsilon}(1+T_1)^A = Z^{C(\beta_*)+E_B(d)-(\beta_*-B)+\varepsilon}(1+T_1)^A,$$

where

$$E_B(d) = a - B + h(z_0 - 1/6) - al_y - \ell/2 + \bar{g}\ell + d(R+\delta/2-z_0), \quad z_0 = 17/50. \tag{18.4}$$

The factor $(1+T_1)^A$ records polynomial growth in the height cutoff T_1 of Section 5; $A \geq 0$ is fixed and finite, as in Section 2. Thus $E_B(d)$ is measured relative to $C(B)$; relative to $C(\beta_*)$, the exponent is $E_B(d) - (\beta_* - B)$. When $l_x + l_y + \ell = 1$ and $B = B(\ell)$,

$$E_B(h) = -\frac{1}{4} + \frac{b}{6} + \frac{5\ell}{4} + (1/2+\ell)\delta + \ell\bar{g} - h(1-R). \tag{18.5}$$

Proof. Use the exact Poisson identity (4.9), which holds for arbitrary positive summation lengths. Move the full holomorphic correction before partitioning it into pointwise amplitude classes. Choose the contour

$$(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (a+16\epsilon_{\mathrm{b}}, 1-a-6\epsilon_{\mathrm{b}}, z_0).$$

For example, first retain the global s line and move w so that $\mathrm{Re}(s+w) \geq 1+14\epsilon_{\mathrm{b}}$; after external height truncation move s to the displayed line, where $\mathrm{Re}(s+w) = 1+10\epsilon_{\mathrm{b}}$. The zero-free separation in Section 5 and the Euler domain (4.11) used in Part I apply.

The joint bound (8.3) applies the conductor-deficit estimate once to the entire product and yields

$$\sum_{u\in\mathcal{B}} |L_F^S(w, \chi_\bullet(u))\mathfrak{H}_{\eta,u,Z}| \ll U^{R+\delta/2+\varepsilon} Z^{\ell(z_0-1/2)+\bar{g}\ell+\varepsilon}(1+T_1)^A.$$

The remaining factors contribute

$$Z^{l_x(1/2-z_0)+a+z_0-1-al_y}\, U^{-z_0}.$$

Subtracting $C(B) = B + l_x/2 - 1 + h/6$ gives (18.4). Algebra using $l_x + l_y + \ell = 1$ gives (18.5). Horizontal joins and discarded heights are bounded by $Z^{B_\eta} T_1^{-N}$: the fixed internal seminorm orders are chosen before the arbitrary external integration-by-parts order N. The class sets need no holomorphic dependence on the contour variables. □

19. Polynomial verification of the rational saving and the remaining norm ranges

In this section, we verify that the nonprincipal exponent (18.5), with the count (18.2), has a strict saving at the rational parameters, uniformly over all zero and amplitude classes. Here $E_B(d)$ is measured relative to the principal exponent $C(B)$, as defined in (18.4), and $h = 1 - l_x + \ell$. The polynomial inequalities and the remaining norm estimates supply the positive exponent savings needed to prove Theorem 15.1 in the next section. The remaining-range estimates in Lemma 19.2 are stated under the general balanced hypotheses, so they are also used at the changed parameters in Parts III and IV.

Lemma 19.1 (Uniform rational saving)**.** *For* (15.1)*,* $R = R_*$ *and* $(\delta, x) \in [0, 5/6]\times[0, 1/2]$*, one has*

$$E_{B_{\mathrm{r}}}(h) \leq -19/320000. \tag{19.1}$$

Proof. At the Part I parameter values $\ell = 1/6, b = 1/8, h = 13/16$, denote the balanced exponent by E_0. Put $y = 1/2 - x$ and

$$j = 185 + 170y + (-138 + 12y + 96y^2)\delta = 108\mathcal{J},$$

$$Q_0 = 10(37+34y) - 8(237+377y+26y^2)\delta + 48(51+131y+94y^2+32y^3)\delta^2, \qquad Q = Q_0 - 96j/5000.$$

Direct expansion gives

$$Q = 10368\mathcal{J}(-E_0 - 1/5000). \tag{19.2}$$

To prove its sign on the whole rectangle, use the Bernstein basis, named after Sergei Natanovich Bernstein, who introduced it in his constructive proof of Weierstrass's approximation theorem [Ber12]; see [Far12] for a modern account. The Bernstein basis polynomials of degree n on $[0, 1]$ are

$$\mathsf{B}_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, \qquad 0 \leq i \leq n. \tag{19.3}$$

They satisfy

$$\mathsf{B}_{i,n}(t) \geq 0, \qquad \sum_{i=0}^{n} \mathsf{B}_{i,n}(t) = 1 \qquad (0 \leq t \leq 1).$$

On $[a_1,a_2]\times[b_1,b_2]$, substitute $\delta = a_1+(a_2-a_1)v$, $y = b_1+(b_2-b_1)w$ and write $Q=\sum_{p=0}^{2}\sum_{q=0}^{3}A_{pq}v^pw^q$. Its representation in the tensor-product Bernstein basis is

$$Q(\delta,y)=\sum_{i=0}^{2}\sum_{k=0}^{3}B_{ik}\mathsf{B}_{i,2}(v)\mathsf{B}_{k,3}(w).$$

The bidegree $(2,3)$ Bernstein coefficients are

$$B_{ik}=\sum_{p\le i,\ q\le k}A_{pq}\frac{\binom{i}{p}\binom{k}{q}}{\binom{2}{p}\binom{3}{q}},\qquad 0\le i\le 2,\quad 0\le k\le 3.$$

Nonnegativity and the partition-of-unity identity give

$$\min_{i,k}B_{ik}\le Q(\delta,y)\le\max_{i,k}B_{ik}\qquad((\delta,y)\in[a_1,a_2]\times[b_1,b_2]).$$

The listed rectangles cover the original rectangle; their complete partition and coefficients are verified by the accompanying exact arithmetic checker. The last column is a coefficient minimum, not a sampled value:

δ interval	y interval	$\min B_{ik}$
$[0,5/24]$	$[0,1/4]$	$1380407/18000$
$[5/24,5/16]$	$[0,1/8]$	$27677/2000$
$[5/16,35/96]$	$[0,1/16]$	$12437/8000$
$[35/96,25/64]$	$[0,1/32]$	$2301/16000$
$[25/64,5/12]$	$[0,1/32]$	$12581/32000$
$[35/96,5/12]$	$[1/32,1/16]$	$2952669/1024000$
$[5/16,5/12]$	$[1/16,1/8]$	$85933/192000$
$[5/24,5/16]$	$[1/8,1/4]$	$53811/2560$
$[5/16,5/12]$	$[1/8,1/4]$	$8213/800$
$[0,5/24]$	$[1/4,1/2]$	$336467/4000$
$[5/24,5/16]$	$[1/4,3/8]$	$946273/24000$
$[5/16,5/12]$	$[1/4,3/8]$	$666871/16000$
$[5/24,5/12]$	$[3/8,1/2]$	$70651/4000$
$[5/12,5/6]$	$[0,1/2]$	$319/125$

Thus $Q>0$ throughout, so $-E_0\ge 1/5000$.

With R_*,δ,x fixed, changing ℓ at fixed b gives

$$\partial_\ell E(h)=-1/4+\delta+\bar g+3R_*/2=5/4+\delta(x-1/2)+\frac{3(\alpha-\delta)\delta P_x}{4\mathcal{J}}\le 45/32.$$

We used $\mathcal{J}\ge\alpha P_x$ and $\delta(\alpha-\delta)\le\alpha^2/4$. Therefore $E(h)\le -1/5000+45/(32\cdot 10000)=-19/320000$. □

The endpoint inequality alone is insufficient: elements of very small and very large norm require different contours. We record a reusable division of the norm ranges.

Lemma 19.2 (Remaining ranges)**.** *Assume the general hypotheses (15.3), $B\le\beta_*\le 7/8$, and $0\le\delta\le 5/6$. Then the following estimates hold, with arbitrarily small fixed losses. Set $d_{\min}=1/100$.*

1. *For elements with $q_u\ll Z^{d_{\min}}$, including unit elements, the normalized exponent relative to $C(\beta_*)$ is at most $13h/75-l_y/2+2d_{\min}$.*
2. *For $d_{\min}\le d\le 1/2$ and $\delta\ge 1/50$, use $R=76/75-2\delta/3$ in (18.4). Its slope in d is $101/150-\delta/6>0$; it is enough to bound $E_B(1/2)$.*
3. *For $1/2\le d\le h$ use R_*. The slope $R_*+\delta/2-z_0$ is at least $4/25$ and at most 2; it is enough to bound $E_B(h)$.*

(4) *The lowest zero class $a = a_0$, without a detector, has $R = 1$ and $\bar{g} \le 1/100$. Its endpoint exponent is at most $-1/4 + b/6 + 5\ell/4 + (1/2 + 3\ell/2)/50$.*
(5) *Extending the third range to $h + \zeta$ increases the exponent by at most 2ζ, provided $\ell/(h + \zeta) > 7/37$. Beyond $h + \zeta$, an absolute contour with sufficiently large fixed $\operatorname{Re} z = z_{\mathrm{new}}$ gives any prescribed power saving.*

Proof. For the small-norm range retain the global reciprocal line $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$ and put $\operatorname{Re} w = 1/2$, $\operatorname{Re} z = z_0$. Use the Euler-product domain (4.11); the inequality $\operatorname{Re}(s + w) - 1 \ge B - 1/2 > 0$. Formula (17.5) and the local table imply $G_p = O(1)$ for $p \nmid u$ and $G_p = O(q_p^{1/2})$ for $p \mid u$. Consequently the full correction is $\ll U^{\varepsilon} Z^{\ell z_0}$: primes not dividing u contribute $P_i^{z_0}$, and primes dividing u contribute at most $U^{\varepsilon} P_i^{z_0 - 1/2}$. The nonprincipal numerator contributes a factor $U^{3/5+\varepsilon}$ on this line, by the growth estimate used in [Ope26a, Lemma 10.6]. Counting $O(U)$ original elements and the factor U^{-z_0} give $U^{63/50+\varepsilon}$.

The exterior factor relative to $C(\beta_*)$ is $Z^{13h/75 - l_y/2 + O(\epsilon_{\mathrm{b}})}$. Since $63/50 < 2$, the first claim follows.

For the intermediate norm range use the count without selected prime factors in Proposition 18.1, allowing an extra $1/75$. For the moderate norm range use its count with selected prime factors. Differentiating (18.4) gives the slopes asserted; for the latter use $R_* \ge 1 - \delta$ and $\delta \le 5/6$. The lowest zero class is bounded trivially by $O(U)$ and $\bar{g} \le \delta/2$. The extension beyond h follows from the slope upper bound. Finally, on $(\operatorname{Re} s, \operatorname{Re} w, \operatorname{Re} z) = (2, 2, z_{\mathrm{new}})$, absolute convergence gives

$$\mathcal{R}(U; Z) \ll Z^{l_x/2 + 1 + l_y + h z_{\mathrm{new}} + \varepsilon} U^{1 + \varepsilon - z_{\mathrm{new}}}.$$

Summing $U > Z^{h+\zeta}$ yields an exponent containing $-\zeta z_{\mathrm{new}}$. Choose z_{new} after ζ is fixed. All statements concern the complete correction $\mathfrak{H}_{\eta,u,Z}$ of (8.2), without division by its local factors, which may vanish. □

For the rational parameters the saving in the small-norm range is at least $78699/1000000$, and the saving for the lowest zero class is at least $1093/192000$. For the intermediate norm range, direct substitution gives

$$\max_{1/50 \le \delta \le 5/6} E_{B_{\mathrm{r}}}(1/2) \le -\frac{49}{14400} + \frac{177}{2000000} = -\frac{59657}{18000000}.$$

Together with Lemma 19.1, these give positive exponent savings independent of the target. Choose

$$0 < \zeta < \min\{5\ell - h,\ (19/320000)/64\}.$$

This also ensures $\ell/(h + \zeta) > 1/5 > 7/37$.

20. Proof of the rational zero-free bound by the continuation criterion

In this section, we assemble the direct bound (16.2), the principal estimate (17.7) and the nonprincipal bounds of Lemmas 19.1 and 19.2 to prove the rational extension by Proposition 3.1. Proposition 20.1 states the resulting comparison (20.1), with its uniformity explicit so that Parts III and IV can apply it with their respective parameters and counts.

Proposition 20.1 (Comparison from strict exponent inequalities)**.** *Assume the balanced hypotheses (15.3). Suppose also that:*

(1) *the direct bound (16.2) with $l_x/2 + b/12 = C_b(B)$;*
(2) *the principal estimates of Proposition 17.2;*
(3) *the exponents for the small and intermediate norm ranges and the lowest zero class in Lemma 19.2 are negative, and $E_B(h) \le 0$ for the count with selected prime factors.*

The chosen count has the form $\#\mathcal{B} \ll U^{R+\varepsilon}(1+T_1)^A$ *with the common-character, coefficient, coprimality, support and height hypotheses of Section 9. Here* $\mathcal{B}$ *is a joint zero and prime-sum size class, as in* (5.1) *and Section 10.1, and* $E_B(d)$ *is defined in* (18.4). *Assume its losses can be made arbitrarily small by fixed length decrements and a sufficiently fine subdivision of the total prime length, chosen independently of* η. *Assume also that* $0 \le \partial_d E_B \le 2$ *in the moderate norm range. If* $\Delta = \beta_* - B > 0$, *then there is a number* $\sigma > 0$, *chosen before the target* η, *such that*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_b(\beta_*)-\sigma}. \tag{20.1}$$

The principal factor is holomorphic on $\operatorname{Re} s > B$ *and satisfies* (17.2).

Proof. Replacing B by β_* in the central exponent subtracts the same Δ in every class. Choose a common positive saving smaller than $\Delta/4$, the strict savings for the small and intermediate norm ranges and the lowest zero class, and $l_y/80, h/2400$. Fix ζ smaller still so that 2ζ is less than a quarter of that saving and all prime-length inequalities remain strict. Choose the requested moment loss, positive gaps in the prime-length restrictions and maximum individual prime lengths smaller than that saving. The chosen count and its underlying moments supply these strict length inequalities and the upper bound on each prime length.

Fix an even K and the actual ℓ_i before the target. Now choose $0 < \kappa_P < B \min_i \ell_i$, decrease the common saving if necessary, and finally choose ϵ_{b} and the remaining smooth-weight and rounding losses so that their finite sum is smaller than a quarter of this saving. All these real choices precede η.

The class count is finite apart from logarithmically many dyadic intervals and admissible polynomial configurations, whose logarithmic factors are absorbed in the chosen exponent losses. Lemmas 19.2 and 17.1, the chosen count, and Propositions 18.2 and 17.2 therefore give a common $\mu > 0$ with

$$|J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}. \tag{20.2}$$

The internal smoothness orders, A_η and B_η are fixed before N. They may depend on the target; μ does not. If $\tau_{\mathrm{det}} > 0$ is the permissible detector height exponent, choose

$$0 < \tau_\eta \le \min\{d_{\min}/100, \tau_{\mathrm{det}}, \mu/[4(A_\eta+1)]\}, \qquad T_1 = Z^{\tau_\eta}.$$

Then choose N so large that $B_\eta - N\tau_\eta < C_b(\beta_*) - \mu/2$. Equation (20.2) proves (20.1) with $\sigma = \mu/2$.

The dependence of the height cutoff on η is harmless: J_η, f_η were fixed in (17.6). □

Proof of Theorem 15.1. Suppose $\Delta_{\mathrm{r}} = \beta_* - B_{\mathrm{r}} > 0$. Part I already gives $\beta_* \le 7/8$, so (18.1) holds. The length inequalities (15.4), the direct bound (16.2), Lemma 19.1 and the remaining-range estimates of Lemma 19.2 verify Proposition 20.1. For any $0 < \omega_0 < \Delta_{\mathrm{r}}$, the normalized direct bound is

$$|J_\eta(Z)| \ll Z^{C_b(B_{\mathrm{r}})+\omega_0}.$$

Apply Proposition 3.1 with $\sigma_0 = B_{\mathrm{r}}$ and $c = -11/16$. Its principal factor is holomorphic and within $1/2$ of one by (17.2); its comparison hypothesis is (20.1). The criterion implies $\beta_* \le B_{\mathrm{r}}$, contradicting $\Delta_{\mathrm{r}} > 0$. The transfer proof of Corollary 14.1 applies with this new positive boundary and gives the Dirichlet assertion. □

Part III. An algebraic extension of the bound

21. The algebraic zero-free bound and its parameters

In this section, we state the stronger bound obtained by varying b and ℓ together: (21.1) gives the bound and (21.2) specifies its algebraic parameters. The following sections verify these choices using the analytic construction of Part II and the character and amplitude notation of Part I. Whereas Part II improved 7/8 by 1/40000 with $b = b_{\mathrm{r}} = 1/8$ fixed, the joint variation gives $7/8 - B_{\mathrm{alg}} = 4.293020083\ldots \times 10^{-5}$.

Theorem 21.1 (Algebraic extension)**.** *Every finite-order Hecke L-function over F, and every Dirichlet L-function, is zero-free in*

$$\operatorname{Re} s > B_{\mathrm{alg}} := \frac{1507 - 2\sqrt{921}}{1653} = 0.874957069799\ldots. \tag{21.1}$$

Principal poles are allowed and the boundary is excluded.

Let $r_{921} = \sqrt{921}$, and set

$$\begin{gathered}
\delta_{\mathrm{c}} = \frac{49 - r_{921}}{48}, \qquad \ell_{\mathrm{alg}} = \frac{33 + 8r_{921}}{1653}, \qquad b_{\mathrm{alg}} = -\frac{4}{29} + \frac{230r_{921}}{26709}, \\
l_{x,\mathrm{alg}} = \frac{1 - b_{\mathrm{alg}} - \ell_{\mathrm{alg}}}{2}, \qquad l_{y,\mathrm{alg}} = \frac{1 + b_{\mathrm{alg}} - \ell_{\mathrm{alg}}}{2}, \qquad h_{\mathrm{alg}} = \frac{1 + b_{\mathrm{alg}} + 3\ell_{\mathrm{alg}}}{2}, \\
B_{\mathrm{alg}} = \frac{11}{12} - \frac{\ell_{\mathrm{alg}}}{4}, \qquad C_{\mathrm{alg}}(s) = s - \frac{4 + b_{\mathrm{alg}}}{6}.
\end{gathered} \tag{21.2}$$

The subscript alg denotes the algebraic extension and its associated lengths and comparison exponent. In particular, $b_{\mathrm{alg}} = l_{y,\mathrm{alg}} - l_{x,\mathrm{alg}}$ is the difference of two real length exponents, whereas b_* in Part I is an arithmetic element. Below we write $b = b_{\mathrm{alg}}$, $\ell = \ell_{\mathrm{alg}}$, $l_x = l_{x,\mathrm{alg}}$, $l_y = l_{y,\mathrm{alg}}$ and $h = h_{\mathrm{alg}}$ when these parameter values are understood. The approximate values, used only for orientation, are

ℓ_{alg}	$0.166838387470\ldots$	b_{alg}	$0.123405436989\ldots$
$l_{x,\mathrm{alg}}$	$0.354878087771\ldots$	$l_{y,\mathrm{alg}}$	$0.478283524759\ldots$
h_{alg}	$0.811960299699\ldots$	δ_{c}	$0.388583712271\ldots$

Every sign and equality below uses the exact values in (21.2).

Theorem 21.1 is an analytic deduction from the cited estimates under all their stated hypotheses. Proposition 25.1 separately proves an algebraic lower bound under three explicit constraints. Part IV proves necessity of these constraints for the explicitly defined uniform exponent comparison system, also allowing the moment parameter to vary.

After the first version of this paper was posted, we became aware of several independent and contemporaneous GitHub/Zenodo posts reporting related improvements of the 7/8 bound: Ignacio Javier Mermet's GitHub repository, Leo Wei's GitHub repository, and Khamzat Chakhkiev's post: doi:10.5281/zenodo.23221313. The two GitHub repositories report the boundary 0.8749570698, and the Zenodo post reports the boundary 0.87496. These bounds use rational choices of b and ℓ and are slightly weaker than the bound in Theorem 21.1.

22. The direct estimate with free total length

In this section, we extend the direct estimate of Proposition 16.2 by allowing $M = l_x + l_y$ to vary independently of ℓ. Proposition 22.2 gives the full direct exponent (22.3) and the resulting lower constraint (22.4). Its individual exponent constraints will be used in Section 25 and in the full optimality proof of Part IV. At the balanced algebraic parameters, the full estimate also gives the direct bound checked in Section 26.

Assume

$$l_x > \ell, \quad l_y > \ell, \quad b = l_y - l_x > 0, \quad h = 1 - l_x + \ell > 0.$$

The length exponents lie in fixed bounded ranges. The principal-term exponent is

$$C(s) = s - 5/6 + l_x/3 + \ell/6. \tag{22.1}$$

In particular the additive constant changes when b changes; the Part I expression $s-11/16$ cannot be retained for the algebraic parameter choice.

Lemma 22.1 (General functional-equation estimate)**.** *Under the common-weight and coefficient hypotheses of* [Ope26a, Lemma 14.3]*, let $B^{\mathcal{I}}_{a,\sigma}$ be the theta sums defined in Section 16, now with outer element-range exponent m and remaining prime-length exponent l. Thus $X'Y' \asymp Z^m$ and $Q = q_{b_*}X'Y'$; the remaining arithmetic, coefficient and smooth-weight data are fixed as in that definition. Their squared norm satisfies*

$$\begin{aligned} &\sum_{q_a \ll Q} |B^{\mathcal{I}}_{a,\sigma}|^2 \ll Z^{m+k(m,l)+\varepsilon}, \\ &k(m,l) = \max\left\{0, \frac{1+3l-2m}{4}, m+l-1, \frac{3l-1}{2}\right\}. \end{aligned} \tag{22.2}$$

The estimate includes both signs of $m + l - 1$.

Proof. Use the local lengths defined in the proof of Lemma 16.1, now with $H \le m - O$, $z_a \le l$. Suppress arbitrarily small offsets arising from the support intervals and logarithmic factors. Then

$$\begin{gathered} O, S_0, B_0, v, l_b, e_\lambda \ge 0, \quad 2A_0 \le O, \quad 0 \le N_0 \le A_0, \\ T_d = 2H + 2A_0 + 2z_a - 1 - l - N_0 - 3B_0, \qquad v + 3l_b + e_\lambda \le T_d, \end{gathered}$$

$$\begin{aligned} E_{\mathrm{ref}} = O/2 + \max(H, v + l_b) - S_0 - B_0 + z_a - u_{\mathrm{hyb}} - l_b - 2e_\lambda/3 \\ - (T_d - v - 3l_b - e_\lambda)_+/2, \qquad u_{\mathrm{hyb}} = \min\{v, z_a, (v + z_a)/3\}. \end{aligned}$$

There are four exhaustive cases. If the maximum is H and $u_{\mathrm{hyb}} = z_a$, then $E_{\mathrm{ref}} \le O/2 + H \le m$. If the maximum is H and $u_{\mathrm{hyb}} \ne z_a$, then $u_{\mathrm{hyb}} \ge v/2$ and the dual terms save at least $T_d/4$.

Substitution gives

$$\begin{aligned} E_{\mathrm{ref}} &\le O/2 + H/2 - A_0/2 + z_a/2 + (1 + l + N_0 + 3B_0)/4 - S_0 - B_0 \\ &\le m/2 + (1 + 3l)/4. \end{aligned}$$

If the maximum is $v + l_b$ and $u_{\mathrm{hyb}} = z_a$, then

$$E_{\mathrm{ref}} \le O/2 + v \le O/2 + T_d \le 2m + l - 1.$$

If the maximum is $v + l_b$ and $u_{\mathrm{hyb}} \ne z_a$, then

$$E_{\mathrm{ref}} \le O/2 + v/2 + z_a \le O/2 + T_d/2 + z_a \le m + (3l - 1)/2.$$

Their maximum is $m + k(m, l)$.

Restore all signed $O(1/\log Z)$ offsets, widths of the support intervals and losses from restricting the summation ranges. If each is bounded by ε_0, a common bound for losses $20\varepsilon_0$ bounds their effect in these four linear inequalities. The homogeneous joint-weight separation precedes squaring; its Gaussian translates have summable seminorms and the complete norm ranges discarded in this partition have arbitrary power saving. The remaining dyadic intervals are logarithmically many.

Choose ε_0 inside the requested ε. This proves the bound with the original paper's coefficient classes and normalization of the outer element sum. □

Proposition 22.2 (Full direct exponent)**.** *The modified average satisfies*

$$|I_{\eta,\mathrm{mod}}(Z)| \ll Z^{L_{\mathrm{low}}+\varepsilon}, \qquad L_{\mathrm{low}} = \frac{l_x}{2} + \max\{0, b/12, b - l_y/2\} + \frac{1}{2}k(M,\ell). \tag{22.3}$$

By (22.1), the condition $L_{\mathrm{low}} \le C(B)$ for the direct estimate is equivalent to $B \ge B_{\mathrm{low}}$, where

$$B_{\mathrm{low}} = L_{\mathrm{low}} + 5/6 - l_x/3 - \ell/6. \tag{22.4}$$

Under the condition $M + \ell = 1$, this simplifies to

$$B_{\mathrm{low}} = \frac{11}{12} - \frac{\ell}{4} + \frac{(8b + 3\ell - 3)_+}{12} + \frac{(5\ell - 1)_+}{8}. \tag{22.5}$$

Proof. Keep all three terms in (7.5); their contribution after applying (4.7) is $\max\{0, b/12, b - l_y/2\}$. For a subset $\mathcal{I}$ with total exponent $d_{\mathcal{I}}$, Lemma 22.1 applies with $m = M - 2d_{\mathcal{I}}$ and $l = \ell - d_{\mathcal{I}}$. After tuple counting and rescaling as in Proposition 16.2, its total exponent is

$$F(d_{\mathcal{I}}) = l_x/2 - d_{\mathcal{I}} + \max\{0, b/12, b - l_y/2 + d_{\mathcal{I}}/2\} + \tfrac{1}{2}k_{\mathcal{I}},$$

$$k_{\mathcal{I}} = \max\left\{0, \frac{1 + 3\ell - 2M + d_{\mathcal{I}}}{4}, M + \ell - 1 - 3d_{\mathcal{I}}, \frac{3\ell - 1 - 3d_{\mathcal{I}}}{2}\right\}.$$

For $d_2 > d_1$, the additive maximum increases by at most $(d_2 - d_1)/2$, and $k_{\mathcal{I}}$ by at most $(d_2 - d_1)/4$. Thus $F(d_2) - F(d_1) \le -3(d_2 - d_1)/8$, so the empty subset controls the direct exponent. Subtract (22.1) to obtain (22.4). With $M = 1 - \ell$, the extra maximum from the functional-equation estimate is $(5\ell - 1)_+/4$, and the extra Gram maximum is $(8b + 3\ell - 3)_+/12$, proving (22.5). □

23. Comparison of the character-count estimates

In this section, we compare the character-count estimates available at the algebraic parameters, allowing for the supply of selected prime factors. The permitted savings are given in (23.1), and (23.2) lists the candidate crossing points. This identifies the count used in Section 24 and records the alternative choices examined in the optimality proof of Part IV.

Retain $\delta = 2a - 1$ from (5.1) and $x = \bar{g}/\delta$ from (10.1). The detecting lengths r, m satisfy (5.3)–(5.4). For elements with $q_u \asymp U = Z^d$, let $z_{\sup} = \ell/d$ be the available total selected-prime exponent relative to U. Put $c_\kappa = 1/(6\kappa)$, $3/4 \le \kappa \le 1$. For $1 \le t \le 3/2$, $t - 1/2 \le r \le 1$ and $m = t - r$, the inverse and unweighted moment estimates give count exponents $1 - \delta F_I(r)$ and $1 - \delta F_S(r)$, respectively, where

$$\begin{aligned} F_I(r) &= r + 2x\min\{(1-r)/2, z_{\sup}\}, \\ F_S(r) &= 2m + 2x\min\{c_\kappa(1-2m), z_{\sup}\}. \end{aligned} \tag{23.1}$$

The count exponent for short detecting polynomials is $1 - \delta \min_r \max\{F_I(r), F_S(r)\}$. The two savings are increasing and decreasing respectively, and their crossing is one of

$$\frac{2t}{3}, \qquad \frac{(2 - 4c_\kappa x)t + 2c_\kappa x - 2xz_{\sup}}{3 - 4c_\kappa x}, \qquad \frac{2t + 2xz_{\sup} - x}{3 - x}, \qquad \frac{(2 - 4c_\kappa x)t + (2c_\kappa - 1)x}{3 - x - 4c_\kappa x}. \tag{23.2}$$

Each expression is used only when the corresponding terms attain the minima in (23.1). If a crossing lies outside the interval, take its appropriate endpoint. The count bound for long detecting polynomials remains $R_{\mathrm{long}}(t)$ from Proposition 18.1.

An additional restriction on the available prime length can only decrease F_I or F_S in (23.1). Increasing κ decreases c_κ and can only decrease F_S. Consequently, among these estimates with $\kappa \in [3/4, 1]$, the smallest count exponent is obtained without an additional prime-length restriction and with $\kappa = 3/4$; optimizing t then gives R_*. With no selected primes, balancing the short- and long-polynomial bounds without selected prime factors instead gives

$$t_N = \frac{\alpha}{\alpha - \delta/3}, \qquad R_N = 1 - \frac{2\alpha\delta}{3\alpha - \delta} \geq R_*.$$

This includes the no-prime alternative in the comparison. Strict length inequalities in the moment estimates, coefficient restrictions and height factors are handled exactly as in Proposition 18.1. These count exponents describe the sufficient bounds furnished by the moment estimates; they are not lower bounds on the numbers of elements.

24. A polynomial proof of the algebraic endpoint inequality

In this section, we derive the algebraic parameters from the critical class of the count (18.2) studied in Section 23, then prove the endpoint inequality on the full rectangle of zero-class and amplitude parameters. The exact discriminant identity (24.4) supplies the central nonprincipal estimate for the final comparison in Section 26. This proves the required endpoint inequality; the separate lower bound in Section 25 addresses optimality and is not needed to establish the zero-free conclusion.

At $x = 1/2$, formula (18.2) reduces to

$$R_*(\delta, 1/2) = \frac{185 - 288\delta + 96\delta^2}{185 - 138\delta}. \tag{24.1}$$

The equation $R_* = 2/3$ has smaller root δ_c, since

$$288\delta^2 - 588\delta + 185 = 0.$$

At that root the coefficient of b in (18.5) vanishes, and $E_B(h) = 0$ forces

$$\ell = \frac{5 - 6\delta_c}{9 + 18\delta_c} = \ell_{\mathrm{alg}}.$$

Finally, the condition $\partial_\delta E_B(h) = 0$ at $(\delta, x) = (\delta_c, 1/2)$, with $R = R_*$, $\ell = \ell_{\mathrm{alg}}$ and b held fixed during differentiation in (18.5), determines $b = b_{\mathrm{alg}}$. These equations locate the candidate; the following identity proves its validity over the full rectangle.

Lemma 24.1 (Algebraic endpoint inequality)**.** *At (21.2), for $(\delta, x) \in [0, 5/6] \times [0, 1/2]$, $\bar{g} = x\delta$ and $R = R_*$,*

$$E_{B_{\mathrm{alg}}}(h_{\mathrm{alg}}) \leq 0. \tag{24.2}$$

Equality occurs only at $(\delta, x) = (\delta_c, 1/2)$.

Proof. Put $y = 1/2 - x \geq 0$ and

$$p(y) = 7 + 18y + 8y^2, \quad j(y, \delta) = 185 + 170y + (-138 + 12y + 96y^2)\delta,$$

so $P_x = p(y)/9$, $\mathcal{J} = j/108$. Write $C_0 = -1/4 + b_{\mathrm{alg}}/6 + 5\ell_{\mathrm{alg}}/4$. Substitution in (18.5) gives

$$\mathcal{Q} := 216\mathcal{J}(-E_{B_{\mathrm{alg}}}(h_{\mathrm{alg}})) = -2jC_0 + j(b_{\mathrm{alg}} + 2\ell_{\mathrm{alg}}y)\delta - 12h_{\mathrm{alg}}(5/6 - \delta)\delta p(y). \tag{24.3}$$

Write $\mathscr{Q} = \mathscr{A}(y)\delta^2 + \mathscr{D}(y)\delta + \mathscr{G}(y)$; these three coefficient polynomials are defined by this identity. For $\mathscr{A}(y) = \sum_{i=0}^{3} A_i y^i$, direct expansion yields

$$\begin{aligned} A_0 &= 31824/551 - 36688r_{921}/169157, \\ A_1 &= 50916/551 + 214096r_{921}/169157, \\ A_2 &= 17352/551 + 347296r_{921}/169157, \\ A_3 &= 2112/551 + 512r_{921}/551. \end{aligned}$$

Every A_i is positive. The exact discriminant identity is

$$4\mathscr{A}(y)\mathscr{G}(y) - \mathscr{D}(y)^2 = y(d_1 + d_2 y + d_3 y^2 + d_4 y^3), \tag{24.4}$$

where

$$\begin{aligned} d_1 &= (3861265032800 - 122715443200r_{921})/279616521, \\ d_2 &= 2386327618300/279616521 - 19093999200r_{921}/93205507, \\ d_3 &= -3389811480400/279616521 + 46823488000r_{921}/93205507, \\ d_4 &= -40221718800/4905553 + 4408611200r_{921}/14716659. \end{aligned}$$

Expansion with $r_{921}^2 = 921$ verifies the identity coefficient by coefficient; the accompanying exact arithmetic checker verifies this identity and the associated parameter inequalities. A rational isolating interval, obtained by squaring its endpoints, is

$$30.347981810 \;<\; r_{921} \;<\; 30.347981811. \tag{24.5}$$

Substitution into these linear expressions gives

$$490 < d_1 < 491, \quad 2317 < d_2 < 2318, \quad 3122 < d_3 < 3123, \quad 892 < d_4 < 893.$$

The same bracket proves $A_0 > 0$; positivity of the other A_i is immediate. All decimal endpoints here are terminating rational numbers. Complete the square:

$$4\mathscr{A}(y)\mathscr{Q} = (2\mathscr{A}(y)\delta + \mathscr{D}(y))^2 + y(d_1 + d_2 y + d_3 y^2 + d_4 y^3) \geq 0.$$

Since $\mathcal{J} \geq 35/54$ on the required rectangle, (24.3) proves the asserted sign. For $y > 0$ it is strict. At $y = 0$, the exact polynomial is $A_0(\delta - \delta_c)^2$, proving the equality assertion. □

25. An algebraic lower bound for the comparison constraints

In this section, we show that three affine comparison constraints force $B \geq B_{\mathrm{alg}}$, even when the total averaging length is free. The constraints are (25.1) and (25.2); Proposition 25.1 proves their lower bound under (25.3). This is the algebraic part of the optimality argument; Section 34 will justify the constraints within the full specified comparison system and allow κ to vary.

Put $M = l_x + l_y$. The two terms 0 and $M + \ell - 1$ in $k(M, \ell)$ of (22.2), inserted into (22.4), give the direct comparison constraints

$$B \geq L_0 := 5/6 + M/12 - \ell/6, \qquad B \geq L_K := 1/3 + 7M/12 + \ell/3. \tag{25.1}$$

At $(\delta, x) = (\delta_c, 1/2)$, a count exponent $R \geq 2/3$ in (18.4), with $h > 0$, also requires

$$B \geq H_c := 1 + \delta_c - (1 + \delta_c)M/2 + \delta_c \ell. \tag{25.2}$$

The following proposition is a statement about these three affine constraints. It assumes the constraints explicitly.

Proposition 25.1 (Algebraic lower bound). *Let $M, \ell, B \in \mathbb{R}$ and $\delta_{\mathrm{c}} = (49 - \sqrt{921})/48$. If*

$$B \geq L_0, \qquad B \geq L_K, \qquad B \geq H_{\mathrm{c}}, \tag{25.3}$$

with L_0, L_K, H_{c} defined above, then $B \geq B_{\mathrm{alg}}$. Equality is attained by $M = 1 - \ell_{\mathrm{alg}}$ and $\ell = \ell_{\mathrm{alg}}$.

Proof. The positive weights

$$w_0 = \frac{4 + 18\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}}, \quad w_K = \frac{2}{9 + 18\delta_{\mathrm{c}}}, \quad w_H = \frac{3}{9 + 18\delta_{\mathrm{c}}}$$

sum to one and satisfy

$$w_0/12 + 7w_K/12 - (1 + \delta_{\mathrm{c}})w_H/2 = 0, \qquad -w_0/6 + w_K/3 + \delta_{\mathrm{c}} w_H = 0.$$

Their weighted sum cancels M and ℓ, giving

$$B \geq \max\{L_0, L_K, H_{\mathrm{c}}\} \geq w_0 L_0 + w_K L_K + w_H H_{\mathrm{c}} = \frac{7 + 18\delta_{\mathrm{c}}}{9 + 18\delta_{\mathrm{c}}} = \frac{1507 - 2r_{921}}{1653} = B_{\mathrm{alg}}.$$

The cancellation proves the lower bound for every real M, ℓ; no numerical enumeration of parameter choices is involved. At $M = 1 - \ell_{\mathrm{alg}}$ and $\ell = \ell_{\mathrm{alg}}$, direct substitution gives $L_0 = L_K = H_{\mathrm{c}} = B_{\mathrm{alg}}$. □

Scope of optimality. Part IV proves necessity of the three constraints in the explicitly defined uniform exponent comparison system. Restricting the moment parameter to $\kappa \geq 3/4$, monotonicity of $F(\kappa)$ in Proposition 34.1 gives $B \geq F(3/4) = B_{\mathrm{alg}}$. The present parameter choice attains that bound. This proves optimality for that comparison problem in the restricted moment range, including free total length and the available estimates without selected prime factors; it does not exclude sharper analytic estimates.

26. Auxiliary estimates and proof of the algebraic zero-free bound

In this section, we combine the endpoint inequality of Section 24 with the direct bound of Proposition 16.2 and the principal-term estimate of Proposition 17.2. Checking the remaining strict auxiliary inequalities permits another application of Proposition 3.1 and proves Theorem 21.1.

Recall that J_η and f_η now use the algebraic lengths in (21.2), as prescribed in (17.6). Their principal exponent is $C_{\mathrm{alg}}(s) = s - (4 + b_{\mathrm{alg}})/6$.

At (21.2), exact substitution gives

$$\begin{aligned} &b_{\mathrm{alg}} > 0, \quad l_{x,\mathrm{alg}} - \ell_{\mathrm{alg}} > 0, \quad l_{y,\mathrm{alg}} - \ell_{\mathrm{alg}} > 0, \quad 1 - 3\ell_{\mathrm{alg}} > 0, \quad \ell_{\mathrm{alg}} < 1/5, \\ &l_{y,\mathrm{alg}} - \ell_{\mathrm{alg}} - 11b_{\mathrm{alg}}/6 > 0, \qquad \ell_{\mathrm{alg}}/h_{\mathrm{alg}} > 1/5 > 7/37, \qquad 5\ell_{\mathrm{alg}} - h_{\mathrm{alg}} > 0. \end{aligned} \tag{26.1}$$

For example these inequalities follow by replacing r_{921} by the rational endpoints in (24.5) after reducing the expressions to $a + cr_{921}$; the denominators are positive. Both additional positive-part terms therefore vanish in (22.5), and

$$|J_\eta(Z)| \ll Z^{C_{\mathrm{alg}}(B_{\mathrm{alg}})+\varepsilon}, \qquad C_{\mathrm{alg}}(B_{\mathrm{alg}}) = l_{x,\mathrm{alg}}/2 + b_{\mathrm{alg}}/12. \tag{26.2}$$

Equation (26.1) verifies the general hypotheses (15.3); the ranges in Lemma 19.2 therefore apply at these lengths. For the lowest zero class $\delta_0 = 1/50$, recall the constant $C_0 = -1/4 + b_{\mathrm{alg}}/6 + 5\ell_{\mathrm{alg}}/4$ from Section 24. The saving is

$$m_F = -C_0 - (1/2 + 3\ell_{\mathrm{alg}}/2)\delta_0 > 0.$$

For the intermediate norm range use $R = 76/75 - 2\delta/3$; the upper endpoint exponent is bounded by

$$E_{B_{\mathrm{alg}}}(1/2) \le \frac{(225\ell_{\mathrm{alg}} + 50 - 75b_{\mathrm{alg}})\delta + 78\ell_{\mathrm{alg}} - 49b_{\mathrm{alg}} - 73}{300} < 0 \quad (1/50 \le \delta \le 5/6).$$

It is affine in δ, and both endpoint values are negative by (24.5). For the small-norm range the saving is

$$m_S = l_{y,\mathrm{alg}}/2 - 13h_{\mathrm{alg}}/75 - 2/100 > 0.$$

For the moderate norm range, $E_{B_{\mathrm{alg}}}(d) \le E_{B_{\mathrm{alg}}}(h_{\mathrm{alg}}) \le 0$ by Lemma 24.1; for $h_{\mathrm{alg}} < d \le h_{\mathrm{alg}} + \zeta$ the increase in the exponent is at most 2ζ. The large-norm range is handled by the same absolute contour as in Lemma 19.2. Equality in the endpoint inequality (24.2) occurs only at $(\delta, x) = (\delta_{\mathrm{c}}, 1/2)$. Under the contradiction hypothesis $\beta_* > B_{\mathrm{alg}}$, measuring the error relative to $C_{\mathrm{alg}}(\beta_*)$ subtracts $\beta_* - B_{\mathrm{alg}} > 0$, so the comparison has a strict saving.

The principal domain also needs no new argument:

$$437/500 < B_{\mathrm{alg}} < B_{\mathrm{r}} < 7/8.$$

Lemma 17.1 and Proposition 17.2 therefore apply with $B = B_{\mathrm{alg}}$ and the new lengths. In particular, the principal factor has no zeros on $\operatorname{Re} s > B_{\mathrm{alg}}$ and is the same factor appearing in the new principal Mellin integral. In the small-norm range, the first-domain condition has the positive gap $\operatorname{Re}(s+w) - 1 \ge B_{\mathrm{alg}} - 1/2 > 0$.

The non-strict inequality (24.2) is covered by Proposition 20.1. Indeed, under the contradiction hypothesis $\Delta_{\mathrm{alg}} = \beta_* - B_{\mathrm{alg}} > 0$,

$$C_{\mathrm{alg}}(B_{\mathrm{alg}}) + E_{B_{\mathrm{alg}}}(h_{\mathrm{alg}}) - C_{\mathrm{alg}}(\beta_*) \le -\Delta_{\mathrm{alg}}.$$

The finitely many contour, moment, subdivision into exponent intervals and normalization losses have fixed finite coefficients. Proposition 20.1 chooses them inside this gap before fixing the target.

Proof of Theorem 21.1. If $\Delta_{\mathrm{alg}} > 0$, the direct bound (26.2), the Euler domain of Lemma 17.1, the strict auxiliary estimates above and Lemma 24.1 verify all hypotheses of Proposition 20.1. Thus for some $\sigma > 0$ independent of the target,

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_{\mathrm{alg}}(\beta_*) - \sigma}.$$

The exponents of the height factors are allowed to depend on η; choose $T_1 = Z^{\tau_\eta}$ and then the tail order as in (20.2). The modified average and principal Mellin integral are unchanged by this choice of height cutoff.

Apply Proposition 3.1 with $\sigma_0 = B_{\mathrm{alg}}$, $c = -(4 + b_{\mathrm{alg}})/6$ and, for example, $\omega_0 = \Delta_{\mathrm{alg}}/2$. Its hypotheses now all hold: the direct bound is at $C_{\mathrm{alg}}(B_{\mathrm{alg}}) + \omega_0$, the comparison saves σ, and the holomorphic principal factor is within $1/2$ of one. The criterion implies $\beta_* \le B_{\mathrm{alg}}$, contradicting $\Delta_{\mathrm{alg}} > 0$. Finally apply the norm-pullback argument of Corollary 14.1 with $B = B_{\mathrm{alg}} > 0$. This gives all Dirichlet L-functions, including ζ, and completes the argument of Part III. □

Part IV. A moment-parameter extension and the optimal comparison bound

27. The limiting zero-free bound and its parameters

In this section, we state the limiting zero-free bound obtained by varying the moment parameter together with the averaging lengths. Theorem 27.1 gives B_{new}, with the explicit root in (27.1) and the limiting parameter values in (33.2). Part III gives a bound strong enough to permit $\kappa < 3/4$, once the unweighted moment has been extended to that range. Sections 28–33 prove it, and Section 34 proves its optimality within the specified comparison system.

Theorem 21.1 gives

$$\beta_* \le B_{\mathrm{alg}} < 7/8, \qquad 2B_{\mathrm{alg}} - 1 < 3/4.$$

We extend [Ope26a, Lemma 18.1] to $13/18 \le \kappa \le 1$ and iterate the resulting length optimization.

Theorem 27.1 (Limiting zero-free bound)**.** *Let*

$$\xi_{\mathrm{new}} = \frac{20}{33} + \frac{\sqrt{1391}}{33}\cos\left(\frac{1}{3}\arccos\frac{28470}{1391^{3/2}} - \frac{2\pi}{3}\right), \tag{27.1}$$

where arccos $\in [0,\pi]$. *Equivalently,* ξ_{new} *is the unique root in* $(0.38858, 0.38859)$ *of* $1188\xi^3 - 2160\xi^2 + 171\xi + 190 = 0$. *Then*

$$\beta_* \le B_{\mathrm{new}} := \frac{18\xi_{\mathrm{new}} + 7}{18\xi_{\mathrm{new}} + 9} = 0.874957019420098946\ldots.$$

Every finite-order Hecke L-function over F, and every Dirichlet L-function, including ζ, is zero-free for $\operatorname{Re} s > B_{\mathrm{new}}$*; principal poles are allowed.*

The separate scalar optimality statement is Theorem 34.4, following the definition of the set $\mathscr{A}$ of admissible parameter choices in Section 34.2.

We retain the original modified average (7.2), the two unweighted factors of (9.4) and the sixth-power inverse amplification (9.3), with $\alpha = 5/6$. The notation is

$$\delta = 2a - 1, \quad x = \bar{g}/\delta, \quad b = l_y - l_x, \quad h = 1 - l_x + \ell;$$

t denotes the detector cutoff and ξ parametrizes the length choices. The exact scalar verifier and the consolidated Lean formalization are available in the accompanying repository [Liu26].

28. Extending the unweighted fourth moment below three quarters

In this section, we extend the proof of [Ope26a, Lemma 18.1] from $[3/4, 1]$ to $[13/18, 1]$; its printed statement alone does not give this range. Proposition 28.1 states the extended estimate (28.2), for the parameter range (28.1) and under its stated coefficient and zero-free hypotheses. This is the analytic estimate needed for the variable-parameter count in Section 29, with its zero-free hypothesis supplied at each step by the preceding bound.

Proposition 28.1 (Extended unweighted product moment)**.** *Retain the notation, character families, fixed twists, common coprimality conditions and bounded length ranges of (9.4). In that notation, n_1, n_2 are the ideal-length exponents, $z_{\mathrm{p}} = \sum_i z_i$ is the total prime-length exponent, and $M = m_0 + q_0$ is the combined range exponent of the moment estimate, where m_0 bounds the element range and q_0 bounds the radical of its variable residue-symbol factors. This M is distinct from the total averaging-length exponent $l_x + l_y$ used in Section 22. The excluded group $\Theta = \langle \eta, \widehat{\mathcal{T}} \rangle$ is the finite group of Hecke characters defined in Section 4.3. The prime polynomials have disjoint underlying prime supports*

before imposing coprimality restrictions and extending the characters by zero. Then (9.4) *holds uniformly for*

$$\frac{13}{18} \le \kappa \le 1, \tag{28.1}$$

in the cases

(28.2)

	length condition	*excluded inducing characters*
$z_{\mathrm{p}} = 0$	n_1, n_2 *arbitrary bounded*	*principal*
$z_{\mathrm{p}} > 0$	$n_1 + n_2 + 6\kappa z_{\mathrm{p}} \le M$	Θ.

If $z_{\mathrm{p}} > 0$ *and* $\kappa < 1$, *assume* $\beta_* \le (1+\kappa)/2$. *The upper bound* ℓ_{mesh} *on individual prime lengths depends only on the bounded real ranges and* ε, *not on the number of prime-polynomial factors. Constants and finite smooth-seminorm and polynomial-height orders may depend on the fixed number of prime-polynomial factors and arithmetic data; they are uniform over the character families, moduli, common coprimality restrictions and fixed outer summation parameters in the prescribed ranges.*

Proof. Put $z = z_{\mathrm{p}}$, the total prime-length exponent, and $A = n_1 + n_2 + z$, the sum of the two ideal-length exponents and z. Here A is a real exponent, unrelated to the primary element A in Section 4. We check the κ-dependent steps of the induction on the combined range exponent in [Ope26a, Section 18]. In these sums, each character remains zero on ideals divisible by primes in its defining modulus or the fixed set S, including primes absent from its primitive conductor. Cancelling residue-symbol factors on ideals coprime to these primes does not remove their displayed prime support from the bound Z^{q_0} in (9.4); compare (2.3). The induction without auxiliary prime-polynomial factors is unchanged.

Prime-sum bounds and coprimality restrictions. For $\kappa < 1$ put $s_\kappa = (1+\kappa)/2$. By [Ope26a, Lemma 4.9], whose real boundary ranges over $[1/2, 1]$,

$$s_\kappa \ge \beta_* \quad \Longrightarrow \quad \frac{L'}{L}(s_\kappa + e + it, \psi) \ll_e \log\bigl[2Q_\psi(3+|t|)^2\bigr].$$

For each character $\vartheta \in \Theta$ occurring in the prime coefficients,

$$\psi \notin \Theta, \quad \psi\vartheta \text{ principal} \quad \Longrightarrow \quad \psi = \vartheta^{-1} \in \Theta,$$

a contradiction. Moving the Mellin contour for the smoothly weighted von Mangoldt sum to $s_\kappa + e$, followed by division of its weight by $\log(Py)$, gives the prime bound. Prime powers and bounded values of P are admissible since

$$s_\kappa \ge 31/36, \qquad P^{1/2+\varepsilon_1} \le P^{s_\kappa+\varepsilon_1} \quad (P \ge 1).$$

Dividing each prime sum of norm length P_i by $P_i^{1/2}$, as in the definition of Q_k in (9.4), gives

$$|Q_{\boldsymbol{\omega}}|^2 \le C Z^{\kappa z+\varepsilon_1} p_j(\boldsymbol{W})^c (1+|\boldsymbol{\omega}|)^h. \tag{28.3}$$

Here p_j is a finite smooth seminorm, $\boldsymbol{\omega}$ comprises normalized norm-twist heights, and j, c, h are finite orders. The contour loss is $2ez$ with bounded total z; derivatives of the test functions or the normalized norm twists affect only these finite seminorm and height orders. For $\kappa = 1$ use absolute prime counting.

Exact removal of the coprimality restrictions and application of the functional equation are independent of κ [Ope26a, Section 18]. Removing the additional coprimality restrictions from the two unweighted ideal sums $S_\psi(n_1)$ and $S_\psi(n_2)$ in (9.4) produces divisor terms with coefficients of absolute value at most $q_d^{-1/2}$. A prime sum with an additional coprimality restriction equals the same sum without that restriction minus its terms at primes dividing the coprimality modulus; the original zero extensions of the character are

retained. The resulting sums of inverse square roots of norms are $O(Z^{\varepsilon_1})$ with arbitrary fixed loss. The retained lengths satisfy

$$A' \le A, \quad z' \le z, \qquad A' + (6\kappa - 1)z' \le A + (6\kappa - 1)z \le M.$$

If $z' = 0$, use the estimate without auxiliary prime-polynomial factors or additional coprimality restrictions. Applying the functional equation preserves $N\mathfrak{f}\, N\mathfrak{r} \ll Z^M$, where $\mathfrak{f}$ is the primitive conductor and $\mathfrak{r}$ is the product, each prime counted once, of primes in the defining modulus that do not divide $\mathfrak{f}$. This bound is uniform in the varying arithmetic parameters. It remains to prove the estimate without additional coprimality restrictions.

Small ranges and comparison of lengths. Let $\rho > 0$ be the lower threshold for the range exponent and $\epsilon_{\mathrm{fr}} > 0$ the allowance for enlarging the supports, independent of the zero-class δ. At $M \le \rho$, absolute counting introduces an exponent loss of at most $\rho + \epsilon_{\mathrm{fr}}$ for the enlarged sum without auxiliary prime-polynomial factors. For $z > 0$,

$$A \ge z, \qquad 6\kappa z \le M, \qquad z \le \frac{3M}{13}, \qquad \kappa z \le \frac{M}{6}. \tag{28.4}$$

Thus the additional exponent loss in this initial range is $\le \rho/6$; the original paper's bound $z \le 2M/9$ is not needed.

For $M > \rho$, first prove the estimate when $A \le 5M/6$. If $A > 5M/6$, set $L = M/4$. Then

$$6\kappa - 1 \ge 10/3, \qquad z < \frac{M}{6(6\kappa - 1)} \le \frac{M}{20}, \qquad A - z > \frac{47M}{60} > 2L.$$

If either original unweighted length exponent n_1 or n_2 is below L, apply the functional equation to the other factor. Otherwise subtract the product of character sums with length exponents $L, A - z - L$, as in (9.5). Estimate this comparison product separately by applying the functional equation to its longer factor. Applying the functional equation to the resulting difference of the two products, including truncation and the fixed enlargement factors for the two length ranges, with aggregate error ξ_{ref} gives

$$A_{\mathrm{comp}} \le \frac{3}{2}M - A + 2z + \xi_{\mathrm{ref}}. \tag{28.5}$$

Hence

$$\begin{aligned} A_{\mathrm{comp}} &\le \frac{23}{30}M + \xi_{\mathrm{ref}}, \\ A_{\mathrm{comp}} + (6\kappa - 1)z &\le \frac{5}{2}M - 2A + 2z + \xi_{\mathrm{ref}} \le \frac{14}{15}M + \xi_{\mathrm{ref}}. \end{aligned} \tag{28.6}$$

Both upper bounds lie below M by at least $M/15 - \xi_{\mathrm{ref}} > 0$ if $\xi_{\mathrm{ref}} \le \rho/30$. For $z = 0$ the gap below M is $M/6 - \xi_{\mathrm{ref}}$. The comparison product is therefore covered by the estimate already proved for smaller total ideal length at the same range exponent M, before taking differences of products.

Poisson summation and reduction of the range. The first Poisson transform, complete support extraction, sixth-power amplification of the Gauss sums, second complete correlation and the full Möbius expansion, followed by inclusion–exclusion for divisibility among the factors, in [Ope26a, Section 18] are independent of κ. Retain the difference of the two products of test functions in (9.5), including its minus sign, until the squared absolute value is formed. The common character, coprimality condition and norm power on each separated side are preserved by [Ope26a, Lemma 18.2]; all displayed moving primes remain counted, and the fixed coprimality restrictions remain common to both unweighted ideal variables and both rectangles. The transformed range exponents satisfy

$$M' = M + J - g - g_2 + t_2 \le M - \sigma, \qquad 0 \le M'_{\mathrm{act}} - M' \le \delta_{\mathrm{fr},2}.$$

Here J is the first-transform length defect, g, g_2 the two complete-support savings, t_2 the second common-support length, and $\sigma > 0$ the decrease in the range exponent, as in [Ope26a, Section 18].

After bounding the norms of the retained Fourier frequencies, expanding the divisibility conditions among the factors, truncating and removing the coprimality restrictions,

$$\begin{aligned} F_{\mathrm{act}} &= (A_{\mathrm{act}} - M'_{\mathrm{act}} + (6\kappa - 1)z_{\mathrm{act}})_+ \\ &\le 6\Delta_{\mathrm{child}} + \delta_{\mathrm{fr},1} + 2\theta_N, \qquad z_{\mathrm{act}} \le z. \end{aligned}$$

Here $\Delta_{\mathrm{child}} \ge 0$ is the support loss in the transformed sum and θ_N is the total error in its length exponents, as defined in the transformed-length estimates of [Ope26a, Section 18]; the inequality uses only the original length condition and $6\kappa - 1 \ge 0$. Apply the pointwise bound (28.3) to whole prime-polynomial factors until the remaining product satisfies the length condition or has no such factors. Removing length d lowers the affine expression by $6\kappa d$ and introduces an exponent loss κd. With $\eta_{\mathrm{mesh}} = \ell_{\mathrm{mesh}}$, if the removed lengths reach $F_{\mathrm{act}}/(6\kappa)$, the first partial sum doing so exceeds that threshold by at most η_{mesh}. Otherwise all factors are removed and $d_z = z_{\mathrm{act}} < F_{\mathrm{act}}/(6\kappa)$. Thus

$$d_z \le \min\left\{ z_{\mathrm{act}}, \frac{F_{\mathrm{act}}}{6\kappa} + \eta_{\mathrm{mesh}} \right\}, \qquad \kappa d_z \le \frac{F_{\mathrm{act}}}{6} + \eta_{\mathrm{mesh}}. \tag{28.7}$$

If all prime-polynomial factors have been estimated pointwise, apply the previously proved estimate without prime-polynomial factors at the smaller range exponent; otherwise apply the estimate with the remaining prime-polynomial factors and the restored length condition. The remaining factors retain their coefficients and disjoint supports. Both frequency errors are included before removing prime-polynomial factors, and Cauchy–Schwarz averages the errors in the two transformed sums.

For terms whose primitive inducing character belongs to the finite group Θ, use the same counting estimate as in [Ope26a, Section 18] and [Ope26a, Lemma 18.3]: the original uncentered sum satisfies $A \le M$, and the centered main terms cancel by the identity $X_1X_2 = Y_1Y_2$ in (9.5). The threshold $5M/6$ is independent of κ.

Uniform parameter choices and finite induction. From the bounded ranges and ε, choose $\rho, \sigma, \epsilon_{\mathrm{fr}} > 0$ with $\epsilon_{\mathrm{fr}} \ll \min(\rho, \sigma)$ and

$$T_{\mathrm{term}} := \rho + \epsilon_{\mathrm{fr}} + \rho/6 + 3\sigma + 5\sigma/3 < \varepsilon/4, \qquad D = 2 + \lceil 2M_{\max}/\sigma \rceil.$$

Since $0 < 6\kappa - 1 \le 5$,

$$|\Delta(A - M + (6\kappa - 1)z)| \le |\Delta A| + |\Delta M| + 5|\Delta z|.$$

A numerical stage constant C_* covers these fixed aggregate operations. Before specifying the number of prime-polynomial factors, choose

$$\begin{aligned} \xi_{\mathrm{ref}} &< \min\{\epsilon_{\mathrm{fr}}/2, \rho/30, \sigma/(4C_*), \varepsilon/(16C_*D)\}, \\ \eta_{\mathrm{mesh}} &< \min\{\sigma/6, \varepsilon/(16C_*D)\}, \qquad \varepsilon_0 < \varepsilon/(16C_*D). \end{aligned}$$

Now fix the family of prime-polynomial factors and its aggregate support bound H_N; impose $\log Z \gg H_N/\xi_{\mathrm{ref}}$ and the thresholds for applying the functional equation and localizing the sums. The losses assigned to the local divisor and prime estimates sum to ε_0 per stage. These losses and thresholds may depend on N, while η_{mesh} does not. For the simultaneous induction over surviving subsets, take

$$\mathcal{E}_d = T_{\mathrm{term}} + (d+1)C_*(\eta_{\mathrm{mesh}} + \xi_{\mathrm{ref}} + \varepsilon_0), \quad 0 \le d \le D-2, \qquad \mathcal{E}_{D-2} < \varepsilon.$$

First establish the estimates without auxiliary prime sums for all range intervals. For estimates involving prime sums, each strict reduction of the range decreases its exponent

by at least $\sigma/2$, while comparisons at a fixed range use only the preceding uncentered estimate. Every application of induction therefore uses an estimate already proved.

Choose the seminorm and height orders needed for the integral kernels, functional-equation transforms, prime sums and initial estimates, working backwards through the D induction steps; then choose the orders for the outer Fourier tails. Full weighted Fourier integrals retain the height factors of (28.3): larger orders raise seminorms and constants, not powers $Z^{J\xi_{\mathrm{ref}}}$. The coprimality moduli remain of norm bounded by a fixed power of Z through these finitely many steps.

This proves the estimate without additional coprimality restrictions; the initial removal of those restrictions gives the proposition. □

Zero-free hypothesis. The required bound $\beta_* \le (1+\kappa)/2$ must precede a smaller fixed κ. It may instead hold by equality when $\kappa = 2\beta_* - 1$ in a contradiction argument. No assertion is made below $\kappa = 13/18$.

29. Counting by simultaneous large values with a variable moment parameter

In this section, we combine the extended moment of Proposition 28.1 with the inverse moment (9.1), its amplified form (9.3) and the simultaneous large values (5.5). Keeping κ variable yields the count exponent $R_\kappa(\delta, x)$ in (29.2), from which Section 30 derives the improved length parameters. Section 34.1 also proves that the counting exponent at the critical class cannot be improved using the specified moment estimates and detecting lengths.

Recall that $\delta = 2a - 1$ comes from (5.1), and $\bar g$ is the weighted mean in (10.1); the variable d satisfies $q_u \asymp Z^d$. Put $x = \bar g/\delta \in [0, 1/2]$ and

$$c = \frac{2}{3\kappa}, \qquad D_{\kappa,x} = 3 - (1+c)x, \qquad P_{\kappa,x} = (2 - cx)(1 - x). \tag{29.1}$$

At $\kappa = 3/4$, $(D_{\kappa,x}, P_{\kappa,x}) = (D_x, P_x)$. The detecting cutoff is t; d remains the element-norm exponent.

Proposition 29.1 (Variable-parameter count)**.** *Retain the detector, character-family, coefficient, smooth-weight and height hypotheses of Proposition 18.1, with the simultaneous large values (5.5) for one common character. Assume*

$$13/18 \le \kappa \le 1, \qquad \beta_* \le (1+\kappa)/2, \qquad \ell/d \ge 1/5 + \rho_{\mathrm{sup}}, \quad \rho_{\mathrm{sup}} > 0.$$

For $0 < \delta \le \alpha = 5/6$ and $0 \le x \le 1/2$, put

$$\mathcal{J}_\kappa = (\alpha - \delta)D_{\kappa,x} + \delta P_{\kappa,x}, \quad t_\kappa = 1 + \frac{\delta P_{\kappa,x}}{2\mathcal{J}_\kappa}, \quad R_\kappa = 1 - \delta + \frac{(\alpha-\delta)\delta P_{\kappa,x}}{2\mathcal{J}_\kappa}. \tag{29.2}$$

For each $\varepsilon > 0$, sufficiently fine fixed prime lengths give

$$\#\mathcal{B} \ll U^{R_\kappa + \varepsilon}(1 + T_1)^A.$$

Here T_1 is the zero-detection height cutoff of Section 5, and $A \ge 0$ is a fixed finite height exponent for the prescribed moment parameters and arithmetic data.

Proof. Apply Proposition 28.1 and the inverse moment (9.1) to the same character and the pair of large values in (5.5). Here r, m are the norm-length exponents of the two detecting polynomials of Section 5, and we set $m = t - r$ in the exponent comparison. The maximal selected prime-length exponents permitted by the inverse and unweighted moments are

$$z_M = (1 - r)/2, \qquad z_S = (1 - 2m)/(6\kappa).$$

The savings divided by δ are

$$F_I(r) = x + (1-x)r, \qquad F_S(r) = 2(t-r) + \frac{c}{2}x\big[1 - 2(t-r)\big].$$

Their crossing is

$$r_\kappa(t,x) = \frac{(2-cx)t + (c/2-1)x}{3-(1+c)x}. \tag{29.3}$$

For $1 \le t \le 3/2$ and $0 \le x \le 1/2$,

$$\partial_t r_\kappa = \frac{2-cx}{D_{\kappa,x}} > 0, \qquad \partial_x r_\kappa = \frac{(2-c)(2t-3)}{2D_{\kappa,x}^2} \le 0,$$

$$\partial_t(t-r_\kappa) = \frac{1-x}{D_{\kappa,x}} > 0, \qquad \partial_x(t-r_\kappa) = -\partial_x r_\kappa \ge 0,$$

$$r_\kappa(3/2,x) = 1, \qquad \frac{6-c}{10-2c} \le r_\kappa \le 1, \qquad \frac{1}{3} \le t - r_\kappa \le \frac{1}{2}.$$

Use the inverse moment for $r \ge r_\kappa$ and the unweighted moment for $r \le r_\kappa$. The admissible prime lengths satisfy

$$z_M \le \frac{4-c}{20-4c} < \frac{1}{5}, \qquad z_S \le \frac{c}{12} < \frac{1}{5}.$$

Decrease each selected length by a fixed $\nu > 0$ and choose the individual prime lengths sufficiently small that selecting whole prime factors introduces a rounding loss smaller than ε. Then

$$1 - r - 2z \ge 2\nu,$$

$$3 - 2r - 8z = 4(1-r-2z) + (2r-1) \ge 8\nu + \frac{1}{5-c} > 0.$$

If no positive prime length remains available, use the moments without selected prime factors.

Substitution of (29.3) gives the short count

$$R_{\kappa,\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_{\kappa,x}}{D_{\kappa,x}}(3/2 - t).$$

The long count and the balancing identities are

$$R_{\mathrm{long}}(t) = 1 - \alpha + (\alpha - \delta)t,$$

$$R_{\kappa,\mathrm{short}}(t_\kappa) = R_{\mathrm{long}}(t_\kappa) = R_\kappa,$$

$$D_{\kappa,x} - P_{\kappa,x} = 1 + x - cx^2 > 0, \qquad 1 \le t_\kappa \le 3/2.$$

At $\delta = \alpha$, $t_\kappa = 3/2$ and $R_\kappa = 1 - \alpha$; use the amplified inverse estimate without selected prime factors, as in Proposition 18.1. All coefficient classes, presentations and smooth separations are unchanged. □

30. Balanced averaging lengths and the critical counting class

In this section, we choose the averaging lengths by balancing the direct estimate of Proposition 16.2 with the critical class of the count in Proposition 29.1. The resulting one-parameter family is defined in (30.2), on the interval (30.1), and contains the Part III parameter choice (21.2); Lemma 31.1 verifies its endpoint inequality for all zero and amplitude classes.

Recall that $b = l_y - l_x$, $\ell = \sum_i \ell_i$, and $h = 1 - l_x + \ell$. The balanced relation $l_x + l_y + \ell = 1$ and the direct comparison give $B = 11/12 - \ell/4$, as in (15.2). For the parameter ξ in

$$I = [38858/100000, 38859/100000] \tag{30.1}$$

define

$$\kappa(\xi)=\frac{18\xi^2-57\xi+20}{6(36\xi^2-78\xi+25)},\qquad \ell(\xi)=\frac{5-6\xi}{9+18\xi},\qquad B(\xi)=\frac{18\xi+7}{18\xi+9},$$
$$b(\xi)=\frac{8(3\xi-1)(12\xi^2-40\xi+15)}{9(2\xi+1)(24\xi^2-20\xi+5)},$$
$$l_x(\xi)=\frac{1-b(\xi)-\ell(\xi)}{2},\qquad l_y(\xi)=\frac{1+b(\xi)-\ell(\xi)}{2},\qquad h(\xi)=\frac{1+b(\xi)+3\ell(\xi)}{2},$$
$$C_\xi(s)=s-\frac{4+b(\xi)}{6},\qquad B(\xi)=\frac{11}{12}-\frac{\ell(\xi)}{4}.$$

(30.2)

At fixed ξ, write b,ℓ,κ,h for $b(\xi),\ell(\xi),\kappa(\xi),h(\xi)$. For $\xi_0=(49-\sqrt{921})/48$, (30.2) recovers (21.2) and $\kappa(\xi_0)=3/4$.

Lemma 30.1 (Critical class and balanced parameters)**.** *For the length parameters* (30.2)*, at* $(\delta,x)=(\xi,1/2)$*,*

$$R_\kappa=2/3,\qquad E_{B(\xi)}(h)=0,\qquad \partial_\delta E_{B(\xi)}(h)=0.$$

The equation $R_\kappa(\delta,1/2)=2/3$ *is equivalent to*

$$(216\kappa-18)\delta^2-(468\kappa-57)\delta+150\kappa-20=0. \tag{30.3}$$

The solution $\delta=\xi\in I$ *is its smaller root.*

Proof. At $x=1/2$, $D=(5-c)/2$, $P=1-c/4$, and

$$R_\kappa-\frac{2}{3}=\frac{(216\kappa-18)\delta^2-(468\kappa-57)\delta+150\kappa-20}{216\kappa\mathcal{J}_\kappa}.$$

Thus (30.3) gives $\kappa(\xi)$. The sum of its roots exceeds two; since $\xi<39/100$, the other root exceeds one. Substitution in (18.5) gives

$$R_\kappa(\xi,1/2)=2/3\quad\Longrightarrow\quad \partial_b E_B(h)=R_\kappa/2-1/3=0,$$
$$E_B(h)=0\quad\Longleftrightarrow\quad \ell=\frac{5-6\xi}{9+18\xi},$$
$$\partial_\delta E_B(h)=0\quad\Longleftrightarrow\quad b=\frac{8(3\xi-1)(12\xi^2-40\xi+15)}{9(2\xi+1)(24\xi^2-20\xi+5)}.$$

The uniform sign is proved below. □

31. A uniform proof of the endpoint inequality

In this section, we prove the endpoint inequality uniformly along the family (30.2): the inequality to be proved is (31.1), including the critical class of Lemma 30.1. The polynomial identity (31.2) and the auxiliary bounds of Lemma 31.2 supply the scalar estimates needed for Proposition 32.1.

Lemma 31.1 (Variable-parameter endpoint)**.** *For every* $\xi\in I$*, with* I *defined in* (30.1)*, use* (30.2) *and the count exponent* $R=R_{\kappa(\xi)}$ *of* (29.2)*. Let* $E_B(d)$ *be the relative exponent in* (18.4)*. Then*

$$E_{B(\xi)}(h)\le 0\qquad (0\le\delta\le 5/6,\quad 0\le x\le 1/2). \tag{31.1}$$

Equality holds only at $(\delta,x)=(\xi,1/2)$*.*

Proof. Put $y = 1/2 - x$, $c = 2/(3\kappa(\xi))$, and

$$D = (5-c)/2 + (1+c)y, \qquad P = 1 - c/4 + 2y + cy^2, \qquad \mathcal{J} = (\alpha-\delta)D + \delta P.$$

Write $E = E_{B(\xi)}(h)$. By (18.5),

$$E = C_0 + (1/2+\ell)\delta + \ell x\delta - h(1-R_\kappa), \qquad C_0 = -1/4 + b/6 + 5\ell/4.$$

Hence

$$Q = -2\mathcal{J}E = -2C_0\mathcal{J} + (b+2\ell y)\delta\mathcal{J} - h(\alpha-\delta)\delta P. \tag{31.2}$$

Write $Q = A(y)\delta^2 + B_q(y)\delta + C_q(y)$ and $\mathscr{D}(y) = 4A(y)C_q(y) - B_q(y)^2$. Use the following polynomial coefficients, with distinct notation from the prime cutoff P_0 and height cutoff T_1:

$$\mathsf{D}_0 = (5-c)/2, \quad \mathsf{D}_1 = 1+c, \quad \mathsf{P}_0 = 1-c/4, \quad \mathsf{T}_0 = -3/2 + c/4, \quad \mathsf{T}_1 = 1-c, \quad \mathsf{T}_2 = c.$$

The coefficients are

$$\begin{aligned}
A_0 &= b\mathsf{T}_0 + h\mathsf{P}_0, & A_1 &= b\mathsf{T}_1 + 2\ell\mathsf{T}_0 + 2h,\\
A_2 &= bc + 2\ell\mathsf{T}_1 + hc, & A_3 &= 2\ell c,\\
B_0 &= -2C_0\mathsf{T}_0 + \alpha\mathsf{D}_0 b - \alpha h\mathsf{P}_0,\\
B_1 &= -2C_0\mathsf{T}_1 + \alpha\mathsf{D}_1 b + 2\alpha\ell\mathsf{D}_0 - 2\alpha h,\\
B_2 &= -2C_0 c + 2\alpha\ell\mathsf{D}_1 - \alpha hc,\\
C_{q,0} &= -2C_0\alpha\mathsf{D}_0, & C_{q,1} &= -2C_0\alpha\mathsf{D}_1.
\end{aligned}$$

The coefficients d_j of $\mathscr{D}$ are

$$\begin{aligned}
d_0 &= 4A_0C_{q,0} - B_0^2 = 0,\\
d_1 &= 4(A_0C_{q,1} + A_1C_{q,0}) - 2B_0B_1,\\
d_2 &= 4(A_1C_{q,1} + A_2C_{q,0}) - B_1^2 - 2B_0B_2,\\
d_3 &= 4(A_2C_{q,1} + A_3C_{q,0}) - 2B_1B_2,\\
d_4 &= 4A_3C_{q,1} - B_2^2.
\end{aligned}$$

The identity $d_0 = 0$ and the following strict bounds hold on I:

quantity	lower bound	quantity	lower bound
A_0	47/100	d_1	41/1000
A_1	121/100	d_2	198/1000
A_2	86/100	d_3	267/1000
A_3	29/100	d_4	76/1000

For $F = N/D$, cancel common factors and normalize the common sign. For a polynomial of degree n, write

$$N(L + (U-L)z) = \sum_{j=0}^{n} a_j z^j, \qquad \mathfrak{b}_i(N) = \sum_{j=0}^{i} a_j \frac{\binom{i}{j}}{\binom{n}{j}},$$

$$L = 38858/100000, \qquad U = 38859/100000.$$

The Bernstein identity (19.3) gives $N(L+(U-L)z) = \sum_i \mathfrak{b}_i(N)\mathsf{B}_{i,n}(z)$. For every table entry, $\mathfrak{b}_i(N), \mathfrak{b}_i(D) > 0$, so

$$F(\xi) \geq \frac{\min_i \mathfrak{b}_i(N)}{\max_i \mathfrak{b}_i(D)}.$$

The exact quotients exceed the table bounds; their coefficients and (31.2) are verified by `combine_v2-2_supplement/check_kappa_endpoint.py`. Therefore

$$A(y) > 0 \quad (y \ge 0), \qquad \mathscr{D}(y) > 0 \quad (y > 0),$$
$$4A(y)Q = (2A(y)\delta + B_q(y))^2 + \mathscr{D}(y) \ge 0, \qquad \mathcal{J} > 0 \implies E \le 0,$$
$$Q(\delta, 0) = A_0(\delta - \xi)^2.$$

Thus equality holds precisely at $(\delta, x) = (\xi, 1/2)$. □

Lemma 31.2 (Uniform positive bounds in the auxiliary inequalities)**.** *All parameter choices in* (30.2) *satisfy*

$$b > 0, \quad 0 < \ell < 1/5, \quad l_x - b - \ell > 0, \quad l_y - \ell > 11b/6,$$
$$h < 1, \qquad 5\ell > h, \qquad 437/500 < B < 7/8.$$

The inequality $l_x - b - \ell > 0$ is stronger than $l_x - \ell > 0$, since $b > 0$. The estimates for the lowest zero class $a = a_0$ of (5.1) *and the intermediate and small norm ranges of Lemma 19.2 have uniform positive exponent savings; the detecting and rounded count formulas have uniformly positive denominators.*

Proof. The exact bounds for the Bernstein coefficients give the lower bound $1/25000$ on I for each expression in

$$\begin{gathered}
36\xi^2 - 78\xi + 25, \quad \kappa', \quad B', \\
\kappa' - 1000B', \quad \kappa - 13/18, \quad 4/5 - \kappa, \quad -p', \\
b, \quad \ell, \quad 1/5 - \ell, \quad l_x - b - \ell, \quad l_y - \ell, \\
l_y - \ell - 11b/6, \quad 1 - h, \quad 5\ell - h, \\
B - 43/50, \quad B - 437/500, \quad 7/8 - B, \\
-C_0 - (1/2 + 3\ell/2)/50, \\
l_y/2 - 13h/75 - 1/50 - (7/8 - B), \\
-\big[(225\ell + 50 - 75b)\delta + 78\ell - 49b - 73\big]/300 \quad (\delta = 1/50, 19/25), \\
c, \quad 1 - c, \quad 1/5 - (4 - c)/(20 - 4c), \quad 1/5 - c/12, \\
(\alpha - 19/25)(5 - c)/2 + (19/25)(1 - c/4) - 3/5, \\
3 - (1 + c)51/100 - 2, \qquad (2 - c51/100)(49/100) - 3/4.
\end{gathered} \tag{31.3}$$

Here $p(\xi) = 1188\xi^3 - 2160\xi^2 + 171\xi + 190$. The 29 side bounds and eight endpoint bounds are recorded in the exact checker. Put

$$m_F = -C_0 - (1/2 + 3\ell/2)/50,$$
$$m_S = l_y/2 - 13h/75 - 1/50,$$
$$m_I(\delta) = -\big[(225\ell + 50 - 75b)\delta + 78\ell - 49b - 73\big]/300.$$

Then

$$m_F > 1/25000, \qquad m_S - (7/8 - B) > 1/25000,$$
$$m_I(1/50), m_I(19/25) > 1/25000 \implies m_I(\delta) > 1/25000 \quad (1/50 \le \delta \le 19/25),$$

since m_I is affine. These are the exponent savings for the lowest zero class and the small and intermediate norm ranges of Lemma 19.2. The initial bound and small fixed class widths give

$$\delta \le 2\beta_* - 1 + O(\epsilon_{\mathrm{b}}) \le 3/4 + O(\epsilon_{\mathrm{b}}) < 19/25.$$

On the enlarged rounding box $0 \le x \le 51/100$, (31.3) gives $D > 2$, $P > 3/4$, and $D - P = 1 + x - cx^2 > 0$. Hence

$$\mathcal{J} = (\alpha - \delta)D + \delta P \ge 2(\alpha - \delta) + (3/4)\delta \ge 43/60 > 3/5,$$

$$3/2 - t_\kappa = \frac{(\alpha - \delta)D}{2\mathcal{J}} \ge \frac{\alpha - 19/25}{2\alpha} = 11/250 > 0.$$

The rational expressions have uniformly bounded derivatives on this compact box, so fixed rounding and class errors are absorbed in the prescribed losses. The detecting cutoff retains its original range. □

32. The one-step zero-free bound via the continuation criterion

In this section, we combine Proposition 29.1 and the uniform endpoint inequality of Lemma 31.1 into one application of Proposition 3.1. Proposition 32.1 gives the new bound $\beta_* \le B(\xi)$ under the preceding-bound hypothesis (32.1); Section 33 repeatedly applies this implication.

Here I is the interval (30.1), and $B(\xi), \kappa(\xi)$ are the functions defined in (30.2); β_* retains its definition (2.3). Fix $\xi \in I$, $\kappa(\xi) \le 3/4$, and use (30.2). The zero-free hypothesis required for the moment estimate is

$$\beta_* \le \frac{1 + \kappa(\xi)}{2}. \tag{32.1}$$

Theorem 13.1 also gives $\beta_* \le 7/8$.

Proposition 32.1 (One step of the iterative improvement)**.** *For $\xi \in I$ and $\kappa(\xi) \le 3/4$,*

$$\beta_* \le \frac{1 + \kappa(\xi)}{2} \quad \Longrightarrow \quad \beta_* \le B(\xi).$$

Proof. Assume (32.1) and $\Delta := \beta_* - B > 0$. Retain the modified average (7.2), with the normalization (17.6), and the coefficients, zero extensions, disjoint supports and finite character expansions of Part I.

The direct estimate. For $d_{\mathcal{I}} = \sum_{i \in \mathcal{I}} \ell_i \le \ell$, Lemma 31.2 gives

$$l_x - d_{\mathcal{I}} > 0, \qquad l_y - d_{\mathcal{I}} - 11b/6 > 0,$$

$$M' = 1 - \ell - 2d_{\mathcal{I}} \ge 1 - 3\ell > 0, \qquad P_a = q_{b_*}^{-1} Z^b, \qquad P_a^2 / Y' \ll P_a^{1/6}.$$

Thus $Q, Y', P_a \ge 1$ for large Z, and the functional-equation and Gram estimates apply to every rescaled subset. The resulting change in the exponent is

$$-d_{\mathcal{I}} + (d_{\mathcal{I}} + 5\ell - 1)_+ /8 \le -7d_{\mathcal{I}}/8 \le 0.$$

Proposition 16.2, with the positive normalization factor for an even number of prime factors, yields

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{C_b(B)+\varepsilon}, \qquad C_b(B) = l_x/2 + b/12, \qquad C_b(s) = s - (4+b)/6. \tag{32.2}$$

Small norms. Take $(s_r, w_r, z_r) = (\beta_* + \epsilon_{\mathrm{b}}, 1/2, 17/50)$, and fix the first-domain gap ε_0 in (4.11) with $0 < \varepsilon_0 < B - 1/2$. For $p \nmid u$, the error exponents in (17.5) are

$$-s_r, \quad -51/25, \quad 49/25 - 5s_r, \quad -77/50 < 0.$$

For $p \mid u$, the boundary exponents are

$$-1/2, \quad 3/2 - 2s_r, \quad (3/2 - 2s_r, 2 - 3s_r), \quad 2 - 3s_r, \quad 3 - 5s_r < 0.$$

Since $s_r \ge B > 437/500$, the quotient-free formula gives

$$G_p \ll \begin{cases} 1, & p \nmid u, \\ q_p^{1/2}, & p \mid u, \end{cases} \qquad \text{full tuple sum} \ll U^\varepsilon Z^{17\ell/50}.$$

The numerator bound, the $O(U)$ choices of u and the Mellin factor contribute $U^{3/5+\varepsilon}UU^{-17/50} = U^{63/50+\varepsilon}$. Hence, for $d \le d_{\min} := 1/100$, the relative exponent is at most

$$\frac{13h}{75} - \frac{l_y}{2} + 2d_{\min} + O(\epsilon_{\mathrm{b}} + \varepsilon) = -m_S + O(\epsilon_{\mathrm{b}} + \varepsilon), \qquad m_S := \frac{l_y}{2} - \frac{13h}{75} - \frac{1}{50} > 0.$$

This includes terms with u a unit and terms for which the character in the denominator L-function is principal; the character in the numerator L-function is nonprincipal. Lemma 31.2 gives $m_S > 7/8 - B$.

The lowest zero class and intermediate norms. The lowest zero class has

$$a_0 = 51/100, \quad \delta_0 = 1/50, \quad R = 1, \quad \bar{g} \le \delta_0/2, \quad E_B(h) \le -m_F < 0,$$

where $m_F = -C_0 - (1/2 + 3\ell/2)/50$ and $C_0 = -1/4 + b/6 + 5\ell/4$. Its positive d-slope controls $d \le h$. For $d_{\min} \le d \le 1/2$, the unchanged unselected count gives

$$R = 76/75 - 2\delta/3, \qquad \partial_d E_B = 101/150 - \delta/6 > 0,$$

$$E_B(d) \le E_B(1/2) \le \frac{(225\ell + 50 - 75b)\delta + 78\ell - 49b - 73}{300} < 0.$$

The last affine expression is negative at $\delta = 1/50, 19/25$ by Lemma 31.2; all actual classes satisfy

$$\delta \le 2\beta_* - 1 + O(\epsilon_{\mathrm{b}}) \le 3/4 + O(\epsilon_{\mathrm{b}}) < 19/25 < \alpha.$$

Moderate and large norms. For $1/2 \le d \le h$, Proposition 29.1 uses the same character and the simultaneous large values (5.5) in both moment estimates. The length selected on the appropriate side of the crossing satisfies

$$z_{\mathrm{sel}} < 1/5 < \ell/h.$$

A fixed length decrement, followed by a sufficiently fine partition into prime intervals, preserves the strict length inequalities for the inverse moment and the original coefficients and supports. Lemma 31.1 gives

$$E_B(h) \le 0, \qquad 0 < 33/50 - \delta/2 \le \partial_d E_B = R + \delta/2 - 17/50 < 2.$$

Thus $E_B(d) \le 0$ for $d \le h$. Choose $\zeta > 0$ with $\ell/(h+\zeta) > 1/5$; extending to $h + \zeta$ increases the exponent by at most 2ζ. On $(s_r, w_r, z_r) = (2, 2, z_{\mathrm{new}})$,

$$\sum_{U > Z^{h+\zeta}} |\mathcal{R}_\eta(U; Z)| \ll Z^{l_x/2+1+l_y+(h+\zeta)(1+\varepsilon)-\zeta z_{\mathrm{new}}+\varepsilon}.$$

Choose $z_{\mathrm{new}} > 2$ after ζ to obtain any prescribed saving. For the moderate norm ranges, measuring the error relative to $C_b(\beta_*)$ rather than $C_b(B)$ subtracts Δ, as in Proposition 18.2. The uniform bound for the reciprocal denominator allows the amplitude-class and rounding errors to be chosen smaller than this positive difference.

The principal term. Since $437/500 < B < 7/8$, Lemma 17.1 gives

$$H_\eta(s) = 1 + O(P_0^{-4/5}), \qquad |H_\eta(s) - 1| \le 1/2 \quad (\operatorname{Re} s \ge B).$$

Choose P_0 independently of η. After fixing η, enlarge S to include the primes dividing its chosen defining modulus; this preserves the bound. The only principal frequency is $u = 1$, with residues at $w = 1, z = 1/6$ and normalizing constant c_S, including the factor $1/6$. Proposition 17.2 supplies the principal term (17.6) and exponent savings

$$l_y/20, \qquad h/600, \qquad 0 < \kappa_P < B \min_i \ell_i.$$

As in Proposition 17.2, bound the approximation error on $\operatorname{Re} s = \beta_* + \epsilon_{\mathrm{b}}$ and move only the main Gaussian integral to $\operatorname{Re} s = 2$.

Application of Proposition 3.1. Let $m_I > 0$ be the saving for the intermediate norm range and choose

$$0 < \mu_0 < \min\{\Delta/4, m_F/4, m_S/4, m_I, l_y/80, h/2400\}, \qquad 2\zeta < \mu_0.$$

The order of the remaining fixed choices is

$$\text{moment loss} \longrightarrow \text{prime-length decrements} \longrightarrow \max_i \ell_i \longrightarrow \text{even } K \text{ and lengths } \ell_i \longrightarrow \kappa_P \longrightarrow \text{class, amplitude and weight losses} \longrightarrow \eta.$$

Decrease the common saving as necessary. Logarithmically many configurations are absorbed in these losses. The proof of Proposition 20.1 uses the chosen count only through its exponent R, the displayed slope bounds and the uniform hypotheses on exponent losses and individual prime lengths, all verified above for R_κ.

Proposition 20.1, with the count of Proposition 29.1, now gives

$$|J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}, \qquad \mu > 0,$$

Here μ is target-independent; A_η, B_η are finite nonnegative exponents, and N is the freely chosen order for the discarded-height estimate, as in (20.2). Fix the arithmetic data and internal Fourier orders before A_η, B_η, and then choose

$$T_1 = Z^{\tau_\eta}, \qquad 0 < \tau_\eta \le \min\{d_{\min}/100, \tau_{\det}, \mu/[4(A_\eta+1)]\},$$

$$B_\eta - N\tau_\eta < C_b(\beta_*) - \mu/2 \quad \Longrightarrow \quad |J_\eta - f_\eta| \ll_\eta Z^{C_b(\beta_*)-\mu/2}.$$

The external order N changes seminorm constants, not the fixed internal degrees; J_η, f_η are independent of this height choice. Apply Proposition 3.1 with

$$(\sigma_0, c, \omega_0, \sigma) = \big(B, -(4+b)/6, \Delta/2, \mu/2\big).$$

The direct bound (32.2) and the bound $|H_\eta - 1| \le 1/2$ imply continuation across a zero sufficiently near β_*. This contradiction proves $\beta_* \le B$. □

33. Proof of the limiting zero-free bound by iteration

In this section, we iterate the one-step improvement and pass to the limit to prove Theorem 27.1. The limiting root satisfies (33.1), its explicit expression is proved in Lemma 33.1, and the limiting parameters are given in (33.2). Proposition 32.1 applies the continuation criterion of Proposition 3.1 at each finite step. Section 34 proves optimality in the stated comparison system.

Put

$$p(\xi) = 1188\xi^3 - 2160\xi^2 + 171\xi + 190. \tag{33.1}$$

The endpoint signs and $p' < 0$ on I give a unique zero $\xi_{\text{new}} \in I$.

Lemma 33.1 (Explicit root)**.** *The root ξ_{new} is given by (27.1).*

Proof. With $u = 66\xi - 40$,

$$242p\left(\frac{u+40}{66}\right) = u^3 - 4173u - 56940.$$

Set

$$\theta = \frac{1}{3}\arccos\frac{28470}{1391^{3/2}} \in (0, \pi/6).$$

For $u = 2\sqrt{1391}\cos(\theta - 2\pi/3)$,

$$u^3 - 4173u = 2\cdot 1391^{3/2}\cos(3\theta) = 56940.$$

Moreover,

$$-\sqrt{1391} < u < 0 \quad \Longrightarrow \quad 0 < \frac{40 - \sqrt{1391}}{66} < \xi < \frac{20}{33} < 1.$$

Since $p(0) > 0 > p(1)$ and p has a root in each of $(-\infty, 0)$, $(0, 1)$ and $(1, \infty)$, this is its unique root in $(0, 1)$, hence the isolated root in I. □

Define

$$\kappa_{\text{new}} = \kappa(\xi_{\text{new}}), \quad B_{\text{new}} = B(\xi_{\text{new}}), \quad b_{\text{new}} = b(\xi_{\text{new}}), \quad \ell_{\text{new}} = \ell(\xi_{\text{new}}). \tag{33.2}$$

The identity

$$\kappa(\xi) - (2B(\xi) - 1) = -\frac{p(\xi)}{18(2\xi + 1)(36\xi^2 - 78\xi + 25)} \tag{33.3}$$

gives $\kappa_{\text{new}} = 2B_{\text{new}} - 1$.

Proof of Theorem 27.1. On I,

$$\kappa'(\xi) = \frac{9(24\xi^2 - 20\xi + 5)}{2(36\xi^2 - 78\xi + 25)^2} > 0, \qquad B'(\xi) > 0, \qquad \kappa' - 1000B' > 0.$$

Let $\xi(\kappa)$ denote the inverse of the strictly increasing function $\xi \mapsto \kappa(\xi)$ on I, and set $g(\kappa) = 2B(\xi(\kappa)) - 1$. Then

$$0 < g'(\kappa) = \frac{2B'(\xi(\kappa))}{\kappa'(\xi(\kappa))} < 1/500,$$

$$\kappa_{\text{new}} < \kappa \le 3/4 \quad \Longrightarrow \quad \kappa_{\text{new}} < g(\kappa) < \kappa,$$

by (33.3) and $p' < 0$. Starting with the known 7/8 bound, define

$$\kappa_0 = 3/4, \qquad \xi_n = \xi(\kappa_n), \qquad B_n = B(\xi_n), \qquad \kappa_{n+1} = g(\kappa_n) = 2B_n - 1.$$

Proposition 32.1 gives

$$\beta_* \le (1 + \kappa_n)/2 \quad \Longrightarrow \quad \beta_* \le B_n = (1 + \kappa_{n+1})/2.$$

Thus $B_0 = B_{\text{alg}}$ and

$$\kappa_n \downarrow \kappa_{\text{new}}, \qquad B_n \downarrow B_{\text{new}}, \qquad \beta_* \le \inf_{n \ge 0} B_n = B_{\text{new}}.$$

At each finite step the preceding bound supplies the zero-free hypothesis required by Proposition 32.1. Passing the real inequalities $\beta_* \le B_n$ to the limit does not require the analytic constants or prime intervals to be uniform in n. Corollary 14.1 supplies the Dirichlet transfer. □

The limiting parameters are

B_{new}	0.874957019420098946…	κ_{new}	0.749914038840197892…
b_{new}	0.123405685419541002…	ℓ_{new}	0.166838588986270882….

Substituting $\xi = (9B - 7)/(18(1 - B))$ into (33.1) gives

$$7884B_{\text{new}}^3 - 18819B_{\text{new}}^2 + 14643B_{\text{new}} - 3686 = 0. \tag{33.4}$$

The choice of root is specified by (27.1).

Corollary 33.2 (An explicit rational bound)**.** *Every L-function in Theorem 27.1 is zero-free for*

$$\operatorname{Re} s > B_{\text{cp}} := \frac{34998281}{40000000} = 0.874957025.$$

Proof. For $\xi_1 = (9B_{\mathrm{cp}} - 7)/(18(1 - B_{\mathrm{cp}})) \in I$,

$$p(\xi_1) < 0 \quad \Longrightarrow \quad \xi_{\mathrm{new}} < \xi_1 \quad \Longrightarrow \quad B_{\mathrm{new}} < B_{\mathrm{cp}}.$$

□

The two improvements are

$$B_{\mathrm{alg}} - B_{\mathrm{cp}} = 4.4799168740\ldots \times 10^{-8},$$
$$B_{\mathrm{alg}} - B_{\mathrm{new}} = 5.03790697938897\ldots \times 10^{-8}.$$

Both require changing the lengths: fixing $\ell = \ell_{\mathrm{alg}}$ leaves $11/12 - \ell/4 = B_{\mathrm{alg}}$.

34. Proof of optimality in the full comparison system

In this section, we prove that the specified exponent comparison system admits no bound smaller than B_{new}. Theorem 27.1 establishes that B_{new} is a valid zero-free bound. Proposition 34.1 first proves the affine lower bound from (34.2) and (34.3). We then show that the permitted moment, length, detector and contour choices cannot evade its constraints, leading to Theorem 34.4.

Recall that $M = l_x + l_y$ is the total averaging-length exponent, $b = l_y - l_x$, and ℓ is the total selected prime length. The comparison exponent $E_B(d)$ is defined in (18.4), with R chosen from the permitted moment estimates. For $13/18 \le \kappa \le 1$, let $\xi(\kappa)$ be the smaller root of (30.3), and set

$$I_0 = [19/50, 39/100], \qquad F(\kappa) = \frac{18\xi(\kappa) + 7}{18\xi(\kappa) + 9}.$$

Writing the critical quadratic as $q_\kappa(\xi)$, one has

	$\kappa = 13/18$	$\kappa = 1$
$\xi = 19/50$	$2776/1875$	$1507/625$
$\xi = 39/100$	$-4003/15000$	$-871/5000.$

Since q_κ is affine in κ, its leading coefficient is positive and $q_\kappa(1) = 19 - 102\kappa < 0$,

$$19/50 < \xi(\kappa) < 39/100 < 1 < \text{the second root}.$$

On I_0, put $D_0(\xi) = 36\xi^2 - 78\xi + 25$. Then

$$D_0' < 0, \qquad D_0 \ge D_0(39/100) = 139/2500 > 0,$$
$$\kappa'(\xi) = \frac{9(24\xi^2 - 20\xi + 5)}{2D_0(\xi)^2} > 0, \qquad B'(\xi) = \frac{36}{(18\xi + 9)^2} > 0,$$
$$p''(\xi) < 0, \qquad p'(\xi) \le p'(19/50) = -597474/625 < 0.$$

Thus $\xi(\kappa)$ inverts the rational function (30.2) on this larger algebraic interval. The uniform endpoint inequality is proved on I.

For arbitrary $M, \ell \in \mathbb{R}$, define

$$L_0 = 5/6 + M/12 - \ell/6, \qquad L_K = 1/3 + 7M/12 + \ell/3, \tag{34.1}$$
$$H_\xi = 1 + \xi - (1 + \xi)M/2 + \xi\ell.$$

Proposition 34.1 (Variable-moment algebraic lower bound)**.** *Let* $13/18 \le \kappa \le 1$ *and* $M, \ell, B \in \mathbb{R}$. *If*

$$B \ge L_0, \qquad B \ge L_K, \qquad B \ge H_{\xi(\kappa)}, \tag{34.2}$$

then $B \ge F(\kappa)$, *with equality at*

$$\ell = \frac{5 - 6\xi(\kappa)}{9 + 18\xi(\kappa)}, \qquad M = 1 - \ell.$$

Under the additional condition on the candidate bound,

$$B \le (1+\kappa)/2, \tag{34.3}$$

one has $\kappa \ge \kappa_{\mathrm{new}}$ *and* $B \ge B_{\mathrm{new}}$. *Equality is attained by* (33.2).

Proof. For $\xi = \xi(\kappa)$, the weights

$$(w_0, w_K, w_H) = \frac{(4+18\xi, 2, 3)}{9+18\xi}$$

are positive and satisfy

$$\begin{aligned} w_0 + w_K + w_H &= 1, \\ w_0/12 + 7w_K/12 - (1+\xi)w_H/2 &= 0, \\ -w_0/6 + w_K/3 + \xi w_H &= 0. \end{aligned}$$

Consequently

$$B \ge \max\{L_0, L_K, H_\xi\} \ge w_0 L_0 + w_K L_K + w_H H_\xi = \frac{18\xi+7}{18\xi+9} = F(\kappa). \tag{34.4}$$

At the displayed M, ℓ, $L_0 = L_K = H_\xi = F(\kappa)$. Moreover (33.3) gives

$$\begin{aligned} F(\kappa) \le B \le (1+\kappa)/2 &\Longrightarrow g(\kappa) = 2F(\kappa) - 1 \le \kappa \\ &\Longleftrightarrow p(\xi) \le 0 \Longleftrightarrow \xi \ge \xi_{\mathrm{new}} \Longleftrightarrow \kappa \ge \kappa_{\mathrm{new}}. \end{aligned}$$

Monotonicity yields $B \ge F(\kappa) \ge F(\kappa_{\mathrm{new}}) = B_{\mathrm{new}}$. The exact identities for this positive weighted sum are checked in the supplement. □

The condition $B \le (1+\kappa)/2$ alone does not justify applying the moment estimate; this requires a previously proved bound for the family. For a known $\beta_* \le \widehat{B}$, the relations are

$$\kappa \ge 2\widehat{B} - 1, \qquad B \ge F(\kappa) \ge F(2\widehat{B} - 1).$$

If $2\widehat{B} - 1 \in \kappa(I)$ and is at most $3/4$, Proposition 32.1 realizes the last boundary. In particular $F(2B_{\mathrm{new}} - 1) = B_{\mathrm{new}}$.

34.1. **The optimal counting exponent in the stated moment system.** In this subsection, we optimize the exponents from the separate moment estimates with zero losses and no additional bound on the total selected prime length. We show that varying the detecting lengths and truncation exponent cannot reduce the supplied counting exponent below $2/3$ at the critical class, as stated in Lemma 34.2. Lemma 34.3 then derives the three necessary comparison constraints, including the nonprincipal constraint used in Theorem 34.4. The length exponents r, m of M_r, S_m in Section 5 and the detecting cutoff t of Proposition 29.1 satisfy

$$t - 1/2 \le r \le t, \qquad 0 \le m \le 1/2, \qquad r + m \ge t, \qquad 1 \le t \le 3/2. \tag{34.5}$$

Here κ is the smallest parameter allowed by the zero-free hypothesis; $\kappa \mapsto (1-2m)/(6\kappa)$ is decreasing. Fix

$$\delta = \xi(\kappa) \in I_0, \qquad x = 1/2, \qquad c = 2/(3\kappa) \in [2/3, 12/13], \qquad \alpha = 5/6.$$

The optimized exponents with combined range parameter $M = 1$ are

$$\begin{aligned} e_I(r) &= 1 - \delta(1+r)/2 && (r \le 1), \\ e_S(m) &= 1 - \delta\big[2m + (c/4)(1-2m)\big] && (0 \le m \le 1/2), \end{aligned} \tag{34.6}$$

and the amplified long exponent is

$$e_L(r) = 1 - \alpha + (\alpha - \delta)r \qquad (r \ge 1). \tag{34.7}$$

The unselected alternatives satisfy

$$1-\delta r \ge e_I(r) \quad (r \le 1), \qquad \{z>0 : r+2z \le 1\} = \varnothing \quad (r>1).$$

For the set $\mathcal{W}_t$ of admissible length pairs in (34.5), put

$$e_{\mathrm{inv}}(r) = \begin{cases} e_I(r), & r \le 1, \\ e_L(r), & r \ge 1, \end{cases} \qquad R_{\mathrm{cert}}(t) = \sup_{(r,m)\in\mathcal{W}_t} \min\{e_{\mathrm{inv}}(r), e_S(m)\}.$$

These are supplied exponents, not lower bounds for actual counts.

Lemma 34.2 (Optimal counting exponent at the critical class)**.** *For the moment estimates and admissible length pairs above, the smallest uniform upper-bound exponent obtainable from these estimates is*

$$\inf_{1\le t\le 3/2} R_{\mathrm{cert}}(t) = R_\kappa(\delta, 1/2) = 2/3.$$

Upper bounds on the selected prime lengths, estimates without selected prime factors and Hölder interpolation between these estimates cannot lower it.

Proof. On $m = t-r$, the crossing in (29.3) satisfies

$$t - 1/2 \le r_\kappa(t,1/2) \le 1, \qquad 1/3 \le t - r_\kappa(t,1/2) \le 1/2,$$
$$e_I' = -\delta/2 < 0, \qquad \frac{d}{dr} e_S(t-r) = \delta(2-c/2) > 0.$$

Hence this admissible split imposes

$$\begin{aligned} R_{\mathrm{cert}}(t) &\ge \min\{e_I(r_\kappa), e_S(t-r_\kappa)\} \\ &= R_{\kappa,\mathrm{short}}(t) = 1-\delta + \frac{\delta P_{\kappa,1/2}}{D_{\kappa,1/2}}(3/2-t). \end{aligned}$$

At $t = 3/2$, $(r,m) = (1,1/2)$; the inverse and unweighted estimates without auxiliary prime factors both give $1-\delta$. This avoids applying an estimate with selected prime factors when its strict length inequalities hold only at equality.

The second admissible split $(r,m) = (t,0)$ imposes $R_{\mathrm{cert}}(t) \ge R_{\mathrm{long}}(t) = e_L(t)$, because

$$e_S(0) - e_L(t) \ge -5/12 + \delta(3/2 - c/4) \ge -5/12 + (19/50)(5/4) = 7/120 > 0.$$

If a discrete presentation omits $m = 0$, take $(r,m) = (t-\epsilon, \epsilon)$ and pass to the scalar limit. The short/long derivation in Proposition 29.1 gives the reverse uniform bound. Thus

$$\begin{aligned} R_{\mathrm{cert}}(t) &= \max\{R_{\kappa,\mathrm{short}}(t), R_{\mathrm{long}}(t)\}, \\ R_{\kappa,\mathrm{short}}' < 0, &\qquad R_{\mathrm{long}}' = \alpha - \delta > 0, \\ R_{\kappa,\mathrm{short}}(t_\kappa) &= R_{\mathrm{long}}(t_\kappa) = R_\kappa(\delta,1/2) = 2/3. \end{aligned}$$

Restoring the upper bounds on selected prime lengths weakens the savings from the separate moment estimates. At each split, for $0 \le \theta \le 1$,

$$\theta e_I + (1-\theta) e_S \ge \min(e_I, e_S).$$

The same minimum includes the estimates without selected prime factors; hence neither choice nor Hölder interpolation lowers this optimal uniform exponent. □

Enlarging the combined range exponent in the ordinary moment estimate also cannot help. For $M' \ge 1$, write

$$I' = M' - \delta r - \delta z, \qquad S' = M' - 2\delta m - \delta z.$$

The affine inequalities for their range exponents give

$$\begin{aligned} &r \le 1,\ z > z_I = (1-r)/2: && I' - e_I(r) \ge (2-\delta)(z - z_I) > 0,\\ &r > 1: && I' \ge r(1-\delta) \ge e_L(r),\\ &z > z_S = (1-2m)/(6\kappa): && S' - e_S(m) \ge (6\kappa - \delta)(z - z_S) > 0. \end{aligned}$$

For selected lengths below these bounds, $M' \ge 1$ and smaller amplitude savings likewise give no improvement. Other displayed unselected exponents satisfy

$$R_{\mathrm{I}}(\delta) \ge 1 - \delta/2 \ge 161/200 > 2/3,$$
$$R_N = 1 - \frac{2\alpha\delta}{3\alpha - \delta} \ge 1 - \frac{2\alpha(39/100)}{3\alpha - 39/100} > 2/3.$$

At $x = 1/2$, each auxiliary prime sum gives a maximal saving $\delta/2$ per unit length. Redistribution, finite unions and separate detector choices introduce no stronger exponent: each detector permits the two adverse splits above. A relation between the large values arising from different detector choices, or a new joint moment estimate, would require an additional estimate.

Lemma 34.3 (Necessary constraints for a putative improvement)**.** *Suppose a construction in the stated comparison system has $B < B_{\mathrm{new}}$, with $l_x, l_y > \ell > 0$, $b = l_y - l_x > 0$, $M = l_x + l_y$ and $h = 1 - l_x + \ell > 0$. Use the full direct majorant (22.4) and uniform nonprincipal comparisons for every permitted class. If $13/18 \le \kappa \le 1$ and the supplied critical count exponent is at least $2/3$, then necessarily*

$$B \ge L_0, \qquad B \ge L_K, \qquad B \ge H_{\xi(\kappa)}.$$

If also $B \le (1+\kappa)/2$, these inequalities contradict Proposition 34.1.

Proof. For $G = \max\{0, b/12, b - l_y/2\} \ge b/12$,

$$\begin{aligned} B \ge B_{\mathrm{low}} &= 5/6 + l_x/6 + G - \ell/6 + k(M,\ell)/2\\ &\ge L_0 + k(M,\ell)/2 \ge \max\{L_0, L_K\}, \end{aligned}$$

using the inequalities $k \ge 0$ and $k \ge M + \ell - 1$. The inequalities for the length parameters consequently imply

$$\begin{gathered} M - 2\ell > b > 0, \qquad B \ge L_0 > 5/6,\\ M - 2\ell \le 12B - 10, \qquad h \ge 6 - 6B + b/2 > 3/4, \qquad h < 1,\\ M < 13/14, \qquad l_y < M, \qquad (1+\xi)/2 < 139/200 < 5/6 < B. \end{gathered}$$

Thus $d = h$ is a moderate norm and the critical zero class is below any upper cutoff supplied by $\beta_* > B$.

At $(d, \delta, x) = (h, \xi, 1/2)$, direct substitution gives

$$E_B(h) = H_\xi - B + h(R - 2/3). \tag{34.8}$$

All z_0 terms cancel, and the lower bound H_ξ is independent of b. Since $h > 0$ and $R \ge 2/3$, $E_B(h) \le 0$ requires $B \ge H_\xi$. Raising the lower zero-class threshold to $\delta_f \ge \xi$ instead gives

$$E_{B,\mathrm{floor}}(h) = H_{\delta_f} - B + h/3,$$
$$H_{\delta_f} - H_\xi = (\delta_f - \xi)(1 - M/2 + \ell) \ge 0, \qquad 1 - M/2 + \ell > 3/4.$$

Its trivial-count comparison therefore requires the stronger $B \ge H_\xi + h/3$. A threshold below ξ retains the critical class.

Global reciprocal contours. For these comparisons assume $B < \beta_* \le 7/8$. Put $s_r = \beta_* + \gamma$, $\gamma > 0$, $w_r = \omega$ and $z_r = \zeta$. For $\omega > 0$, $\zeta > 1/6$ and $s_r + \omega > 1$, the local error exponents for $p \nmid u$

$$-s_r - \omega,\ -s_r - 6\zeta,\ -\omega - 6\zeta,\ 4 - 6s_r - 6\zeta,\ 1 - s_r - \omega - 6\zeta$$

are < -1 and the local error exponents for $p \mid u$ are negative. The local table and quotient-free selected formula give

$$G_p = O(1)\ (p \nmid u), \qquad G_p \ll q_p^{\max(1-\omega,0)}\ (p \mid u),$$
$$\text{full positive tuple sum} \ll U^{\varepsilon} Z^{\ell\zeta}, \qquad P_i^{\zeta - \min(\omega,1)} \le P_i^{\zeta}.$$

For $0 < \omega \le 11/10$, the strip bound [Ope26a, Lemma 4.8] contributes $U^{3/5+\varepsilon}$, giving the relative dyad exponent

$$\gamma + h(\zeta - 1/6) + l_y(\omega - 1) + d(8/5 - \zeta) + O(\varepsilon).$$

At $d = h$, it is bounded below by

$$\begin{aligned}\gamma + 43h/30 + l_y(\omega - 1) &\ge 43h/30 - \beta_* l_y + \gamma(1 - l_y)\\ &> 43/40 - (7/8)(13/14) = 21/80 > 0.\end{aligned}$$

For $\omega > 11/10$, Euler convergence gives instead $\gamma + 5h/6 + l_y(\omega - 1) > 0$.

For $-1/100 \le \omega \le 0$, admissibility requires $s_r + \omega > 1$ and $\zeta \ge 17/50$. Even when the additional nonnegative losses are omitted, the endpoint is $> 21/80$; the selected-divisor factor adds $-\omega\ell \ge 0$ and deleted numerator factors add only nonnegative conductor powers.

Absolute and principal contours. For any eligible fixed $s_r, w_r > 1$ and sufficiently large z_r, Euler convergence and the same local table give an endpoint exponent relative to $C(B)$ of

$$(s_r - B) + l_y(w_r - 1) + 5h/6 > 1 - B + 5h/6 > 3/4.$$

It remains positive relative to $C(\beta_*)$, since $\beta_* - B < 1/24$. At $d = h$ all z_r terms cancel. Principal contour changes affect only the exponent savings for the error terms: the residues $w = 1$, $z = 1/6$ and the exponent of the principal term are fixed. Thus the displayed central, unselected, global and absolute comparisons cannot evade $B \ge H_\xi$. Only the stated local, strip-growth and Euler-convergence estimates are included; new estimates or residue extractions change the system. □

34.2. **The optimality theorem for the specified comparison system.** In this subsection, we specify the full comparison system and combine the preceding lower bounds to prove Theorem 34.4. The admissible set $\mathscr{A}$ is defined immediately below; the theorem proves $\inf_{\mathfrak{C}\in\mathscr{A}} B(\mathfrak{C}) = B_{\text{new}}$.

Let $\mathscr{A}$ be the set of parameter choices satisfying the following comparison inequalities, with all exponent losses set to zero:

$$l_x, l_y > \ell > 0, \quad b = l_y - l_x > 0, \quad M = l_x + l_y, \quad h = 1 - l_x + \ell > 0,$$
$$13/18 \le \kappa \le 1, \qquad B \le (1 + \kappa)/2,$$
$$B \ge B_{\text{low}}, \qquad E_B(d) \le 0 \quad \text{for every permitted zero and amplitude class.}$$

Here $E_B(d)$ is the exponent in (18.4), and the zero and amplitude parameters come from (5.1) and (10.1). The supplied count choices are those of Section 34.1; the contour choices, lower zero-class thresholds and norm ranges are those examined in the proof of Lemma 34.3. There is no additional bound on the total selected prime length, and all positive losses are suppressed. Analytic applications retain the original estimates' coefficient, coprimality, support and domain conditions. Uniform scalar comparisons include

classes whose nonemptiness is unknown; a fixed positive gap $\beta_* - B$ does not replace the requirement $E_B(d) \le 0$.

Theorem 34.4 (Optimality of the displayed exponent comparison)**.** *For the admissible parameter choices $\mathfrak{C} \in \mathscr{A}$,*

$$\inf_{\mathfrak{C}\in\mathscr{A}} B(\mathfrak{C}) = B_{\mathrm{new}}.$$

Every such choice satisfies $B(\mathfrak{C}) \ge B_{\mathrm{new}}$; equality is attained when all exponent losses are set to zero, using (33.2). This includes free M, all detecting cutoffs in $[1, 3/2]$, the specified estimates with and without selected prime factors, and the auxiliary choices above.

Proof. If $B < B_{\mathrm{new}}$, Lemmas 34.2 and 34.3 give $B \ge \max(L_0, L_K, H_\xi)$. Proposition 34.1 and $B \le (1+\kappa)/2$ then give $B \ge B_{\mathrm{new}}$, a contradiction. Conversely, at (33.2),

$$B_{\mathrm{low}} = B_{\mathrm{new}} = L_0 = L_K = H_{\xi_{\mathrm{new}}}, \qquad B_{\mathrm{new}} = (1+\kappa_{\mathrm{new}})/2.$$

Lemma 31.1 gives every central comparison; Lemma 31.2 supplies the remaining strict auxiliary inequalities. Hence these parameters satisfy every defining inequality of $\mathscr{A}$ when the exponent losses are set to zero. Their analytic conclusion is the limit of the finite steps justified by the preceding zero-free bounds in Theorem 27.1, not a moment application with an unproved endpoint hypothesis. □

Corollary 34.5 (Arbitrary successive admissible constructions)**.** *At each improving step of this system, allow arbitrary admissible choices of the length parameters, the zero-detector truncation exponent t, the moment parameter κ and the other auxiliary parameters. If each moment hypothesis comes from the preceding proved boundary, every resulting bound and every real limit is at least B_{new}. The sequence in Section 33 attains this infimum.*

Proof. For an improving step justified by the preceding zero-free bound,

$$\beta_* \le \widehat{B}, \qquad \widehat{B} \le (1+\kappa)/2, \qquad B \le \widehat{B} \Longrightarrow B \le (1+\kappa)/2.$$

The theorem applies independently to each step; its closed lower bound persists under real limits. □

The optimality statement concerns the uniform counting and comparison exponents obtained from the estimates specified above. An estimate excluding a critical class, a relation between the large values arising from different detector choices, or a stronger analytic estimate changes the system.

34.3. **Rational approximation and the scope for further improvements.** In this subsection, we give a one-step improvement with rational parameters in Proposition 34.6, with the exact inequalities in (34.9), and determine the infimum when the parameters are required to be rational. The rational construction gives an explicit approximation to the algebraic limiting bound without changing its infimum. We then distinguish variation of parameters already covered by Theorem 34.4 from stronger analytic estimates that could improve the bound.

Proposition 34.6 (A one-step bound with rational parameters)**.** *Recall the rational boundary B_{cp} of Corollary 33.2. Put*

$$(\kappa_{\mathrm{cp}}, b_{\mathrm{cp}}, \ell_{\mathrm{cp}}) = \left(\frac{74991414}{10^8}, \frac{617}{5000}, \frac{5005157}{30000000}\right), \qquad 11/12 - \ell_{\mathrm{cp}}/4 = B_{\mathrm{cp}}.$$

With the coefficient, coprimality, support and smooth-weight hypotheses of the modified average (7.2) and its normalization (17.6), these parameters give

$$\beta_* \le B_{\mathrm{alg}} \quad \Longrightarrow \quad \beta_* \le B_{\mathrm{cp}}.$$

Thus the explicit rational bound also follows in one step from Part III.

Proof. Suppress the subscript cp and set $l_x = (1-b-\ell)/2$, $l_y = l_x + b$, $h = 1 - l_x + \ell$ and $c = 2/(3\kappa)$. The moment hypothesis follows by squaring positive quantities:

$$q := \frac{1507 - 1653(1+\kappa)/2}{2} = \frac{6069596329}{200000000} > 0,$$

$$921 - q^2 = \frac{402989723759}{40000000000000000} > 0 \quad \Longrightarrow \quad B_{\mathrm{alg}} < (1+\kappa)/2, \qquad 13/18 < \kappa < 3/4.$$

Use (31.2) at this rational triple, without imposing the critical-class equations, and write

$$Q = A(y)\delta^2 + B_q(y)\delta + C_q(y), \quad A(y) = \sum_{j=0}^{3} A_j y^j, \quad 4A(y)C_q(y) - B_q(y)^2 = \sum_{j=0}^{4} d_j y^j.$$

Let $\mathcal{S}_{\mathrm{cp}}$ consist of the 24 expressions in (31.3) after omitting $36\xi^2 - 78\xi + 25$, κ', B', $\kappa' - 1000B'$ and $-p'$, and substituting the rational parameters above into the remaining expressions. Exact substitution gives the following *system of exact rational inequalities* $\mathcal{C}_{\mathrm{cp}}$:

$$A_j > \frac{1}{4}\ (0 \le j \le 3), \qquad d_j > \frac{1}{10^7}\ (0 \le j \le 4), \qquad s > \frac{1}{25000}\ (s \in \mathcal{S}_{\mathrm{cp}}). \tag{34.9}$$

The accompanying exact-arithmetic certificate file [Liu26] records these lower bounds and all 33 exact values in its `rational_checkpoint` entry. Consequently, for $y = 1/2 - x \ge 0$,

$$4A(y)Q = (2A(y)\delta + B_q(y))^2 + \sum_{j=0}^{4} d_j y^j > 0, \qquad E_B(h) = -Q/(2\mathcal{J}_\kappa) < 0$$

on $[0, 5/6] \times [0, 1/2]$.

The auxiliary inequalities give the balanced direct hypotheses, $\ell/h > 1/5$, the principal domain and all required exponent savings for the lowest zero class, the intermediate and small norm ranges, the prime lengths and the rounding errors used in Section 32. Proposition 29.1 therefore applies with the preceding moment hypothesis, and its exponent satisfies $0 < \partial_d E_B < 2$ on the actual class range. If $\Delta = \beta_* - B > 0$, the same small-, intermediate- and large-norm estimates, with $\mathcal{C}_{\mathrm{cp}}$ in place of the two ξ-family certificates, verify Proposition 20.1. Together with the direct and principal estimates, this gives

$$|J_\eta(Z)| \ll_\eta Z^{C_b(B)+\Delta/2}, \qquad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C_b(\beta_*)-\sigma}, \qquad \sigma > 0,$$

with σ independent of η and $|H_\eta - 1| \le 1/2$ on $\mathrm{Re}\, s > B$. Proposition 3.1 is the desired contradiction; Corollary 14.1 transfers the bound to Dirichlet L-functions. □

At the limit,

$$\ell_{\mathrm{new}} = 11/3 - 4B_{\mathrm{new}}, \qquad 53835237b_{\mathrm{new}}^3 + 381756141b_{\mathrm{new}}^2 + 94706196b_{\mathrm{new}} - 17602208 = 0.$$

The B-cubic (33.4) and this b-cubic reduce modulo 7 to

$$2X^3 - 3X^2 - X + 3, \qquad X^3 - 3X^2 - 3X - 1,$$

with value lists

$$(3, 1, 5, 6, 2, 5, 6), \qquad (6, 1, 3, 4, 3, 6, 5).$$

Their leading coefficients are nonzero and neither has a root; Gauss's lemma proves irreducibility. Thus

$$[\mathbb{Q}(B_{\mathrm{new}}) : \mathbb{Q}] = [\mathbb{Q}(\ell_{\mathrm{new}}) : \mathbb{Q}] = [\mathbb{Q}(b_{\mathrm{new}}) : \mathbb{Q}] = 3.$$

The supplement also verifies the eliminations.

Rational parameters have the same infimum. Given $\epsilon > 0$, choose, after finitely many iterations, a bound $B_n < B_{\mathrm{new}} + \epsilon/2$ and a slightly larger rational bound $B_n < \widetilde{B} < \min\{B_{\mathrm{new}} + \epsilon, (1+\kappa_n)/2\}$. This makes the direct and central inequalities strict. On the compact class rectangle their exponents are continuous; a rational $\widetilde{\kappa} > \kappa_n$ sufficiently close to κ_n preserves the preceding hypothesis, and sufficiently close rational lengths preserve all strict exponent comparisons and auxiliary inequalities. Hence

$$\inf_{\mathscr{A} \text{ with rational parameters}} B = \inf_{\mathscr{A}} B = B_{\mathrm{new}}.$$

The first infimum is not attained: positivity of the weights forces $L_0 = L_K = H_\xi = B_{\mathrm{new}}$ at equality, so $\ell = \ell_{\mathrm{new}} \notin \mathbb{Q}$.

The total length M is already free, balance occurs at equality, and t is already minimized. The variables δ, x, d are class coordinates requiring uniform control. Upper bounds on selected prime lengths and the losses from rounding and smoothing can only weaken the relaxed comparison; the central and remote contour coordinates cancel at $d = h$. A stronger analytic estimate can change the obstruction. For example, with the present D, P, formally vary α and put

$$R = 1 - \delta + \frac{(\alpha - \delta)\delta P}{2\mathcal{J}}, \qquad \mathcal{J} = (\alpha - \delta)D + \delta P.$$

Then

$$\partial_\alpha R = \frac{\delta^2 P^2}{2\mathcal{J}^2} > 0, \qquad \partial_\alpha e_L(r) = r - 1 > 0 \quad (r > 1).$$

A smaller *proved* amplification exponent than $5/6$ improves these upper-bound exponents. Larger admissible prime lengths in the moment estimates, a joint moment estimate for the detecting polynomials, better functional-equation or Gram estimates, or restrictions on actual critical classes could likewise alter the comparison. Such changes require new results with the original coefficient, coprimality and height hypotheses; they are not additional free parameters in the theorem.

Appendix A. Theta coefficient conventions

In this section, we match the theta coefficients of Section 4.1 with the cited cusp expansions, recording the conjugations and fixed normalizing factors explicitly. This identifies the same arithmetic normalization throughout Parts I–IV.

The coefficient formulas, support and normalization are those of [DR24, equations (5.7)–(5.16) and Appendix A], with the conjugation convention in [Ope26a, Proposition 5.1, proof, cusp expansion and infinity coefficient]. Precisely, if d_i^{DR} denotes the coefficient in [DR24, equation (5.12)], then our three functions are

$$(d_0, d_1, d_2)(\mu) = \big(\overline{d_1^{\mathrm{DR}}(-\mu)}, \overline{d_{19}^{\mathrm{DR}}(-\mu)}, \overline{d_{10}^{\mathrm{DR}}(-\mu)}\big);$$

the corresponding coefficient formulas are rows $1, 19, 10$ of [DR24, Appendix A], reproducing [Pat77, Table III, p. 151]. The matrices H_j are the representatives $\gamma_1, \gamma_{19}, \gamma_{10}$ specified in [DR24, Section 5.1, before equation (5.9)]. These locators refer to the version specified in the bibliography.

For clarity, write τ^{DR} for the infinity coefficient in [DR24, equation (5.7)], and $g^{\mathrm{DR}}(1,c) = \sum_{x \bmod c} (x/c)_3 \check{e}(x/c)$ for its unnormalized cubic Gauss sum. The fourth case of that formula, with its integer exponent equal to zero, gives

$$\tau^{\mathrm{DR}}(\pm\lambda^{-3} c n^3) = 3^{5/2} \frac{|n|}{|c|} \overline{g^{\mathrm{DR}}(1,c)}.$$

[DR24, equation (5.16), p. 19] uses the conjugated coefficient at $-\mu$, while [DR24, equation (5.7), p. 17] conjugates the Gauss sum in the infinity coefficient; hence, since $|c| = \sqrt{q_c}$,

$$d_0(\lambda^{-3}cn^3) = \overline{\tau^{\mathrm{DR}}(-\lambda^{-3}cn^3)} = 3^{5/2}|n|\widetilde{g}_3(c).$$

Finally, [DR24, equations (5.6), (5.15), (5.16)] gives the constant term $3^{5/2}v^{2/3}/2$ for the representative γ_1, and zero for γ_{19}, γ_{10}, exactly as in Section 4.1.

Appendix B. Index of cited results

In this section, we list the cited results used in Parts I–IV and indicate their roles in the argument. The first table indexes the cited results and deductions in Part I. The second records how the same results are used at the new parameters in Parts II and III. The third records the additional results and deductions in Part IV. Each citation identifies the proved result used in that step.

Reference	Function in the argument
[Ope26a, Proposition 2.1]	Holomorphic continuation with a uniform exponent.
[Ope26a, Lemmas 4.1–4.4]	Ray characters, Gauss identities and reciprocity phases.
[Ope26a, Lemmas 4.5–4.10]	Smooth separation, transform decay and Hecke bounds.
[Ope26a, Proposition 5.1; Lemmas 5.2–5.8]	Cubic theta functional equation and unweighted mean square.
[Ope26a, Lemmas 6.1–6.2; Proposition 6.3]	Balanced additive mean square and $Z^{1/4+\varepsilon}$ estimate.
[Ope26a, Lemma 7.1]	Exact scalar Euler quotient and holomorphic correction.
[Ope26a, Lemmas 8.1–8.2; Proposition 8.3]	Zero-free rectangles and simultaneous large character polynomials.
[Ope26a, Lemma 9.1; Proposition 9.2]	Sextic sieve and preliminary character count.
[Ope26a, Lemmas 10.2–10.6]	Contour tails, exponent accounting and principal residues.
[Ope26a, Lemma 11.1; Proposition 11.2; Theorem 3.1]	Height choice, first comparison and 11/12 conclusion.
[Ope26a, Lemmas 13.1–13.4]	Fixed-ray prime asymptotics and complete correlations.
[Ope26a, Corollary 14.1; Lemmas 14.2–14.3]	Divisibility-restricted theta functional equation and mean square.
[Ope26a, Lemma 15.1; Propositions 15.2–15.3]	Modified theta and additive norms; $Z^{3/16+\varepsilon}$.
[Ope26a, Proposition 16.1; Lemma 16.2]	Joint local errors and absolute tuple bounds.
[Ope26a, Lemmas 17.1–17.3, 17.5–17.6; Corollary 17.4]	Möbius-weighted moment, finite induction and sixth-power amplification.
[Ope26a, Lemmas 18.1–18.3]	Unweighted fourth moment and cancellation under $X_1X_2 = Y_1Y_2$.
[Ope26a, Lemma 19.1; Proposition 19.2]	Prime sum bounds and refined character counts.
[Ope26a, Lemmas 20.1–20.2; Proposition 20.3]	Final exponent inequality and uniform parameter choices.

Reference	Function in the argument
[Ope26a, Proposition 11.3]	Transfer to Dirichlet functions.

Cited result	Use in Parts II and III
[Ope26a, Lemma 14.3]	General exponents from the functional equation: Lemmas 16.1 and 22.1.
[Ope26a, Proposition 15.2]	Direct bounds with the full additive majorant: Propositions 16.2 and 22.2.
[Ope26a, Lemma 7.1]	Extended principal Euler region: Lemma 17.1.
[Ope26a, Lemma 13.1]; [Ope26a, Lemmas 10.2–10.6]	Normalization, residues after changing the averaging lengths and tails: Proposition 17.2 and Lemma 19.2.
[Ope26a, Proposition 8.3]; [Ope26a, Lemmas 17.1, 17.6, 18.1, 19.1]	Fixed-κ count with strict length inequalities: Proposition 18.1.
[Ope26a, Proposition 16.1]; [Ope26a, Lemma 16.2]	Joint conductor deficit and nonprincipal exponent for the new lengths: Proposition 18.2.
[Ope26a, Lemmas 4.5–4.10, 10.2–10.4, 11.1]	Smooth separation and combination of height estimates: Proposition 20.1.
Proposition 3.1 and Corollary 14.1	Continuation contradiction and Dirichlet transfer at both new boundaries.

Cited result	Use in Part IV
[Ope26a, Lemma 4.9]; [Ope26a, Section 18, proof of Lemma 18.1]	Prime displacement, induction on the combined range exponent, and the extended moment: Proposition 28.1. The proof in the original paper is extended, rather than its printed parameter range invoked unchanged.
[Ope26a, Proposition 8.3]; [Ope26a, Lemmas 17.1, 17.6, 19.1]	The original common detector and inverse amplification, combined with the extended unweighted moment in Proposition 29.1.
(18.5) and the exact Bernstein calculation of Section 31	Critical length parameters, the universal endpoint inequality, and strict auxiliary inequalities: Lemmas 30.1, 31.1, and 31.2.
Propositions 16.2, 17.2, 18.2; Lemma 17.1	Direct and transformed representations for the new lengths, principal residues, and every norm range in Proposition 32.1.
Proposition 3.1; Corollary 14.1	One-step continuation contradiction, iteration of the zero-free bounds, and Dirichlet transfer in Theorem 27.1.

References

[Ber12] S. Bernstein, Démonstration du théorème de Weierstrass fondée sur le calcul des probabilités, *Communications de la Société mathématique de Kharkow*, 2nd series, **13** (1912), no. 1, 1–2. Original paper and bibliographic record.

[BGL14] V. Blomer, L. Goldmakher, and B. Louvel, *L-functions with n-th-order twists*, Int. Math. Res. Not. IMRN **2014**, no. 7, 1925–1955. doi:10.1093/imrn/rns257.

[DR24] A. Dunn and M. Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, Ann. of Math. (2) **200** (2024), no. 3, 967–1057. doi:10.4007/annals.2024.200.3.3. Equation and appendix locators in this paper refer to arXiv:2109.07463v3, 14 May 2024.

[Far12] R. T. Farouki, The Bernstein polynomial basis: A centennial retrospective, *Computer Aided Geometric Design* **29** (2012), no. 6, 379–419. doi:10.1016/j.cagd.2012.03.001.

[GZ20] P. Gao and L. Zhao, *Moments of quadratic Hecke L-functions of imaginary quadratic number fields*, J. Number Theory **209** (2020), 359–377. doi:10.1016/j.jnt.2019.09.002. The equation locator refers to arXiv:1707.00091v3.

[GZ24] P. Gao and L. Zhao, *Moments and one level density of sextic Hecke L-functions*, Funct. Approx. Comment. Math. **70** (2024), no. 1, 7–28. doi:10.7169/facm/2057. The lemma locator refers to arXiv:2201.01885v2.

[GL13] L. Goldmakher and B. Louvel, *A quadratic large sieve inequality over number fields*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 2, 193–212. doi:10.1017/S0305004112000370.

[GM26] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) **203** (2026), no. 2, 623–675. doi:10.4007/annals.2026.203.2.6. The theorem locator refers to arXiv:2405.20552v2, 7 April 2026.

[HB95] D. R. Heath-Brown, *A mean value estimate for real character sums*, Acta Arith. **72** (1995), no. 3, 235–275. doi:10.4064/aa-72-3-235-275.

[HB00] D. R. Heath-Brown, *Kummer's conjecture for cubic Gauss sums*, Israel J. Math. **120**, part A (2000), 97–124. doi:10.1007/s11856-000-1273-y.

[Hux68] M. N. Huxley, *The large sieve inequality for algebraic number fields*, Mathematika **15** (1968), no. 2, 178–187. doi:10.1112/S0025579300002540.

[IK04] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004. doi:10.1090/coll/053.

[Kub69] T. Kubota, *On automorphic functions and the reciprocity law in a number field*, Lectures in Mathematics, Department of Mathematics, Kyoto University, no. 2, Kinokuniya Book-Store, Tokyo, 1969. Kyoto University repository.

[Liu26] B. Liu, *Lean formalizations of the rational and algebraic zero-free extensions*, accompanying source-code repository, 2026, Lean 4.34.1. GitHub repository for the paper and Lean formalizations. Paths below are relative to the repository root. The consolidated entry for Parts I–III is `combine_v2-2_lean/PartsI_III.lean`, with public namespaces `CombinedPaper`, `RHZeroFreeExtension` and `RHV2`; the Part IV entry is `combine_v2-2_lean/PartIV.lean`, with final conclusions in namespace `CombineV22`. The latter namespace exports `beta_star_le`, `hecke_nonzero`, `dirichlet_nonzero` and `zeta_nonzero`. The verifier `combine_v2-2_lean/check.py` replays the eight consolidated OAI foundation files and both manuscript files with `-trust=0`, then checks the explicit final signatures. The audited transitive axioms are `propext`, `Classical.choice` and `Quot.sound`.

[MP24] J. Maynard and K. Pratt, *Half-isolated zeros and zero-density estimates*, Int. Math. Res. Not. IMRN **2024**, no. 19, 12978–13014. doi:10.1093/imrn/rnae191. The appendix locator refers to the arXiv manuscript.

[M+17] A. Meurer et al., *SymPy: symbolic computing in Python*, PeerJ Comput. Sci. **3** (2017), e103. doi:10.7717/peerj-cs.103.

[Mil20] J. S. Milne, *Class Field Theory*, course notes, version 4.03, 6 August 2020. Author's notes.

[Ope26a] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\operatorname{Re} s > 7/8$, 30 September 2026. Preprint, 30 September version. Public manuscript. All result and equation locators for the original paper refer to this dated version.

[Ope26b] OpenAI, *The Quasi-Riemann Hypothesis (alternate* 11/12 *proof)*, preprint, 5 October 2026. Public manuscript.

[Ope26c] OpenAI, *The quasi-Riemann hypothesis: scope of the Lean formalization*, repository documentation, accessed 7 October 2026. Scope and comparator links.

[Pat77] S. J. Patterson, *A cubic analogue of the theta series*, J. Reine Angew. Math. **296** (1977), 125–161. doi:10.1515/crll.1977.296.125.

[Rie16] M. Riesz, *Sur l'hypothèse de Riemann*, Acta Math. **40** (1916), 185–190. Original paper and bibliographic record.

[TZ19] J. Thorner and A. Zaman, *A unified and improved Chebotarev density theorem*, Algebra Number Theory **13** (2019), no. 5, 1039–1068. doi:10.2140/ant.2019.13.1039.

Department of Mathematics, Purdue University, West Lafayette, IN, 47907, USA
Email address: liu2053@purdue.edu